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Anthrax

(**Syn:** Splenic fever, Milzbrand, Charbon, Wool sorters disease and Malignant carbuncle)

It is a peracute and acute septicaemic disease of cattle and sheep leading to sudden death.

Etiology

Bacillus anthracis

It is a Gram positive, non motile, capsulated, aerobic, spore forming organism. Which appear as straight large shaped rods arranged in small short chains with truncated ends

The organism produces spores when exposed to external environment and can live for a prolonged period outside the host in the environment

Epidemiology

The disease world wide in distribution

The disease is enzootic in India

The prevalence of the disease depends upon the nature of soil and climatic condition, because, the climatic parameter helps to forecast the “ **anthrax year**”

If the infection is restricted to a particular area is called as “ **anthrax belts**” or **enzootic region**

The **outbreak** is mostly occurring in **summer months**, after heavy rainfall following a draught

Economic impact s

Mortality, reduced milk yield and the expenses incurred for vaccination leads to severe economic loss

Predisposing factors and source of infection

In wet soil, and in stagnant water, the spores are concentrated and suspended

Alkaline soil (pH 9.0) preserves the vegetative bacilli

Blood transfusion

Contaminated bone meal, pasture contaminated with tannery effluents, fodders grown on infected soil, turning up of soil, blood and other discharges from infected animals are the major source of infection

Transmission

Ingestion, inhalation and contamination of wound by spores are prime mode of spread

Ingestion is the main mode of transmission in an animals

Inhalation is the main mode of transmission in human

Mechanical transmission by Tabanid fly

Host affected

Cattle and sheep are highly susceptible as compared to goat and horses

Human is less susceptible than cattle

Pigs, dogs and cats are relatively resistant

But once affected the case fatality rate is high in farm animals and pigs

Hippopotamus, Cape buffalo and elephants die in large numbers due to anthrax

Predators act as carrier for this infection

Zoonotic potential

B.anthraxis is an agent of bio-terrorism and listed as category A disease by OIE

Occupational hazard by inhalation of spores occurs mostly while sorting out wool and processing of hides.

Pathogenesis

Upon ingestion, the spores are proliferate in the regional draining lymph node, subsequently, travels via lymphatic vessels and reach the blood stream and cause septicemia followed by massive invasion into all body tissues produce **lethal toxin** and **edema factor** causes edema, tissue damage, and death as a result of **shock, acute renal failure and terminal anoxia**

In pigs, localization of the organism occurs in the lymph node of the throat following invasion through upper part of the digestive tract eventually lead to a fatal septicemia

Clinical finding

Incubation period 3-7 days (range:1-14 days)

Cattle and sheep

Peracute form

Sudden death without any premonitory signs

Fever, muscle tremor, dyspnoea, congestion of mucous membrane, staggering, trembling, terminal convulsion and collapse

Animal collapses and die due to terminal convulsions

After death, **oozing of blood from nostrils, mouth, anus and vulva and will not clot**

Acute form

Incubation period of the disease is 1-2 weeks

The course of this form is about 48 hours

Fever (42° C), severe depression, listlessness, congestion and hemorrhage of deep mucous membrane, respiratory and cardiac distress, open mouth breathing due to oxygen hunger, staggering, convulsions and death

Abortion, decrease in milk yield, blood stained or deep yellow color milk, diarrhea, local **edema of the tongue, throat, sternum, perineum and flank region**

Pigs

Sudden death

Oropharyngeal anthrax is characterized by rapid progressive swelling of throat causes suffocation and death

Acutely and sub-acutely infected animals show high temperature, appears dull, anorectic with distinct inflammatory edema in throat and face

Painless swelling causes obstruction which leads to difficulty in swallowing and respiration

Blood stained froth present at the mouth

Intestinal involvement- anorexia, vomition, diarrhoea and constipation

Pulmonary form

In baby pigs, it develops lobar pneumonia and exudative pleurisy through inhalation of dust which leads to death in one or two days

Occasionally, affected individuals may be alive for several days after infection

Horse

There is a high fever, severe depression, dyspnoea due to swelling of the throat and colic due to intestinal irritation

Painful, edematous and subcutaneous swellings appear in throat and dependent parts of the body

Necropsy finding

Absence of rigor mortis with gaseous decomposition, quick assumption of the characteristic saw horse posture

Un-clotted tarry colored blood oozes out through natural orifices

There is a soft and enlarged spleen with a black berry jam consistency (**Splenomegaly**) is important feature of anthrax

The **carcass** suspicious of anthrax **should not be opened before confirmation** made by peripheral blood smear examination

In horses and pigs enlargement of local lymphnodes, lesions in the soft tissues of the neck and pharynx are important and it is called as pharyngeal form of anthrax

Diagnosis

Based on clinical signs and necropsy findings

Isolation and identification of organism by culture

Gelatin stab culture: The growth resembles an 'inverted fir tree appearance' with slow liquefaction commencing from the top

The square ended blue bacilli (chain) with metachromatic pink capsule detected by microscopic examination after staining with polychrome methylene blue stain is called as “ **McFadyean's reaction**”

Fluorescent antibody technique using monoclonal antibody can be done to detect the disease

Isolation of bacilli from infected soil is difficult but real time quantitative PCR and Nested PCR have been found highly sensitive

Ascolis thermoprecipitation test: To demonstrate the antigen in severely decayed carcass and the skin in which tube containing positive serum is overlaid with trincturaed suspected sample develops a white precipitation ring at the junction of the two liquids

Sample collection

Unopened carcasses, blood, edema fluid in sealed leak proof containers and blood smear from tip of the ears

In live animals, organism may be detected in a stained peripheral blood smear, edema fluid, lymph node needle puncture smear, amputated tail or ears

If anthrax is suspected then shipping of diagnostic samples via air or courier is not permitted

Differential diagnosis

Lightening strike

Peracute blackleg

Malignant edema

Bacillary hemoglobinuria

Hypomagnesemia

Treatment

Recovery occurs when the infection is identified in early stages and treated with penicillin 20,000 IU /kg b.wt twice daily for 3 days

Long acting antibiotics can be in early stage, which give better results

Oxytetracycline 5-10mg /kg/day IV or IM can be given

Antiserum is given daily for 5 days however, it is expensive

Antibiotics should not given while vaccination, which interfere antibody development

Prevention

Sterne vaccine- Live spore vaccine contain uncapsulated, avirulent strain of *B. anthracis* is used in all animals in the endemic area

Cattle Sheep and goat 1 ml s/c caudal fold region

Elephant and horse 1 ml, s/c followed by a booster 3ml, s/c one month following primary vaccination

Following outbreak the animals can be consecutively vaccinated for three years for the prevention of further outbreak in the region

Antibiotics should not be used within 1 week of vaccination

Control

Anthrax suspected carcass should not be opened

Plugging of natural orifices with cotton to prevent exposure and spore formation

Carcass must be disposed by burning or deep burial

Burial should be done in 2 meter deep with adequate application of quick lime, liberal application of 5% formaldehyde to control the spread of the spores

Strong disinfection with 5% Lysol for 2 days, 5-10% solution of formalin or 2% sodium hydroxide is highly effective

Peracetic acid 3% solution of 8L/m² is considered as **very good sporicide** and effective sterilant

Maintain suspected animals for 2 weeks under quarantine

Vaccination of survivors, administration of hyper immune serum to contact animals should be carried out

Livestock and animal by products like meat, milk should not be taken out of the farm during quarantine period

Clostridial infections

The clostridial infections are of major importance in farm animals. All clostridial species produce potent exotoxins. The toxins of the different organisms vary in their effects and gain entry into circulation.

Botulism: Ingestion of preformed toxin in the feed

Enterotoxaemia- Absorption of toxin from gut after abnormal proliferation in the alimentary tract

Tetanus, black leg, braxy, black disease and bacillary haemoglobinuria-Elaboration of toxin from localized minor lesions

Clostridium sordelli can cause fatal myositis and can be identified as blackleg or malignant edema. It has been found to be a cause of fatal hepatitis in newborn lambs, enteritis with hemorrhagic diarrhoea in calves.

a) Botulism

(Syn: Lamziekte, Limber neck, Shaker foal syndrome)

Botulism is a serious potentially fatal intoxication usually acquired by **ingestion of preformed toxin** produced by *Clostridium botulinum* in decomposing animal tissue and sometimes in plant material.

Etiology

Clostridium botulinum is a spore (oval, sub terminal endospores) forming, gram positive rods, appears as single/paired/short chain

Clostridium botulinum, produces **potent neurotoxins** during their vegetative transformation from spores. The Spores survive in the environment for long period

There are seven *Clostridium botulinum* types are available namely Type A-G

***Clostridium botulinum* type B, C and D causes disease in farm animals**

Clostridial multiplication is inhibited at >pH 4.5

A-E type toxins are pathogenic for mammals

Type C toxin are pathogenic in ducks, pheasants, chickens

Epidemiology

Prevalence of infection

The disease is distributed worldwide, including India

The **incidence of disease is more frequent in chickens** and relatively low in cattle and horse

Host affected

Cattle and sheep are susceptible for **type C and D** while **horses** are susceptible for **type B toxin** and also for type C and D (Forage contamination)

Pigs, dogs and cats found to be resistant

Human is also susceptible for botulinum toxins

Predisposing factors

Heat and moisture favours the germination of spores

Phosphorus deficiency in cattle

Protein or energy deficiency in sheep

Source of infection

The proliferation of organism occurs in decaying animal or plant materials

The toxin is stable particularly in bones

Phosphorous deficient cattle chew and ingest dead carcass, bones, debris and get highest possible infection

In sheep, due to dietary deficiency of protein or net energy

Meat and milk containing toxin causes public health hazard

Transmission

The spread of infection by **ingestion of preformed toxin** in decaying plant or animal tissue

Cattle also get infection while grazing on pasture spread with poultry manure

Spread of infection through injury also

Based on transmission botulism is classified into four types

Carrion associated botulism

Wound botulism

Toxico-infectious botulism

Forage botulism

Carrion associated botulism

Multiplication of the organism in the decaying carcass and produce toxin, which is mostly consumed by phosphorous deficient and protein deprived animals.

Wound botulism

The organisms produce toxins directly in the wounds.

Toxico-infectious botulism

The organism grows and produce toxin in alimentary tract.

Forage botulism

Most cases of botulism is associated with the ingestion of preformed toxin in forages (Hay or silage contaminated with toxin containing carcasses of birds or mammals and poultry litter fed to the cattle, big bale silage-fermentation failure to produce low and stable acidic pH.

Zoonotic potential

Botulism toxin is an agent of bio-terrorism

Meat and milk containing toxins should not be used for human consumption.

Human get infections by eating improperly cooked meat, its products and milk.

Pathogenesis

Botulism toxins are neurotoxins absorbed by the intestinal tract or the wound and traverse through the blood stream to peripheral cholinergic nerve terminals and neuromuscular junction, post ganglionic parasympathetic nerve endings and peripheral ganglia

The toxin bind to the neuron surface receptors and **block the release of acetyl choline** from synaptic vesicle at the neuromuscular junction causes **flaccid paralysis** and death

Clinical signs

Cattle and horses

Incubation period is 3-17 days

Peracute

The animal die without premonitory signs

Acute

The disease is not accompanied by fever

Progressive symmetric muscular paralysis of the limb muscles and the muscles of jaw, tongue, throat and generalized progressive muscle weakness
Muscle weakness, an obvious muscle tremor and fasciculation may be enough to cause limb tremor

Decreased tongue tone

Hanging of tongue from mouth

Difficulty in chewing and swallowing and drooling of saliva is due to paralysis of tongue

Disturbed vision

Defaecation and urination is not affected although may be constipated

Sternal recumbency progressed to lateral recumbency just before death

Paralysis of chest muscle result in terminal abdominal type of respiration

Colic in horse is an important sign

Sub acute

Restlessness, respiratory distress, disinclination to rise, **pronounced roaring sound** and persists up to 3 months in some animals

In-coordination, stumbling, knuckling, ataxia, inability to rise or to lift the head, mydriasis and ptosis occur early

Difficulty to take hay for 3 weeks

Mydriasis is prominent in type-C botulism in horses

Sternal recumbancy, turning of head on the ground or into the flank

Reduced tongue tone, paralysis, tongue hangs out from mouth, difficulty in chewing, or swallow and drooling of saliva is prominent

Depression of rumen motility and constipation present

Paralysis of chest muscles results in terminal abdominal type respiration

Animals are alert and consciousness until at the end or death

Foals

Toxico-infectious botulism in foal is caused by type-B toxin

In young foals of up to 8 months **highest prevalence occur at 3-8 weeks** of age

The multiplication of organism in the intestine and production of toxins causes the disease called “ **shaker foal syndrome**”

Type– B toxin has been isolated from faeces of naturally infected animals

Muscle tremor is a prominent early signs

If the **foal walks, show stiff and stilted gait and toes are dragged**

Inability to stand for more than 4- 5 minutes

If the **foal sucks milk drools from mouth**

Attempts to eat hay but regurgitate it through nostrils

Constipation occurs consistently

Mydriasis and frequent urination

As the disease progresses dyspnoea with extension of head and neck, tachycardia, respiratory arrest and death

Rapid progression to severe muscular weakness, prostration, loss of condition and unable to rise

Prostrate foals are bright and alert have normal sense, pain perception

Death occurs in 72 hours after the onset of signs due to **respiratory failure**

Sheep

Affected sheep **do not show typical flaccid paralysis**

Stiffness while walking, in-coordination and excitability in the early stages

Head held on one side or bobbed up and down while walking (Limber neck)

Lateral switching of tail, salivation and serous, nasal discharge are common

Terminal abdominal respiration, limb paralysis followed by rapid death

Pigs

Affected pigs show staggering, recumbancy, vomiting, pupillary dilatation, flaccid paralysis of the muscle and anorexia

Dogs

In dogs, symptoms develop more quickly, within 12-36 hours after exposure to the toxin

There is generalized weakness, paralysis which spread from the hind limbs to the forelimbs

In addition, increased respiratory effort, facial nerve paralysis, and difficulty in swallowing

The disease progresses to fatal respiratory failure

Poultry

Leg weakness and paresis that progress to flaccid paralysis of legs, wings, neck and eyelids

Paralysis of neck muscle is called as limber neck

Feathers are easily pulled out

Laboured breathing

Necropsy findings

Non specific sub endocardial, sub epicardial haemorrhages and congestion of intestines

Non specific perivascular haemorrhages in the corpus stratum, cerebellum and cerebrum

In shaker foal syndrome- **pulmonary edema**, congestion and excessive **pericardial fluid containing free floating strands of fibrin**

Samples collection

Suspected contaminated feed material, liver, rumen contents and serum etc

Formalin fixed brain

Diagnosis

Based on clinical signs and necropsy findings

Demonstration of toxin in serum, feed, intestinal contents and from wounds

In foals arterial blood analysis show acidemia, hypercapnia, hypoxemia and desaturation of haemoglobin

Repeated arterial blood gas analysis should be conducted during first 48 hours of treatment

Demonstration of spores of *Cl.botulinum* and their toxins in feed

Mice: Injected with toxin and neutralized with polyvalent antitoxin

But the sensitivity is low in mice as compared to ruminants and horses to botulism toxin

In chronic/sub-acute condition, ELISA is used for the detection of type C and D toxin in cattle

Differential diagnosis

Ruminants

Parturient paresis in cattle
Hypocalcaemia in sheep
Tick paralysis
Rabies
Organophosphate or carbamate poisoning
Louping ill

Horses

Equine encephalomyelitis
Hepatic encephalopathy
Rabies

Treatment

Early administration of antitoxin before complete recumbency
Supportive therapy includes fluid therapy, parenteral feeding, nasal in-sufflation with oxygen and mechanical ventilation
Although it is expensive, a single dose of specific or polyvalent antiserum administered early in course with 30,000IU for a foal and 70,000IU for adult horses may increase the survival rate
Antimicrobials for treating secondary complications such as aspiration pneumonia
Avoid the use of drugs that may deplete the neuromuscular junction of acetylcholine such as neostigmine and those procaine penicillin, tetracyclines, aminoglycosides potentiate neuromuscular weakness
Muzzling is essential to prevent aspiration pneumonia
Frequent turning to prevent the development of muscle necrosis and decubital ulcers

Prevention

In enzootic area animals vaccinated with type specific combined C and D toxoid for occasions at bi-weekly intervals
Stocks should be rigorously vaccinated with a toxoid on 3 occasions at 2 weeks intervals
Carrion verses non-carrion associated botulism is an important factor when considering prophylactic vaccination programme

Control

Hygienic disposal of carcasses
In horses as the disease is sporadic caused by accidental contamination of food and water must be curtailed

Range animals - supplementation of feeds with phosphorous and/or protein

b.Tetanus

(Syn: Lock jaw)

It is a non contagious, non-febrile, infectious disease of mammals affected by toxins released by *Clostridium tetani* and clinically characterized by spasmodic contractions of skeletal muscles.

Etiology

Clostridium tetani is a gram positive, a long, slender, rod shaped anaerobe with rounded ends

The organism develops terminal spores and has a drumstick appearance

It lives in the intestine of herbivores as a normal inhabitant

The organism grows mainly under strict anaerobic condition

Spores are resistant and persists in soil for many years

The organism releases three types of toxins under anaerobic condition namely

Tetanospasmin and Tetanolysin

Epidemiology

Prevalence of infection

The disease is more common in tropical countries than in temperate

It is a sporadic disease in India

Case fatality rate in young cattle may be over 80%

Mortality rate may be 50% in animals die of acute infection

Predisposing factors

Infected nature of soil and climatic conditions

Wound or break in the skin, and accidental injury

Shearing, docking, parturition period, favors the infection

Sources of infection

Infected soil and alimentary tract (normal inhabitant)

Transmission

The organism present in soil gain access into animals during parturition, shearing, docking, castration, spaying and any other unprotected surgical and accidental injury.

Spread can also occurs through ingestion

Host affected

Farm animals are probably most affected

Horses and mules are highly susceptible

Cattle are relatively less susceptible than man and horses

Sheep, goats and pigs are more susceptible compared to cattle

Pathogenesis

The organism enters through any broken skin or injury does not cause tetanus

For the tetanus to develop need some favourable conditions such as;

- ☒ Access of spores into body tissues. Tissue damage causes localized impairment in blood supply oxygen creates anaerobic condition
- ☒ **Anaerobiosis leads to germination of spores**
- ☒ Following germination of spores vegetative growth occurs rapidly and releases toxins; **tetospasmin, (neurotoxin)**, tetanolysin (haemolysin), and fibrinolysin (muscle and tissue cell lysis)
- ☒ **All these toxins are released by cellular autolysis** and get absorbed from site of production and spread into blood stream, exert neuro-toxic effect in peripheral nervous system by retrograde intra axonal transport, via motor nerves to central nervous system via brain and spinal cord, along the nerve route
- ☒ Among these toxins **tetospasmin** is considered **very important and potent toxin**
- ☒ Through axonal terminal and intra axonal transport mechanism the toxin gain the central nervous system
- ☒ From the CNS toxins reach dendrites and transferred to neurons that synapse with cells
- ☒ Generally, **toxin can't cross the blood brain barrier**
- ☒ The released toxin binds with a "protagon", which is made of cerebroside and oligosaccharides like N-Acetyl galactosamine and galactose. The toxin acting on the **inhibitory** synapses interfere with action of the **inhibitory transmitter** thus cause hyperirritability and thereby leads to spasm
- ☒ As the toxins enter the blood stream causes **spastic paralysis**, and invasion into nerve endings develops flaccid paralysis
- ☒ **Death usually occurs due to asphyxiation** as a result of fixation of respiratory muscles

Clinical signs

Horses

Initially affected animals are very hypersensitive to the external stimuli

Inability to open the mouth

Animals continue to be with spasm for several days with mild muscle spasm

Saw horse stance, prolapse of the third eye lid, erection of ears, immobility of the ears seen as predominant signs

Head, neck, and leg are in extended position (opisthotonus)

Spasm of facial muscle, masseter muscle leads to trismus

Mastication impairment, drooling of saliva because of spasm and swallowing difficulty

Very sensitive to light and noise, trembling and profuse sweating

At the end, affected animals unable to stand and die due to asphyxia in 3-10 days.

Cattle and Buffaloes

Affected animals exhibits **mild problems** and identified at the initial level recover from the infection

Affected animals **restrict its movement, exhibits muscle stiffness**, difficulty in walking and decrease milk yield

Lock Jaw, hyper irritability, **prolapse of the 3rd eyelid, pump handled position of the tail** is more common and unique sign of tetanus in these cattle

Lateral recumbency, extend rigidity and opisthotonus

Suspension of rumination, drum like appearance of the rumen and left abomasal displacement

Animal dies due to asphyxiation

Sheep and goats

Dullness, head is drawn on one side or backward, opisthotonus, hypersensitivity to noise and bending of neck

Kids and lambs fail to suck milk due to tonic contraction of masseter muscle

Death occurs by asphyxiation

Dogs

The clinical signs occur as localized and generalized form

Localized

Rigidity of one leg, hypersensitivity, erection of ears, wrinkling of forehead, hypertonicity, opisthotonus, masseter muscle spasm, caudal retraction of the lips

Generalized

Clinical signs may be prominent in 10 days of infection

Dyspnoea, laryngeal spasms, miosis and stiff gait

Out stretched tail, difficulty in rising or lying is the most common

Constipation, retention of urine due to spasm of the sphincter

Affected animals die of respiratory muscle failure, muscle spasm, laryngeal dysphagia, decrease secretion from respiratory tract and death occurs in 5 days

Necropsy finding

Rigor mortis develop very quickly

Congestion of spinal cord, medulla

Hyperaemia of lungs due to asphyxia

Sample collection

Impression smear from wound, Necrotic tissue from wound and serum

Diagnosis

Based on clinical signs

Gram staining of wound impression reveal *Cl.tetani* spore

Real time PCR for detection of neurotoxigenic gene in bacteria

Analysis of toxin in serum

Mice inoculation test

Analysis of toxin in serum is the confirmatory test

Treatment

Principles of treatment

Neutralization of free toxin

Destruction of tetanus organism

Relaxation of muscle to prevent asphyxiation

Supportive care

a. Neutralization of free toxin by Tetanus Anti Toxin (TAT or ATS)

TAT may be mixed in saline and administered through I/V over a period of 30 min.

It can be given in mares following, foaling and new born foals

It is given during last week of pregnancy

It can be administered following castration, docking and other operations

For prophylaxis (1500IU) is given either subcutaneous or intramuscular route, which provides transient immunity ie. only for 21 days but elicits quicker action

Therapeutic Use:

The large doses 100,000 - 200,000 IU of antitoxin should be given very slowly by intravenous route followed by smaller weekly doses intramuscularly as required. Some physicians prefer a smaller initial dose of 20,000 IU or so intramuscularly. Active immunisation may be initiated when the patient is cured of the illness.

b. Neutralization of toxin by Tetanus Toxoid (TT)

It is a toxoid which establishes active immunity. Vaccination with tetanus toxoid is recommended for healthy domestic animals as preventive measures.

Tetanus toxoid can be given at the same time with anti toxin provided in different sites

Destruction of organism

Early tetanus in cattle responds to the treatment. In very early cases high doses of Penicillin may be helpful, particularly if combined with local treatment of the infected site

Penicillin G 22,000 IU/kg bw 3-4 times I/M

Inj. Metronidazole should be given in dog

Surgical De-bridement

Wound is cleaned, **debrided with H₂O₂** followed by administration of antitoxin/antibiotics at the site of wound

Topical hydrogen peroxide infiltration increases the local oxygen tension and prevent bacterial growth

Relaxation of muscle

Affected animals should be kept in dark place and muscle relaxant must be given

Large animals

Diazepam 0.01-0.4mg through I/V

Dogs

Diazepam 1.0mg/kg bw 1-4 hours intervals I/V

Phenobarbitone 2-6mg /kg bw 6-12 hours I/V or I/M

Management

Nasogastric tube feeding can be given in horse

Enema and use of catheter to remove stagnant urine and faecal matters

Tympany is relieved by rumenotomy

Use of indwelling tube should be done to avoid disturbances caused during repeated passing of stomach tube

Provision of moist and soft feed

Animals should be provided clean water, feed and bedding material to avoid injury

Prevention

Tetanus toxoid is used to inducing active immunity. Vaccination with tetanus toxoid is recommended for healthy domestic animals and human being as preventive measures

Pregnant animals can be vaccinated at the advance stage of pregnancy (cattle at seventh and eighth month, sheep and goats third and fourth month of pregnancy)

Tetanus toxoid is to be administered before all surgical operations like docking, shearing and castration

Dose: Tetanus toxoid - 0.5ml s/c

Control

All surgical operations should be carried out aseptically in a hygienic area

Spores are resistant to many disinfectants but can be destroyed by 115°C for 20-60 minutes.

Vegetative form can be destroyed by heat and disinfectants but spores are not

c. Black Quarter

(Syn: Symptomatic Anthrax, Quarter ill)

It is an acute, infectious myositis of cattle, sheep and goats is characterized by development of gas gangrene with emphysematous swelling in the affected muscle and systemic toxemia leads to fatal death.

Etiology

It is caused by *Clostridium chauvoei*

There are 23 strains of *C.chauvoei* identified so far around the world

It is a **gram positive, spore forming, rod shaped bacteria** measuring 0.6µ in dia and 3-8µ in length

Spores produced by this organism persists in soil for many years and is highly resistant to disinfectants, dryness and hot

Epidemiology

Prevalence of infection

Disease is endemic in both developed and developing countries

In India, the **disease occurs sporadically**, however, in some region, the disease is endemic

Predisposing factors

Occasional outbreak: Damage to tissues caused by **administration of formalized vaccines**, accidental injury of muscle and injury made by chemical agents

In sheep outbreak of disease following enterotoxaemia vaccine (formalized vaccine injure the muscle tissue)

Excavation or turning up of soil expose the spores to the top layer of soil and spreads the infection

Alternate hot and dry seasons followed by heavy rainfall favors the dispersion of spores for long distance

Sources of infection

It is an soil borne infection

Organism present in decomposed carcass of animals affected with black leg and persists in soil for long period

Organism exists as normal inhabitant in spleen, liver and alimentary tract of healthy animals

Sheep: Infection mostly spreads via wound developing during **shearing, docking, lambing , navel infection at birth and fighting of rams and navel infection at birth** etc., hence, it is called as traumatic BQ

Transmission

Cattle: It is transmitted by **ingestion, inhalation of spores and absorption of toxins through wound**

Sheep and goat: through skin wounds at **shearing, docking , lambing , navel infection at birth and fighting of rams**

Occasional outbreak occur in sheep after vaccination against enterotoxemia

Host affected

The disease is highly fatal in cattle

Cattle, 6 months to 2 years and well nourished animals are mostly affected

In sheep all age groups are susceptible

It is reported also in deer, elephant and horses

Both, *C. chauvoei* and *C. septicum* influence the development of gas gangrene in pigs

Organism has also been isolated from wounds of dogs and cats

Pathogenesis

Black quarter is a soil borne infection

Spores ingested would be taken via alimentary mucosa and then distributed to tissues and skeletal muscles via the circulation.

Injured muscles favors spores germination in the wound that provides anaerobic condition for growth of organism and produce exotoxin causes necrotizing myositis

Mostly, the organism affects **muscles of loin, shoulder and gluteal muscle.**

In addition, muscles of diaphragm may be affected.

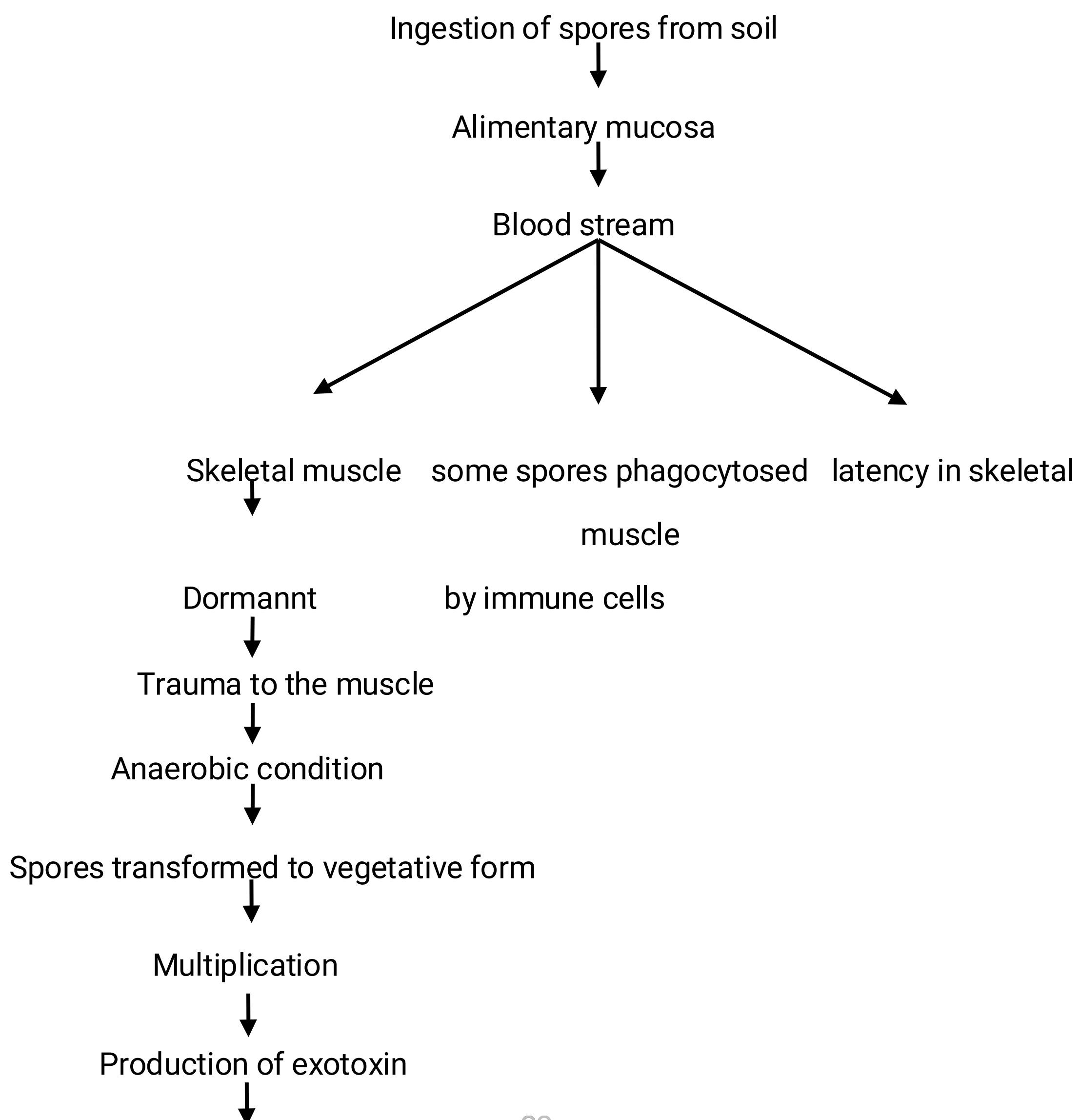
If animals consume the **preformed toxins while grazing, ingested toxins is directly absorbed and spread along the systemic circulation which leads to fatal toxemia.**

Some spores are directly phagocytosed by the immune cells, the remaining assumes latency in the muscle, followed by transformation as vegetative form once the muscle is injured

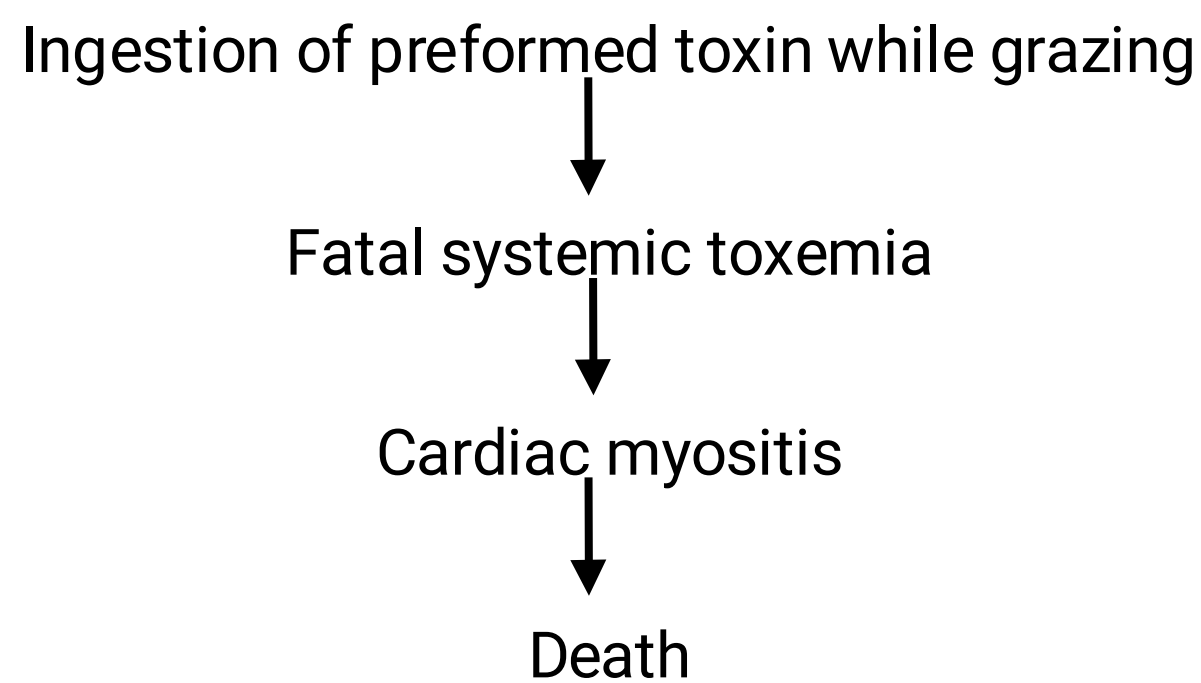
C.septicum is a potent exotoxin producer as compared to ***C.chauvoei* because virulent form of *C.chauvoei* only able to produce exotoxin.** This exotoxin is responsible for the development of edema and pulmonary congestion

Toxins released by the organism produces a severe necrotizing myositis in skeletal muscles and a systemic toxemia followed by death

In cattle and sheep during atypical outbreak, sudden death follows attributable to clostridial cardiac myositis



Necrotizing myositis



Clinical signs

Cattle

Peracute form

Affected animals show sudden death

Acute form

Incubation period is 2 to 5 days

Disease follows a rapid and fatal course of infection following ingestion of preformed toxins

Prominent swelling on the upper part of the affected limbs leads to marked lameness

Animal is found to be very depressed, complete anorexia, ruminal stasis with high temperature and pulse rate (100-120/min)

Pyrexia is not present in all cases

In the early stage the swelling is hot and painful to touch, later becomes of cold and painless

Edema, emphysema, crepitation of affected heavy muscles on thigh with muscle stiffness

Discolored, gangrenous to black, dry and cracked skin

A thin sanguineous fluid containing bubbles of gas from affected sites

Swelling of throat, tongue and protrusion of the tongue

Death follows in 12 to 36 hours of infection

Cattle with cardiac myositis are found dead

Sheep

No palpable crepitation swelling as in cattle

Stiff gait, disinclined to move, severe lameness in one or most commonly in several in several limbs

Painful walking

Subcutaneous emphysema is not common

Gaseous crepitation cannot be felt before death

Discoloration of skin with no necrosis and gangrene

Extensive local lesion in vulva and vagina occurs through skin wounds

Severe local swelling of head due to edema

Bleeding from nose

There is high fever, anorexia, depression and quick death

Sheep with cardiac myositis are usually found dead

Horses

Pectoral edema, stiff gait and incoordination

Necropsy finding

Cattle

Bloat and putrefaction occurs quickly and blood stained fluid oozes out from the natural orifices such as nostrils, anus and oral with **rapid clotting of blood**

Freshly cut surfaces of muscles are often dry and may have a metallic sheen

Affected muscles become dark red to black, swollen and emanates a rancid odour

The thoracic cavity and pericardial sac may contain excess blood stained fluid with fibrin

Serositis is often get confused with pleuropneumonia

Lungs are usually congested and is atelectatic

Sheep

Muscle lesions are more localized and deeper

Sample collection

Muscle samples should be collected in an air tight container.

Air dried impression smears obtained from surface of freshly cut lesion for bacteriology, 10% formalin fixed samples from suspected muscles for histopathology.

Impression smears from heart blood and liver

Diagnosis

Based on clinical signs and necropsy findings

Isolation of organism by culture of heart blood, peritoneal fluid, muscles and liver is important for both *Cl.chauvoei* and *Cl.septicum*

Muscle bundle separation by gas

Tube agglutination test, counter immuno electrophoresis, double immuno diffusion test used to detect humoral antibody, Immunofluorescence, Fluorescent antibody test to detect antigen

Biological inoculation: Filtrate containing spores obtained from infected animals heated at 60° C for 30 minutes and injected one ml into gluteal muscle of guinea pig leads to death in 48hrs

Differential diagnosis

Malignant edema

Anthrax

Lightening stroke

Bacillary haemoglobinuria

Treatment

Penicillin 10,000 IU/kg IM is a drug of choice for 5-6days, (Large dose of 40,000IU/kg bw) can be given following administration of Crystalline Penicillin via IV

Oxytetracycline in high doses i.e. 5-10 mg/Kg body weight IM or IV

Tease the swelling and drain off

B.Q. antiserum in large doses, if available

Injection of antihistamines can be given. It can also be sprayed directly on affected tissues

Prevention

Cattle

Maternal immunity persists for atleast 3 months, so until this time calves should not be vaccinated and this may interfere with the development of vaccinal immunity

In enzootic areas, annual vaccination of all cattle between **3-6 months** must be done with 2 vaccinations at 4 weeks interval followed by annual booster vaccination

Best time of vaccination is before onset of monsoon

In an outbreak area, all cattle should be vaccinated immediately and treated with penicillin at 10, 000 IU/kg bw I/m or in combination with penicillin and benzathine penicillin

Sheep

In enzootic areas, ewes should be vaccinated twice

Vaccination should be given one month before lambing and subsequently yearly booster before lambing to prevent infection of lambs and ewes and umbilical infection at birth

In outbreak

Prophylactic injections of penicillin and antiserum may be effective

Vaccination of lambs before they go into pasture is recommended

Clostridial vaccines have poorer antigenicity in sheep and goats than in cattle, so, combined vaccine which contain antigen of *C. chauvoei*, *C. septicum*, *C. novyi* is essential

Vaccines

Formalin inactivated and alum precipitated vaccine contain strain 49 is used for immunization

Dose of vaccine

Cattle and buffaloe-3 ml s/c

Sheep and goats-1-2 ml s/c

Immunity last for 6 months

Control

Isolation of affected cattle

Constant surveillance and early treatment is necessary

Disposal of carcasses by deep burial or burning to soil contamination is essential

The organism can be destroyed by **3% formalin for 15 minutes** is advisable

d. Enterotoxemia

It is a sporadic disease affecting all group of animals

It is caused by *C.perfringens* isolates are type A-E as based on their ability to produce alpha, beta, epsilon, and Iota toxins. Each serotype have different pathogenicity and causes disease

It is a gram-positive, non spore-forming, anaerobic rods

It is a normal inhabitant of alimentary tract and also found in soil

It can grow in blood agar and produce double zone of haemolysis around the colonies

Enterotoxaemia -Type A

(Syn: Yellow lamb disease)

It is a sporadic disease affecting all groups of animals, characterized by diarrhoea, occasionally sudden onset of weakness, coma and death.

Etiology

It is caused by *Clostridium perfringens* type A

It is a gram-positive, spore-forming, anaerobic bacterium, produce alpha toxin under favourable condition, which is responsible for pathogenesis.

Epidemiology

Prevalence of infection

C. perfringens type A has been suggested as a cause of jejunal hemorrhage syndrome in beef and dairy cattle and calves in most of the **temperate countries** predominantly in Canada, Belgium, in the western United States including in India.

Predisposing factors

It is also occasionally associated with wound contamination and gas gangrene

Sources of infection

Organisms found in soil, faeces and intestinal tract of animals and humans

Transmission

Contact with infected group of animals

Ingestion of contaminated feed and water materials with organism

Host affected

All animals such as cattle, sheep and goats, horses, pigs are affected

Young one of the above species are highly susceptible to the disease

The disease mostly occur in calves at 2-4 weeks of age

Pathogenesis

C. perfringens type-A produce alpha toxin. It possesses phospholipase C and sphingomyelinase activity and consequently have haemolytic action. The presence of Beta 2 toxin in these strains may contribute to the pathogenicity leading to high temperature, jaundiced mucous membrane, haemoglobinuria

Affected animals die in 12 hours

Clinical conditions

Adult horse-Enteritis is characterized by acute profuse diarrhoea and die following strenuous exercise

Pig-Enteritis is characterized by watery yellow diarrhoea in piglets less than 5 days of age

Calves and adult cattle- Haemorrhagic enteritis

Lambs- Yellow lamb disease is due to haemolytic activity of toxin

Foals- Haemolytic enterotoxemia

Goat-hemolytic disease

Clinical signs in haemolytic disease

An acute increase in body temperature 41°C , severe depression, pale mucous membrane, jaundice, hamoglobinuria, dyspnoea and severe abdominal pain.

Affected animals collapse and die in 12 hours

Necropsy finding

Horse

Cecitis and colitis

Pigs

Mild enterocolitis and villous atrophy

Cattle, sheep and goats- hemolytic disease

Liver is pale and enlarged

Infarction in the kidney

Hydropericardium and pulmonary edema

Extensive necrosis of the intestine

Diagnosis

Isolation of organism from young and adult animals feces with enterocolitis

The organism can be isolated from young one even when they are 3 days old

Identification of Beta 2 toxin in feces (enterotoxin gene). Beta toxin detection in gut contents. It may cause protease degradation

Culture of the organism from the calf is not sufficient to confirm a diagnosis of disease due to *C. perfringens*. So culture results should be confirmed with clinical signs, lesions in the tissues, and, in some cases, toxin identification, to obtain a true diagnosis. Tissue samples from calves should be collected soon after death and preserved

Differential diagnosis

Chronic copper poisoning.

Treatment

In subacute and chronic cases, antitoxin along with Tetracycline orally at the rate of 11 to 22 mg/kg bodyweight will usually be effective

Sulpha group of drugs as injection or tablets may be given

Sometimes sulpha drugs in combination with trimethoprim can be tried

Prevention

Type A antiserum is very effective in preventing disease in calves

Formalised vaccine has some effects in controlling

Control

Prevention of enterotoxemia due to *C. perfringens* infection focuses on three areas

- ☒ Minimize exposure
- ☒ Enhance immunity in the young calf through giving of adequate colostrums
- ☒ Management practices

Enterotoxemia- Type B, C and E

(Syn: Lamb dysentery and Pine in older lamb)

The disease is mostly occurring in animals under first few days of life, with the exception of struck in adult sheep. Their predominance in very young animals may be due to immaturity of alimentary tracts and ready colonization by *Cl.perfringens* in gut.

Etiology

Cl.perfringens type B produce alpha, beta and epsilon toxin, which causes lamb dysentery

Cl.perfringens type C produce alpha and beta toxin, which causes enterotoxaemia in calf, foal and goat, necrotic enteritis in pig and poultry and struck in adult sheep

Cl.perfringens type E produce alpha and Iota toxin rarely cause disease

Epidemiology

Lamb dysentery

Incidence of disease in group of lambs 20-30%

Outbreak of the disease is usually occur at 1-4 days of age

The disease in older lambs occur at 2-3 weeks of age

Case fatality rate approaches 100%

Type C Enterotoxemia in lambs and goat is prevalent in cold weather conditions

Type C Enterotoxemia

It is prevalent in lambs and goat during cold weather conditions

Single born lambs are highly affected and restricted to 2 hrs to 4 days of age

Necrotic enteritis

The disease most commonly occur in piglets upto 7 days of age

Case fatality during severe outbreak is 80%

Enterotoxemia in foal and calf

It mostly occur under 7 days of age

Struck

It common in adult sheep on good pasture in spring season and manifested as sudden death

Predisposing factors

Poor management factors predisposes the disease

Well nourished animals are affected more

Transmission

Ingestion of organism from soil or faecal contamination on the surface of the dam's udder

Pathogenesis

The organism enters the host through ingestion. It proliferates and attaches to the surface of the intestinal villus but toxin production and mucosal damage may precedes the attachment.

The toxigenic strain of *Cl.perfringens* type B and C produce both alpha and beta toxin

The alpha toxin (lethal toxin) contain phospholipase, hydrolysis the membrane phospholipids in erythrocytes, platelets, leukocytes and endothelial cells results in cell lysis or other forms of cytotoxicity

The beta toxin (necrotizing toxin) causes increased capillary permeability may facilitate uptake of toxin in intestine, which leads to damage of microvilli with eventual destruction and desquamation of intestinal epithelium and production of haemorrhagic enteritis

Clinical signs

Lambs

Per acute

Sudden death without any signs

Acute form

It is more common, the affected lambs shows loss of suckling, severe abdominal pain, bleating, stretching and looking at the abdomen

Lambs pass brown colored feces and sometimes with blood

Defecation is often accompanied with painful straining

Death usually occurs after a period of recumbency and coma and within 24 hours of onset of illness

Chronic form

It occur in older lambs causes abdominal pain, reluctance to suck, and without diarrhoea

Necrotic enteritis in piglets

Rapid death in young ones and more prolonged disease in older piglets

Dull, depression, diarrhoea, dysentery and reddening of mucous membrane
Watery yellow faeces initially, and sometimes haemorrhagic and red brown in color
Feces contains necrotic debris
In old pigs yellow coloured diarrhoea without any evidence of blood in faeces
Rapid dehydration, hypoglycemia, hypothermic and comatose
Clinical course is less than 24 hours

Foal enterotoxemia

Depression, pronounced toxemia, marked abdominal pain and bloody feces
Subnormal body temperature, rapid pulse and respiratory rate
Death within a few hours after colic
In sub acute cases, recovery is slow and requires 10-14 days

Necropsy finding

The kidneys are swollen, dark brown in color, and may contain infarcts; the liver is pale and swollen and there may be hydropericardium and pulmonary edema.
There is extensive necrosis of the small intestine
Intestinal mucosa are dark red with large ulcers (upto 2.5 cm dia)
Intestinal contents are blood stained and may contain fibrin clots
Peritonitis, sub endocardial and sub epicardial haemorrhage in heart

Diagnosis

Isolation of organism from feces
Stained intestinal content shown large numbers of clostridium organism
Demonstration of toxin in intestinal content
ELISA and Rapid Passive Latex Agglutination Test
Multiplex PCR for detecting gene encoding major toxin

Differential diagnosis

Salmonellosis
Enteric colibacillosis
Cryptosporidiosis
Isospora suis infection
Transmissible gastro enteritis
strongyloidesis

Treatment

Fluid and supportive therapy
Hyper immune serum is a specific therapy
Oral and parenteral administration of penicillin prevents the further proliferation of organisms

Control

Vaccination with type specific toxoids or bacterins

Vaccination of pregnant animals with two injection at 1 month prior to parturition

Administration of antiserum immediately after birth

Vaccination of sows 5 and 3 weeks before farrowing

If vaccination done at mating booster vaccination should be done 2-3 weeks before farrowing

Feeding of bacitracin (300mg/kg of feed), salinomycin 60 mg/kg feed) to the sow for 2 weeks before farrowing reduce the incidence of disease

Cleaning and disinfection of premises

Enterotoxaemia- Type D

(Syn: Over eating disease, Pulpy kidney disease)

It is an acute fatal toxemia of ruminants characterized by sudden death.

Etiology

It is caused by *C.perfringens* type D, which proliferate in intestine produce Alpha, beta, **epsilon**, and Iota toxins

Gram-positive, non spore-forming, anaerobic bacteria

Epidemiology

Prevalence of infection

The disease prevails mostly in sheep and goats reared on young grass routes and shoots

The disease has been widely recorded from many countries

Predisposing factors

Upset in gut flora

Sudden changes in diet

Carbohydrate rich diet

Heavy grain feeding

Grazing on lush rapidly growing pasture or cereal crops

Lambs on well fed heavy milking ewes

Source of infection

The organism found in soil and intestine

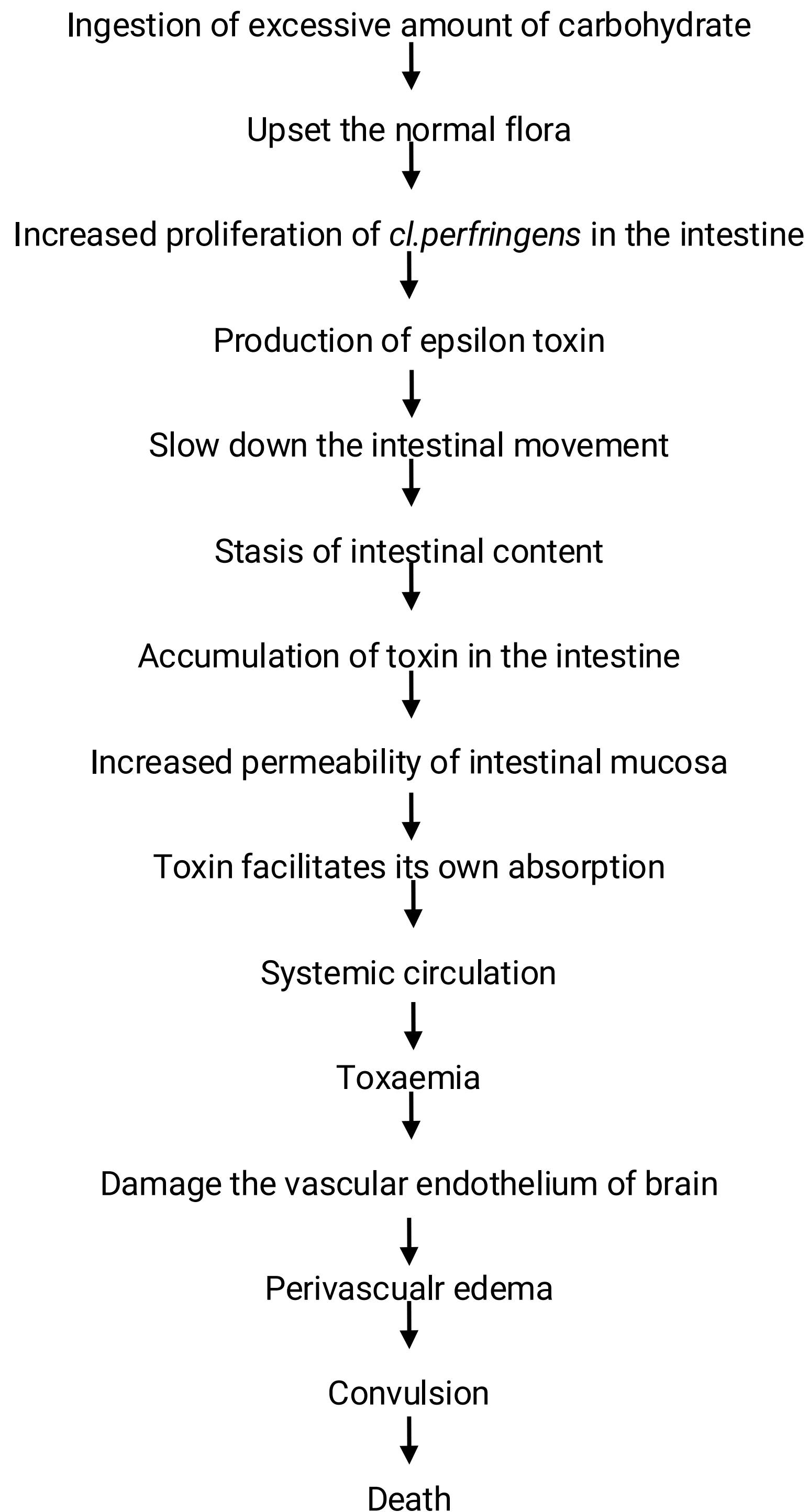
Host affected

Sheep, calves, goats and horses are affected

Sheep are highly susceptible than other animals

Lambs 3 weeks to 10 weeks old and suckling lambs are commonly affected
Single lambs are more susceptible than twins
Weaned lambs upto 10 months of age are second most susceptible
Feeder lambs are most commonly affected soon after they introduced into feedlots

Pathogenesis



Clinical signs

Toxin facilitates its own absorption, toxaemia with no enteritis, dullness, retraction of the head, opisthotonus, convulsions and sudden death

Lambs

Course of the illness is very short

Many lambs found dead

On close observation the flock shows dullness, depression, yawning facial movements and loss of interest in feed

Acute cases **severe chronic convulsion** is higher **with frothing at the mouth and rapid death**. Cases survive for a few hours show a **green, pasty diarrhoea, staggering, opisthotonus and recumbency**

Temperature is usually normal but may be elevated if convulsions are severe

Death occurs during the period of convulsions

Adult sheep

They usually survive for a long period up to 24 hours

Lagging behind the flock staggering and knuckling

Champing of the jaw, salivation, rapid shallow, irregular respiration, convulsion, muscle tremor and bruxism

There may be bloat in the terminal stages

Calves

Similar to that of adult sheep

sudden onset of bellowing, mania, convulsion persists until death

Goats

Peracute form- common in young kids

Acute form-common in adult

Diarrhoea is a prominent sign

Severe abdominal pain

Dysentery

Death occurs 4-36 hours in kids and 2-4 days in adult

Chronic cases- anorexia, intermittent diarrhoea, epithelial shreds in the faeces, anemia and eventual emaciation

Clinical pathology

Hyperglycemia (Glucose level 150-200 mg/dl)

Glucosuria

Necropsy finding

Rapid decomposition of carcass

Excessive amount of clear straw coloured pericardial fluid clots on exposure to air

Many petechia are present in epicardium and endocardium

Pulmonary oedema

Soft pulpy kidney within few hours of death

Symmetrical area of haemorrhage, oedema and liquefaction in the brain

Sample collection

Intestinal contents should be preserved by adding 1 ml of chloroform to 10 ml of contents which have been collected in glass within 12 hours of death

Diagnosis

Based on clinical signs, and post mortem examinations

Bacteriological examination by culture, (isolation and identification)

Analysis of toxins in filtrate of intestinal contents

ELISA is used for diagnosis of toxin in serum

Glucosuria

Mouse neutralization test

Differential diagnosis

Coccidiosis

Salmonellosis

Rumen impaction

Treatment

Benzathine penicillin and amoxicillin at birth

Bacitracin 300 mg/kg feed

Salinomycin 60mg/kg feed

Treatment is ineffective against the peracute and acute cases

Administration of 50 ml of specific hyperimmune serum intravenously every 4 to 8 hours in the valuable animal

Prevention

Vaccination of lambs and kids above 3 months of age before onset monsoon season. Cross protection occurs between Type C and B

Inactivated vaccine (Formalin inactivated alum adsorbed vaccine) are available for vaccination

The pregnant ewes can also be safely vaccinated and the second injection is being given about 3 weeks before lambing

Regular vaccination once in 6 months is recommended. Adult sheep 5 ml and lambs 2-3 ml S/C. The immunity is expected to last for 6 months

Control

Avoid excessive feeding of grain diet
Avoid grazing of animals on lush pasture for longer period

e. Bacillary haemoglobinuria

Etiology

***Clostridium haemolyticum* (C. novyi type D)**

It is a soil borne anaerobe

The longevity of spores in soil is unknown, but the **organism has been isolated from bones a year after the death of an animal** from bacillary haemoglobinuria. **In infected areas the organism is often found in the liver of healthy cattle.** Under anaerobic conditions the organism grows and produces a **necrotic and haemolytic beta toxin** responsible for the clinical disease.

Damage to the liver from telangiectasis, necrobacillosis caused by Fusobacterium necrophorum and fasciolasis have been suggested as precipitating causes.

Epidemiology

It has been reported principally from the western parts of the USA, although the disease has also been observed in the southern states, Canada, Mexico, Venezuela, Chile, Turkey, Australia, New Zealand, UK and Ireland.

The cattle are usually affected, occasionally sheep and rarely pigs are affected. In many clostridial diseases, animals in good condition are more susceptible. **The disease is common in summer and autumn season.**

The highest incidence of bacillary hemoglobinuria occurs in irrigated or poorly drained pasture, especially if the soil is alkaline.

The disease spreads from infected to non-infected areas by flooding, natural drainage, contaminated hay from infected areas or carrier animals.

The carriage of bones or meat by dogs or other carnivores also spreads the infection.

Contamination of pasture may occur from faeces or decomposing cadavers.

Pathogenesis

In endemic areas invasion occurs from the alimentary tract after ingestion of contaminated materials.

Migrating flukes are the primary factor leading to liver necrosis and the establishment of anaerobic conditions in the liver that will lead to the that will lead to multiplication of *clostridium haemolyticum* in the liver

As in black disease of sheep, the bacteria are carried to the liver and lodge there until damage to the parenchyma of the liver and the resulting hypoxia create conditions favourable for proliferation

The **development of an organized thrombus in a subterminal branch of the portal vein produces large anaemic infarct** is the characteristic feature of disease

Most of the bacteria are to be found in this infarct. Under anaerobic conditions, the necrotoxic and haemolytic beta toxin is released systemically results in toxemia, generalized vascular damage and intravascular hemolysis

Clinical manifestation

Cattle

The illness is short duration, **cattle may be found dead at pasture**

More often there is sudden onset, anorexia, complete cessation of rumination, lactation and defecation

Abdominal pain is evidenced by disinclination to move and an arched back posture

Grunting may be evident on walking

Brisket Edema

Respiration is shallow and labored

Pulse is weak and rapid

Fever in early stage, the temperature subsides to subnormal before death

Faeces are dark brown; there may be diarrhoea with mucus and some blood

Urine is dark red

Jaundice is present but is not very obvious

Abortion in pregnant animals

Severe dyspnoea is evident before death

Sheep: The clinical symptoms are similar to that of Cattle

Clinical pathology

Blood profile: Anaemia and leukocytosis

Blood culture during the acute stage of the disease may be positive

Serum agglutinin against *C. haemolyticum* may be detectable at low levels (1:25 or 1:50) during the clinical disease and in recovered animals it may rise to (1:50 to 1:800)

Positive agglutination test is not conclusive evidence for the presence of the disease

Necropsy findings

Rigor mortis develops more quickly

Perineum is soiled with blood stained urine and faeces

Subcutaneous gelatinous edema, which tend to become crepitant within few hours

Excessive amount of clear to blood stained fluid in the pleural, pericardial and peritoneal cavities

Ischemic infarct in the liver

Generalized subserous haemorrhages

Haemorrhagic abomasitis and enteritis are accompanied by the presence of blood stained ingesta

Sample collection: Liver tissue from infarct area in air-tight container and impression smear from lesions

Diagnosis

PCR test to identify toxin-producing genotypes of *Clostridium* sp

Fluorescence antibody technique on impression smears taken from the hyperaemic zone around the liver infarct and stained with FITC labelled rabbit immunoglobulin (Specific for *Clostridium haemolyticum*)

Treatment

Immediate use of penicillin or tetracyclines at high doses

Penicillin G sodium/potassium @15,000 IU/kg IV every 6– 8 hr

Oxytetracycline @15– 20 mg/kg IV every 24 hr

Antitoxic serum is administered in the early stage of disease

Supportive treatment: Blood transfusion, parental fluid and electrolyte solutions

Avoid undue excitement and exercise

Bulls should not be used for Service at least 3 weeks after recovery because of the danger of rupture of liver

For hemopoiesis: provide micro minerals (Iron, Copper and Cobalt)

Prevention and control

A formalin killed whole culture adsorbed on aluminium hydroxide give protection for one year

Vaccination is carried out 4-6 weeks before and expected occurrence of the disease

Annual vaccination of all animals above 6 months in enzootic areas

The carcass should be disposed off by deep burial or burning

f. Malignant edema

(Clostridial myonecrosis)

Etiology

Acute wound infection associated with *Clostridium septicum*, *C. chauvoei*, *C. perfringens*, *C. sordellii*, and *C. novyi*

C. sordellii has been associated chiefly with malignant edema of cattle and swelled head in sheep

In ram, malignant edema are restricted to the head is most commonly associated with *C. novyi* infection

In a retrospective study of 37 horses with **clostridial myonecrosis**, ***C. perfringens* was isolated from 68%**, *C. septicum* from 16%, and the remainder were mixed infections with these two species. *C. chauvoei*, *C. novyi*, and *C. fallax*

Epidemiology

All ages and species of animals are affected

The disease occurs sporadically, affecting individual animals, except in circumstances where a management procedure in a group of animals results in an outbreak

The **swelled head** occurs in young rams between 6 months to 2 years old and fight among themselves

Source of infection: Soil- borne

Transmission

In most cases a wound is the portal of entry

Deep puncture wounds accompanied by severe trauma provide the most favourable conditions for growth of anaerobes

Infection may occur through surgical or accidental wounds following vaccination, intramuscular injection of drugs, venipuncture or through umbilical cord in the new born

Dormant spores of *C. perfringens* and other clostridial species can be found in the normal muscle of horses, and it is possible in some cases that these may be activated by anaerobic conditions produced by the injected material

In horses, intramuscular injection of drugs, commonly in association with the treatment of colic, is the common precipitating factor

Perivascular leaking of drugs is also a precipitating cause in horses

Outbreaks can occur in sheep after management practices such as shearing and docking, or following lambing

Outbreaks have also been observed **in cattle following parturition, sometimes associated with lacerations of the vulva**

An unusual method of infection occurs when crows that have eaten infected carrion carry the infection to live

Pathogenesis

Potent necrotoxins are produced in the local lesion and cause death when absorbed into the bloodstream. Locally the exotoxins cause extensive edema and necrosis followed by gangrene.

Clinical manifestation

Acute onset with fever and toxæmia, clinical signs appear within 6 to 48 hours of

Infection

There is always a local lesion at the **site of infection** consisting of a soft, doughy swelling with marked local erythema accompanied by severe pain on palpation

At a later stage the swelling becomes tense and the skin dark and taut. Emphysema may or may not be present, depending on the type of infection, and may be so marked as to cause extensive frothy exudation from the wound

With *C. novyi* infections, there is no emphysema

Systemic illness

A high fever (41 to 42° C; 106 to 107° F)

Affected animals are depressed, are weak, and show muscle tremor and usually **stiffness or lameness**

The mucosae are dry and congested and have very poor capillary refill

The illness is of short duration, and affected animals die within 24 to 48 hours of the first appearance of signs

New cases continue to appear for 3 to 4 days after shearing or other precipitating cause

When infection occurs at **parturition**, **swelling of the vulva accompanied by the discharge of a reddish-brown fluid occurs** within 2 to 3 days

The **swelling extends to involve the pelvic tissues and perineal region**

The local lesions are accompanied by a profound toxemia, and death occurs within 1 to 2 days

In “ swelled head” of rams, the edema is restricted initially to the head

It occurs first under the eyes and spreads to the subcutaneous tissues of the head and down the neck

In **pigs**, the lesions are usually restricted to the axilla, limbs, and throat and are edematous with very little evidence of emphysema

In horses, emphysema detected by palpation or ultrasound

Necropsy findings

Edema fluid varies from thin serum to a gelatinous deposit

It is usually blood stained and contains bubbles of gas except in *C. novyi* infections when the deposit is gelatinous, clear and contains no gas

A foul, putrid odor is often present in infections with *C. perfringens* and *C. sordellii*

In “ **swelled head**” of rams, the edema of the head and neck may extend into the pleural cavity and also involve the lungs

Sample collection: Facial tissue and edema fluid in air tight container

Diagnosis

Examination of gram stained oedma fluid or the wound swab

Clinical pathology: Elevated muscle enzymes such as CPK, AST, and LDH

Blood-gas analysis conducted in affected horses revealed severe acidemia and metabolic acidosis

PCR for rapid identification and differentiation of the clostridium sp associated with malignant edema

Treatment

Administration of penicillin (high doses of crystalline penicillin @ 40,000 IU/Kg intravenously, repeated at 4- to 6-hour intervals) or a broad-spectrum antibiotic.

Antitoxin aids in controlling the toxemia but is expensive and must be given very early in the course of the disease

Surgical incisions to drain and flush with dilute hydrogen peroxide

Flunixin meglumine@2.2mg/Kg IV 24hr or other NSAIDs may be used

In horses, early and aggressive treatment with myotomy and fasciotomy, repeated if indicated, coupled with IV potassium penicillin is reported to allow recovery rates approaching 70%.

The success rate in treating horses with infections with *C. perfringens* was higher than that with *C. septicum*

Prevention and control

Vaccination with a specific or multivalent clostridial bacterin toxoid will prevent the occurrence of the disease in enzootic areas

Penicillin can be given prophylactically to animals at risk

Maintenance of hygiene at lambing, shearing, castration and docking

g. Black disease

(Infectious necrotic hepatitis)

An acute toxemia of sheep, cattle, and sometimes pigs and horses caused by the toxin of *Clostridium novyi* type B produced in damaged liver tissue. Outbreaks usually associated with fasciolosis

Epidemiology

Adult sheep in good condition are mostly affected

Seasonal prevalence related to the migration of immature liver fluke in the liver and snail population

The disease occurs in sheep and cattle and rarely in pigs and horses

The disease is always fatal in both sheep and cattle

Field outbreaks are usually precipitated by **migration of immature flukes or local hepatic injury or cyst of *Cysticercus tenuicollis*, trauma from liver biopsy and *Thysanosoma actinoides***

Outbreaks are most common in summer and autumn months

C. novyi type A and B have been incriminated as a cause of sudden death in sows and most common in spring season

Source of infection

Faecal contamination of the pasture by carrier animals

Cadaver of sheep died of the disease may cause heavy contamination

Pathogenesis

Spores of *C. novyi* are ingested and carried to the liver in the lymphatic system; the organism can be isolated from the liver of normal animals. Under local anaerobic conditions, such as occur in the liver when migrating flukes cause severe tissue destruction, the organisms already present in the liver proliferate, liberating α -toxin, which is necrotoxic and causes local liver necrosis and more diffuse damage to the vascular system. The nervous signs observed may be caused by this general vascular disturbance or by a specific neurotoxin.

Clinical manifestation

Sheep

Sheep have a rapid clinical course and are usually found dead

Affected sheep commonly die during the night time

Initially, fever (40– 42° C, 105– 107° F), which subsides to a premortal (subnormal) level

Some animals show hyperesthesia; respiration is rapid and shallow

The sheep remains in sternal recumbency and often dies within a few minutes while still in this position without evidence of struggling

Short clinical course in cattle and horses with profound depression and toxaemia abdominal pain noticed on deep palpation, and peritonitis and periorbital oedema

Necropsy finding

Rapid autolysis

Blood stained froth may exude from nostrils

Engorgement of subcutaneous vessels with edema

Dark appearance of the inside of the skin, particularly noticable on drying

liver has small areas of yellow-colored necrosis surrounded by a zone of hyperaemia

Blood stained serous fluid is present in body cavities

Evidence of recent invasion by liver fluke with channel of damaged liver tissue

Diagnostic confirmation

Isolation of *C. novyi* in the typical liver lesion

Fluorescent antibody staining identifies *C. novyi* but not the type

Differential diagnosis

Acute fasciolosis

Black leg

Malignant edema

Anthrax

Treatment

Parenteral penicillin may be attempted

Control

Vaccination with an alum-precipitated toxoid is highly effective and can be performed during the course of an outbreak

The mortality begins to subside within 2 weeks

On an affected farm the initial vaccination is followed by a second vaccination 4 to 6 weeks later and subsequently by annual vaccinations.

control of the liver fluke and snail

Pasture contamination from cadavers should be minimized by burning the carcasses

h. Braxy

Braxy is an acute infectious disease of sheep characterized by **inflammation of the abomasal wall, toxæmia with high fatality rate.**

Etiology

Clostridium septicum

It is a soil borne-organism and normal inhabitant of ovine intestinal tract

Epidemiology

The disease occurs only in midwinter with heavy frost and snow in pasture

Weaner and yearling sheep are usually affected

Case fatality rate is about 50%

Pathogenesis

Abomasitis associated with the ingestion of frozen grass or other feed permits invasion by *C. septicum* resulting in fatal toxæmia.

Clinical manifestation

Sudden onset of illness

Affected sheep segregated from the group

High fever

Abomasum may be distended with gas and abdominal pain

Sheep become recumbent, comatose and dies within few hours of illness

Necropsy findings

Localized area of edema, necrosis and ulceration of abomasal wall

Congested intestinal mucosa and few subepicardial petechiae

Diagnosis

Clostridium septicum can be isolated from the cut surface of abomasal wall or culture from heart, blood and other organ of fresh carcasses

Bacteriological examination of tissues must be carried out within an hour of death

Fluorescent antibody test for identification of organism in tissue sample

Differential diagnosis

Infectious necrotic hepatitis

Treatment: No treatment has been found to be of value

Control

Sheep should be yarded at night

Animal should be fed with hay before allowed to frosted pasture

Vaccination with a formalin-killed whole culture of *Cl.septicum*, preferably 2 injections at 2 weeks interval

Listeriosis

(**Syn:** Silage disease, circling diseases)

It is a sporadic disease of ruminants, particularly in sheep produce syndrome of encephalitis, spinal myelitis, abortion, septicaemia, uveitis, enteritis and mastitis. Occasionally septicemic form affects horses and pigs.

Etiology

L. monocytogenes, a gram positive non spore forming extremely resistant coccobacilli ubiquitous in the environment

Based on somatic (O) and flagellar (H) antigens *L.monocytogenes* is divided into 16 serovars

Virulent strains multiply in macrophages and monocytes and produce haemolysin and **listeriolysin O toxin**

Listeria spp. can grow and reproduce at temperature from 4°C to 45° C. At low pH and low temperature reduces the growth rate of the organism. Grow at pH ranges from 4.5- 9.6

Natural reservoir of *L. monocytogenes* is soil and mammalian G.I tract

Epidemiology

Prevalence of infection

Disease is more common in temperate countries as compared to tropical and subtropical countries

It is most common during the winter months

Predisposing factors

Reduced host animal's resistance

Poor nutrition, sudden changes of weather such as very cold and wet weather

Stress due to late pregnancy, parturition and transport

Long periods of flooding and silage feeding

Source of infection

The organism is ubiquitous in the environment

The organism commonly found in animal faeces, human faeces, farm slurry, sewage sludge, soil, farm water trough and surface water etc

Transmission

The highest prevalence of the disease due to feeding of silage has been recorded

Ingestion of contaminated food, water, milk, silage feeds etc

Animal to animal transmission through faeco-oral route

Host affected

It is primarily a disease of ruminants, particularly in **sheep** and occasionally in horses and pigs

Introduction of virulent strains through migratory carrier animals, birds, sea gulls also possible

Organism persist for **more than 3 months in sheep faeces** and survive in soil for 1-2years

Pathogenesis

Listeria is a facultative intracellular organism; infect intestinal cells by direct endocytosis

The organism is taken up by macrophages and monocytes but the phagocytic activity is prevented by bacterial superoxide dismutase and produce listeriolysin O, which disrupts lysosomal membranes and permit organism to grow in cytoplasm

Upon ingestion, organism reaches the intestine by penetration and develops in-apparent/sub-clinical bacteremia which may be cleared by development of immunity

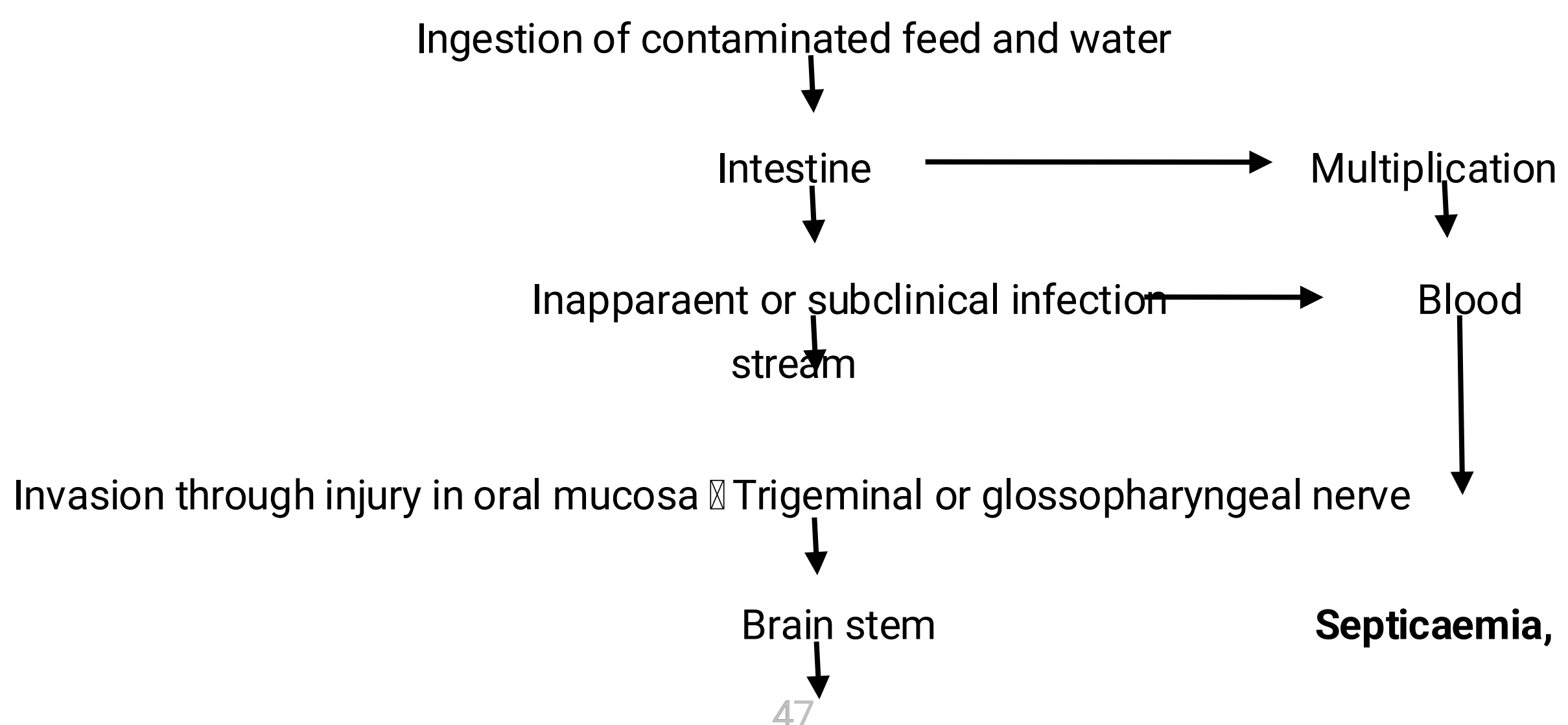
Bacteremia is frequently sub-clinical and may be accompanied with excretion in milk

Septicemic listeriosis develops with or without meningitis, most commonly in neonatal ruminants and in adult sheep and goats

In pregnant animals, invasion into placenta and foetus occurs in 24 hours of onset of infection

Edema and necrosis of placenta leads to **abortion**. **Infection in late pregnancy** results in still birth or birth of young ones that quickly develop a **fatal septicaemia**

Invasion of bacteria through injury in buccal mucosa, reach the brain stem through trigeminal or glossopharyngeal nerve causes acute inflammation of brainstem unilaterally leads to **encephalitis**



Acute inflammation



Encephalitis

Clinical signs

Sheep and goats

Encephalitis

Incubation period 14-40 days

In early infection, affected sheep separate from flocks with hunched stance

Fever

Affected animals develop in-coordination, head deviation, head tilt, walking in circles, unilateral facial paralysis and hypoalgesia, paralysis of lips and face

Keratitis, corneal ulceration, strabismus, nystagmus, panophthalmitis with pus evident in anterior chamber of one or both eyes

Paresis of muscles of the jaw, dropped jaw which ensues difficulty in prehension and mastication

More acute death in 2-4 days of infection

Abortion

Abortion at 12th week of pregnancy onwards (Third months)

Retained placenta

Blood stained vaginal discharge for several days

Septicaemic listeriosis

Incubation period 2-3 days

Most common in lambs and occasionally in pregnant sheep

High case fatality rate in newborn lambs and kids and also in periparturient sheep and goats

Depression, weakness, emaciation, pyrexia and diarrhea in some cases

Corneal opacity is accompanied by dyspnea, nystagmus and mild opisthotonus

Spinal myelitis

Uncommon

It may occur possibly as a result of ascending infection in the sensory nerves of the skin following dermatitis from prolonged wetting of fleece

Fever, ataxia, initial knuckling of hind limbs progressing to hindlimb weakness, and paralysis

In some cases, paresis and paralysis of the forelimbs. There is no evidence of CN involvement

Animals affected appear to be alert, bright and eat well

Ophthalmitis/Uveitis

It can occur as outbreak in sheep and cattle as result of ocular contact with contaminated silage

Iritis, swelling of iris, constriction of pupil. Advanced cases, pannus and corneal opacity may be uni/bilateral

Enteritis

It occurs in weaned sheep following feeding of big bale silage

Initially sheep found dead

The affected animals shows lethargy, anorexia, passing of loose green coloured faeces

Mastitis

It is found to be rare

The infection of udder may single or both quarter and occur as chronic

Poor response to treatment

Increased somatic cell count, but milk appears normal

Cattle

Encephalitis

The clinical signs of encephalitis is same as that of sheep, but clinical course is longer

Abortion

Abortion or stillbirth occurs sporadically in the last trimester of pregnancy

Septicemic listeriosis

Acute septicaemia - *L. monocytogenes* is not common in adult ruminants. No nervous signs but depression, weakness, emaciation, pyrexia, diarrhoea, hepatic necrosis and gastroenteritis

Corneal opacity, dyspnoea, nystagmus and mild opisthotonus

Death occurs in 12 hours following infection

Necropsy finding

Encephalitis

There is no gross lesions in brain

Histopathological examination of nervous tissue shows **microabscess**

Septicaemic listeriosis

Visual lesions occurs as **multiple foci of necrosis in the liver, spleen, myocardium** and in septicaemic form

Enteritis

Ulcerative abomasitis, typhlo-colitis

Histopathology - Micro-abscesses throughout the intestine

Sample collection

Choice of sample: brain, aborted placenta, foetus, occasionally spinal fluid, nasal discharge, urine, faeces and milk of clinically ill animals For histopathological examination- Brain sample should be collected in 10% formalin

Diagnosis

Based on clinical signs and necropsy findings

Listeriosis is confirmed only by isolation and identification of *L.monocytogenes*

Cerebrospinal fluid has increased protein (0.6-2g/L) and Pleucocytes composed of large mononuclear cells

Isolation of organism by Blood agar and cold enrichment technique especially in brain

Agglutination and Complement fixation tests can be performed to detect antibody

Molecular diagnosis by PCR

Micro-abscess present in CNS is important

Gram staining of paraffin embedded tissue may permit confirmation

Flourescent antibody technique and immunoperoxidase test is essential

Serology is not used routinely used for diagnosis, because many healthy animals have high listeria titre

Differential diagnosis

Pregnancy toxaemia in sheep

Nervous ketosis, rabies, kid, polioencephalomalacia, middle ear disease

Trichomoniasis

Brucellosis

Toxoplasmosis

IBRT

Salmonellosis

Contagious ophthalmia

Bovine spongiform encephalopathy

Polioencephalomalacia

Lead poisoning

Coenurosis

Brain abscess

Vestibular disease

Treatment

Procaine penicillin G @ 200000 IU/Kg IV initial loading dose

Inj. Penicillin G-44000 IU/kg BW, I/M daily for 10-14 days
Inj. Oxytetracycline @ 20 mg/Kg once a day for 10-14 days
Thiamine @ 10mg/Kg slow IV
Combination of ampicillin and gentamicin antibiotics are more effective
Inj. Flunixin and meglumine @ 1.0 mg/Kg IV
High dose of dexamethasone 1 mg/ kg b.wt
Supportive therapy including fluid and electrolyte therapy
For iritis , systemic antibiotics coupled with subpalpebral corticosteroid and atropine to dilate the pupil
Live attenuated vaccine shown to induce protection
Commercial killed vaccine available

Control

It is difficult to control because of ubiquitous nature of organism
Avoid feeding of poor quality silage to pregnant animals
Proper care in silage preparation
Avoid soil contamination and mole hills
Avoid eye contact with silage
Disinfection of premises with common disinfectants like bleaching power and sodium hypochlorite

Swine Erysipelas

(Syn: Diamond skin disease)

Erysipelas is an infectious disease of growing or adult swine, clinically characteristic by diamond shaped skin lesions, arthritis and endocarditis.

Etiology

It is caused by *Erysipelothrix rhusiopathiae* (*insidiosa*) is a pathogenic organism where as *E. tonsillarum* is non-pathogenic found in tonsil
It is a gram positive non spore forming facultative anaerobic rods
Around 32 serotypes have been identified, among which 15 serotypes are affecting pigs
Cattle strains are pathogenic for pigs
It can survive for long time in frozen or refrigerated media and in alkaline soil
The organism are resistant to environmental influences and disinfectant like 0.2% phenol

Epidemiology

Worldwide in distribution

The disease most commonly occur in unvaccinated growing pigs >3 months and adult animals

Erysipelas also occurs in lambs and turkeys

It occurs occasionally in other poultry and many kinds of wild birds

Prevalence of infection in carrier pigs ranges from 3-98%

About 20-40% infected pigs are acting as carriers

Morbidity (30%) and case fatality rate (7.5%)

Predisposing factors

Age, health and intercurrent disease like hog cholera

Piglets from unvaccinated dam are affected at <2weeks of age

Heat and cold stress

Sudden change in diet

Poorly constructed floors and cohabitation with other carrier species such as birds

Source of infection

Soil contamination through faeces of infected or carrier pigs

The organism can survive in faeces contaminated soil for several months

Clinically normal animal carry the organism in tonsil

Effluent from commercial piggeries

Cattle and lambs are infected by grazing on pasture contaminated with piggery effluent

Transmission

Ingestion of contaminated feed and water

Skin abrasion

Flies also transmit the disease mechanically

Zoonotic implications

Veterinarian, abattoir workers and butchers are increased risk and usually causes swollen finger and erysipeloid like lesions

Pathogenesis

Infection is acquired through ingestion or skin wound

The organism invade the blood stream of all infected animals and develops septicemia in 1-7 days

Vascular damage is characterized by swelling of endothelial cells, adherence of monocytes to vascular wall and widespread hyaline microthrombus formation

Localization of bacteria in chronic form commonly occur in skin, joints and on heart valve during haematogenous spread causes diamond skin lesions, arthritis and endocarditis

Disc spondylitis also occur in association with poly arthritis due to erysipelas

Acute form

Incubation period is 1-7days

Sudden onset of high fever, severe prostration, complete anorexia, thirst, occasional vomiting, reluctant to rise, in-coordination during walking, dyspnoea, conjunctivitis with ocular discharge

Affected animals may be active and continue to eat, eventhough their temperature are high

Severely affected pigs show marked **red to purple discoloration of the skin of jowl, ventral surface of the whole body and become cyanotic**

Abortion in pregnant animals

Skin form

Lesions are pathognomonic: **Classical diamond shaped, red urticarial plaque**, about 2.5-5 cm² or more diffuse edematous eruption present

Mostly, lesions seen in belly, inside the thigh, throat, neck and ears. After 2-4 days, the pig recovers or dies as a result of diarrhoea, dyspnoea, and cyanosis

Skin lesions disappear in about 10 days

Chronic form (Vegative endocarditis and arthritis)

Alopecia

Sloughing of tail and tip of the ear

Dermatitis in the form of hyperkeratosis of the skin, back, shoulder and legs

Retarded growth

Joint lesions are commonest in the elbow, hip, hock, stifle and knee joint causes **lameness and stiffness**

Joints are usually enlarged, hot and painful to touch

Paraplegia is due to involvement of intervertebral joint

In **endocarditis**, affected animals die suddenly especially at times of **exercise such as mating or movement between pens**

Progressive emaciation, dyspnoea and cyanosis

Increased cardiac impulse

Loud murmur audible on auscultation

Necropsy finding

Internal petechial and ecchymotic haemorrhage mainly in pleura, peritoneum, beneath the renal capsule, heart, liver and spleen

Non suppurative proliferative arthritis in limbs and intervertebral joint

Serofibrinous amber coloured fluid in joints

Friable vegetative lesions on the heart

Sample collection

Whole blood, heart blood smear, swab from heart valve, spleen and kidney

Diagnosis

Based on clinical signs and necropsy finding

Isolation and identification of organism in blood agar

Fluorescent antibody technique

Differential diagnosis

Septicaemia

Septicaemic salmonellosis

Hog cholera

Streptococcal septicaemia

Streptococcal endocarditis

Arthritis

Glassers disease

Mycoplasma arthritis

Ricket/chronic zinc poisoning

Foot rot

Leg weakness

Treatment

Procaine penicillin and anti-erysipelas serum 5-20 ml provide protection for 1-2 weeks

Mild case- Penicillin alone effective

Inj. Oxytetracycline can also be used

Prevention

Vaccine with sero-types 2 and 10 give protection

Single vaccine between 6 and 10 weeks and booster 2-4 weeks intervals behind axilla region can be followed

Crystal violet vaccine can be used very widely

Control

Isolation of infected animal

Maintenance of good hygiene

Implementation of strict bio security measures

Reduce stress

Slaughter of all infected animals

Suckling piglets in endemic areas give antierysipelas serum 10 ml/day during first week at monthly intervals until vaccination is done

Disinfection of premises with caustic soda and sodium hypochlorite

Leptospirosis

(Syn: Weil's disease, Canicola fever, Seven day fever, Rat scratch yellow fever and Paddy field fever)

Leptospirosis is a transmissible disease of animals and humans caused by infection with any of the pathogenic members of the genus *Leptospira*. It causes broad range of clinical effect from mild, subclinical to acute severe septicaemic infection leading to multiple organ failure and death.

Etiology

It is caused by *Leptospira interrogans* containing 212 serovars arranged in 23 serogroups

It is a gram negative spiral shaped motile organism. One or both ends of the spirochete are usually hooked

Common *Leptospira* serovar in animals

L. pomona is the commonest infection in all farm animals

Bovine- *L. Hardjo-bovis*

Swine- *L. pomona* and *tarassvi*

Horse - *L. pomona*, *L. Bratislava*

Dog-*Leptospira interrogans* serovar *Canicola*, *L. Icterohaemorrhagiae*, *L. Grippotyphosa*, *L. Pomona*, and *L. Bratislava*

Pig, cattle, racoon, opossum, skunk- *L. Pomona*

Maintenance host

Leptospira serovars are generally adapted to one or more mammalian or marsupial maintenance hosts, which may or may not develop clinical signs

Dogs are reservoir hosts for *L. Canicola*

Pigs – *L. Bratislava* and *L. Pomona*

Horses may also maintain *L. Bratislava*

Cattle are the primary reservoir hosts for *L. Hardjo*, however this serovar are maintained in red deer, sheep and goat

Rodents – *L. Icterohaemorrhagica*

Incidental host- Leptospiral serovar are highly pathogenic and tendency to cause acute severe disease

Sheep and Man- *L. hardjo*

Sheep cattle and pig- *L. Grippotyphosa*

Cattle and pig- *L. Icterohaemorrhagica*

Horse – *L. Bratislava*

Epidemiology

Prevalence of infection

World wide in distribution

The disease is highly prevalent in India

Morbidity rate- 10-30%

Case fatality rate- 5%

Host affected

All animals including human beings are susceptible

Cattle and lambs are highly susceptible

Predisposing factor

Moisture and dampness increase the incidence of leptospirosis

In high rainfall area *L.hardjo* and in low rainfall areas *L.pomona* are prevalent

There is a direct correlation exists between incidence and rainfall

Numerous outbreaks of leptospirosis have occurred following heavy rainfall events and floods

Stagnant water preserves more organism than the flowing water

Access of cattle to contaminated water supplies such as streams, rivers, flood,or drainage water

Source of infection

Urine of infected animals

Rodents- excrete leptospiral organism in urine

Contaminated flood water with urine

Aborted foetus

Genital secretions of persistently infected animals. *L. interrogans* serovar Hardjo is excreted from the genital tract of aborting cows for as long as 8 days after abortion or calving

Ground surface moisture increases the persistence of organism in the environment

Wildlife (deer, skunk)-common source of *L.Pomona*

Sheep and horses are not common sources of infection because of low-grade and intermittent leptospiuria

Transmission

Direct contact with infected urine, placental fluid or milk

Penetration via abraded/injured skin, mucous membrane and eyes

Transplacental or venereal transmission is also possible

Through licking of urine of infected animals

Through rodents

Pathogenesis

Leptospire invade the host across mucosal surfaces or softened skin. They have the ability to bind to epithelial cells and attach to the constituents of the extracellular matrix through an active process involving surface proteins. Pathogenic leptospire are found extracellularly between cells of the liver and kidney. Release of lymphokines such as TNF- α from monocytes through the endotoxic activity of the leptospiral LPS may be an important virulence mechanism. Induction of TNF- α release may help explain the damage to endothelial cells with resultant hemorrhage seen in severe leptospirosis.

Leptospirosis can occur as an acute and severe disease caused by **septicemia with evidence of endotoxemia such as hemorrhages, hepatitis, nephritis, meningitis**; as a **subacute moderately severe disease with nephritis, hepatitis, agalactia, and meningitis**; or as a **chronic disease characterized by abortion, stillbirth, and infertility**. ***L. Pomona* produce hemolysin** causes intravascular hemolysis and interstitial nephritis ***L. hardjo* does not produce hemolysin and does not cause interstitial nephritis**, but it does cause clinical infection in sexually mature, lactating or pregnant females

In the acute phase of the disease, the **animal may die of septicemia or hemolytic anemia or both**. Subsequently, the animal may die of uremia caused by interstitial nephritis. **Focal chronic interstitial nephritis, also called white-spotted kidney**, is a common finding in clinically healthy cattle at slaughter and has frequently been assumed to be related to current or prior infection with *Leptospira* spp.

Localization of leptospirae in nervous tissue is common in sheep and goats and may result in the appearance of signs of **encephalitis**.

Pulmonary haemorrhage

Respiratory manifestation of leptospirosis associated with severe respiratory distress, pulmonary hemorrhage, and high case fatality has been described in humans, dogs, and horses

Clinical signs

Incubation period- 4- 20 days

Leptospira infections may be asymptomatic, mild or severe, and acute or chronic. The illness tends to be milder in the reservoir host and more severe when the serovar is not adapted to the host species. The clinical signs are often related to **kidney, liver and reproductive tract**, but other organs can be affected.

Cattle

The clinical signs may occur as acute, subacute and chronic

Acute leptospirosis occurs mainly in calves

Fever, anorexia, conjunctivitis and diarrhoea, and in more severe cases, **jaundice, hemoglobinuria** as a result of extensive intravascular hemolysis, **hemolytic anemia, pneumonia or signs of meningitis** (e.g., incoordination, salivation, muscle rigidity)

Localization in kidney causes interstitial nephritis and persistence in these lesion

Animal may die of uremia caused by interstitial nephritis

Some infections can be fatal, and the survivors may be unthrifty after recovery

In adult cattle- abortions, stillbirths, and occasionally, weak calves with increased neonatal mortality

Sudden onset of agalactia or decreased milk production.

Mastitis -the milk are red coloured or contain blood clots

Repeat breeding and infertility

Severe lameness is due to tenosynovitis

Flabby bag syndrome - (Sudden drop milk syndrome)

Subacute

Fever, anorexia, dyspnoea and haemoglobinuria

Marked drop in milk production and appearance of blood stained or yellow orange thick milk in all four quarters without apparent physical changes in the udder

Chronic

Abortion during last trimester of pregnancy

Sheep and goats

Leptospirosis in sheep and goats is similar to the disease in cattle

Swine

The **chronic form of leptospirosis is most common**

Most of the abortion occurs 2-4 weeks before term and high incidence of still birth, mummified or macerated fetuses, and increased neonatal mortality. Fever, decreased milk production and jaundice may also be seen

Horse

Abortion

Periodic ophthalmia (uveitis in horse), moon blindness and recurrent iridocyclitis is a late **complication** of systemic leptospirosis

Recurrent episodes of ocular discharge including photophobia, lacrimation, conjunctivitis, keratitis, pericorneal coma of blood vessels, hypopyon and iridocyclitis

Agglutinins are present **in the aqueous humor in greater concentration** than in the serum

Dogs

Acute kidney injury has been the most common presentation for canine leptospirosis in recent years

Affected dogs may present with lethargy, **anorexia, vomition, abdominal pain, and history of polyuria, oliguria, or anuria.**

Dogs that survive acute renal failure may return to baseline or progress to chronic kidney disease.

Leptospirosis should also be considered in any dog with previously diagnosed chronic kidney disease

Acute liver disease may accompany acute renal failure in dogs with leptospirosis, or it may occur alone. Affected dogs may be icteric, and serum biochemistry analysis reveals increased bilirubin and alkaline phosphatase. ALT is typically less markedly increased than **alkaline phosphatase**

Less common manifestations of canine leptospirosis include bleeding disorders characterized by petechial hemorrhages, epistaxis, melena, and hematemesis

Additional clinical signs reported in dogs with leptospirosis include **vomiting, diarrhoea,** weight loss, fever, hypothermia, oculonasal discharge, **lymphadenopathy,** effusions, and edema

The less common finding in canine leptospirosis; however, cough or dyspnea, or radiographic abnormalities have been noted in a number of affected dogs

Zoonosis

Human can get infection by direct contact with infected urine (and also by ingestion of food and water contaminated with urine of infected animals)

Spread of infection during convalescent period is rare

Swimmers, slaughter house workers, veterinarians, butchers, sewage clearing persons, farm workers at risk

Clinical signs - Human

Incubation period is 4-14days.

Disease is biphasic in nature.

First phase: Flu like fever, chills, **myalgia, intense headache** and become asymptomatic

Second phase: Flu like signs with muscle pain, abdominal pain, **vomition, diarrhoea, rash, red eyes** renal failure, meningitis, liver damage, jaundice and highly lethal pulmonary haemorrhage

Myalgia is commonly reported in human leptospirosis and is associated with the septicemic phase of the disease

In people, **acute kidney injury due to leptospirosis is often nonoliguric and can be associated with hyponatremia and hypokalemia**

Pulmonary hemorrhage is now one of the most common clinical signs in outbreaks of human leptospirosis.

Hypotension, irregular pulse-ECG reveals the type of arrhythmia

Increased neck rigidity and cells in cerebrospinal fluid

Necropsy finding

Generalised pale and icteric carcass

Liver is enlarged, mottled, and pale-red to yellow

Kidneys are swollen and edematous with pale, radiating streaks in both cortex and medulla

Kidneys may have white foci on the subcapsular surface

Mucoid exudates in chorionic surface of placenta

petechial or ecchymotic hemorrhages on any organ, pleural, or peritoneal surface

lungs may be wet, heavy, and discoloured

Samples Collection

Live animals- **Serum** during acute (leptospiremic phase) and **urine** during chronic phase (Leptospiuric phase), body fluids, blood, milk, cerebrospinal fluid, thoracic and peritoneal fluids and cervico-vaginal mucus

Dead animals- Liver, lung, brain, kidney, body fluids like thoracic and peritoneal fluids, cervico-vaginal mucus

Diagnosis

Based on clinical signs and necropsy findings

neutrophilia, lymphopenia, monocytosis, and mild anemia, thrombocytopenia

Coagulation abnormalities may include increased fibrin degradation products and prolonged prothrombin time (PT) or activated partial thromboplastin time (APTT).

Urinalysis may reveal hyposthenuria, isosthenuria, or hypersthenuria, depending on the degree of renal involvement. Other changes may include **proteinuria, glucosuria, cylindruria, hematuria, and pyuria**

Increased serum creatinine and blood urea nitrogen are due to acute or chronic renal damage

Liver damage-Serum bilirubin, with normal or mildly elevated SGPT, ALT, SGOT and CPK

Lung: Chest X-ray shows lower and mid zone opacities

Isolation of organism by culture using Fletchers medium, Stuarts medium and **EMJH** medium

Identification of organism in serum or urine by using dark field microscope (DFM)

Detection of antibody titre by microscopic agglutination test (MAT). **MAT titre is 1:100 is considered positive**

Molecular diagnosis by PCR

Microscopic agglutination test (MAT) is the standard serological test for detection of acute infection. Antigens selected for use in the MAT should include representative strains of the serogroups known to exist in the region

MAT may not useful for diagnosis of chronic infection

A commercially available **ELISA** and the **ImmunoComb Leptospirosis Kit**, which detect *L. hardjo* antibodies, have been compared with the MAT.

Measurement of aqueous humor antibody titers against leptospire in horses

Differential Diagnosis

Haematuria Babesiosis , post parturient haemoglonuria and bacillary haemoglobinuria

Brucellosis , IBRT

SMEDI, (Still birth, mummification, early embryonic death, and infertility, etc).

Treatment

Cattle

Inj. Dihydrostreptomycin @12 mg/kg IM twice for 3 days

For the elimination of leptospiruria in **cattle and pigs**, a **single dose of dihydrostreptomycin @ 25 mg/kg IM for 3-5 days**

Inj.Oxytetracycline@ 20mg/Kg IM for 3-5 days or single dose of long acting Oxytetracycline @ 20mg/Kg IM

Inj. Tylosin @ 18mg/kg for 3-5days

In an outbreak in cattle, the **simultaneous treatment of all animals with dihydrostreptomycin (25 mg/ kg IM as single dose) and vaccination** has been successful in preventing new cases and abortion when pregnant cattle are involved

Dihydrostreptomycin G (25 mg/kg IM for 1 day, 3 days, or 5 days)

For outbreaks of **leptospiral abortion in horses**, treatment of pregnant mares with **dihydrostreptomycin (50 mg/kg)** IM daily for 3 to 5 days

Swine

Oxytetracycline (40 mg/kg IM daily for 3 days or 5 days), tylosin (44 mg/kg IM daily for 5 days)

For periodic ophthalmia in horse-**Atropine eye ointment** is also usually applied three times daily to **maintain dilatation of the pupil**

Dog

Current recommendations are to treat the affected animals with **Doxycycline 5 mg/kg/day, IV or PO) for 2 week**

For dogs that cannot tolerate doxycycline, initial therapy with penicillin 20,000 IU /kg bwt is appropriate, but this should be followed by a 2-wk course of doxycycline to eliminate the renal carrier phase of infection

Dogs recently exposed to leptospirosis may be treated prophylactically with oral doxycycline for 14 days

Supportive therapy to maintain normal fluid, electrolyte, and acid-base balance

Supportive measures may include antiemetics, GI protectants, phosphate binders, and hepatic supportive medications

Renal replacement therapy (Dialysis) in anuric animals

Commercial bacterins for dogs are available for serovars *Canicola*, *Icterohaemorrhagiae*, *Grippotyphosa*, and *Pomona*

Corticosteroid (eg.prednisolone 30-60mg for 7-10 day)

Prevention

Prevention by maintenance of appropriate hygiene, sanitation and management.

Vaccine

Vaccine: Trivalent *Leptospira* (inactivated vaccine for bovines) - This vaccine contains three Leptospiral serovars respectively, ***L.pomona*, *L.hardjo* and *L.grippotyphosa***. It is a **killed vaccine** containing, thiomersal inactivated, aluminium hydroxide gel adsorbed vaccine which confers specific immunity against leptospirosis in cattle. Available as 30ml (5 doses) and 6 ml in single doses.

Canine leptospira killed vaccine containing serovars ***L.icterohaemorrhagiae*, *L.canicola*, *L.pomona* and *L.grippotyphosa*** are currently used

Control

Pasteurization of milk

Rodents control measures

Filling of stagnant water place with sand or removal of water should be done

Garbages should be discarded / covered

Disinfection of infected premises

Isolation and identification of renal carriers for controlling of infection

Spirochetosis

(Syn: Avian Borreliosis)

It is an acute septicemic bacterial disease that affects a wide variety of bird' s mainly young fowls. The spirochetosis are divided into

1. Intestinal spirochetosis

2. Blood spirochetosis

Etiology

It is caused by *Borrelia anserina*, a helical spiral shaped motile organism

Epidemiology

The disease is prevalent in North Africa, the Middle East and Asia

Morbidity can approach 100% and mortality may be 33-77%.

Chickens, turkey, geese, ducks, pheasants, grouse and canaries are affected

Predisposing factor-The disease is associated with deep litter system or free range

Source of infection-infected droppings, infected ticks and infected birds

Transmission

Transmitted by soft tick *Argus persicus* (biological vector)

Other vectors like mosquitoes, ticks, lice and inanimate object mechanically transmit the disease

The disease also transmitted through contact with infected birds and infected materials such as faecal dropping, blood and tissue

Ingestion of contaminated feed or water with infected fecal droppings

Cannibalism during spirochetemia

Clinical signs

Blood spirochetosis

Anorexia, pyrexia, listlessness, depression, somnolence, moderate to marked shivering and increased thirst

Cyanosis of head

Young birds are more severely affected than older birds

Anaemia, pale comb, diarrhoea and neurological signs

Bile-stained faeces observed followed by paralysis

Intestinal spirochetosis

Chronic diarrhoea with wet greasy droppings
Green or yellow diarrhoea with increased urates
Feathers around vent and eggs soiled
Egg production in layers or breeders may be reduced by 5%– 10%, with a higher number of small eggs
Weakness and progressive paralysis
The course of the disease is 1-2 weeks

Necropsy finding

Spleen is enlarged, with petechial or ecchymotic hemorrhages, appearing dark or mottled
Occasionally, the liver may be swollen and contain focal areas of necrosis
Kidneys may be enlarged and pale

Sample collection

Live birds-Droppings, blood and serum
Dead birds-Scrapings from caecal mucosa

Diagnosis

Based on clinical signs and necropsy finding
Demonstration of actively motile *Borrelia* sp in blood by Dark field microscopy
Demonstration of organism in Giemsa stained blood smear
Silver impregnation staining
Culture on Barbour-Stoenner-Kelly medium, embryonating duck or chick embryos or in young ducks or chicks
Polymerase chain reaction (PCR)
Increased numbers of immature RBC due to the anemia
Agar gel immunodiffusion test and haemagglutination test also used for diagnosis of borreliosis

Differential diagnosis

Salmonellosis
Pasteurellosis
Ranikhet disease

Treatment

Several antibacterial agents are effective
The most widely used are penicillin derivatives, but streptomycin, tetracycline, and tylosin are also effective
Control of ticks and immunization of animals

Prevention and control

Vaccines comprising inactivated organisms have been used

Whole-egg propagated bacterins are usually given I/M 1 or 2 times

Tick control measures

Hyperimmune serum unable to reduce the infection in ticks

Birds recovered from one type of *Borrelia* can confer immunity for more than one year for homologous types of infection

Bordetellosis

(Infectious Tracheobronchitis, Kennel cough)

Tracheobronchitis is an acute or chronic inflammation of the trachea and bronchial airways; It is a mild, self-limiting disease but may progress to fatal bronchopneumonia in puppies or to chronic bronchitis in debilitated adult or aged dogs. The illness spreads rapidly among susceptible dogs housed in close confinement (eg, veterinary hospitals or kennels). Canine infectious tracheobronchitis (kennel cough) is often secondary to viral infection of the respiratory system.

Etiology

Bordetella bronchiseptica may act as a primary pathogen, especially in dogs <6 mo old; however, it and other bacteria (usually gram-negative organisms such as *Pseudomonas* sp, *Escherichia coli*, and *Klebsiella pneumoniae*) may cause secondary infections after viral injury to the respiratory tract. Concurrent infections with several of these agents are common. Other causes of tracheobronchitis in dogs include parasites, eg, *Aelurostrongylus abstrusus* (also in cats), *Capillaria aerophila*, *Crenosoma vulpis*, and *Oslerus osleri*.

Canine parainfluenza virus, canine adenovirus 2 (CAV-2), or canine distemper virus can be the primary or sole pathogen involved in tracheobronchitis.

Clinical Findings:

The prominent clinical sign is paroxysms of harsh, dry coughing, which may be followed by retching and gagging and partial anorexia. Body temperature and WBC counts usually remain normal. Development of more severe signs, including fever, purulent nasal discharge, depression, anorexia, and a productive cough, especially in puppies, indicates a complicating systemic infection such as distemper or bronchopneumonia. Stress, particularly due to adverse environmental conditions and improper nutrition, may contribute to a relapse during convalescence.

Sample collection: Specimens collected by transtracheal aspiration or bronchoscopy

Diagnosis:

Tracheobronchitis should be suspected whenever the characteristic cough suddenly develops 5– 10 days after exposure to other susceptible or affected dogs.

Severity usually diminishes during the first 5 days, but the disease persists for 10– 20 days.

Tracheal trauma secondary to intubation may produce a similar but generally less severe syndrome.

Thoracic radiographs are essential to determine the severity of disease and to exclude other causes of cough.

Treatment

Provide good nutrition, hygiene, and nursing care, as well as correction of predisposing environmental factors, hasten recovery.

Cough suppressants containing codeine derivatives, such as hydrocodone (0.25 mg/kg, PO, bid-qid) or butorphanol (0.05– 0.1 mg/kg, PO or SC, bid-qid), should be used only as needed to control persistent nonproductive coughing.

Antibiotics are usually not needed except in severe chronic cases; cephalosporins, quinolones, chloramphenicol, and tetracycline are preferable because they reach effective concentrations in the tracheobronchial mucosa.

kanamycin sulfate (250 mg) or gentamicin sulfate (50 mg) diluted in 3 mL of saline may be administered by aerosolization bid for 3 days for those animals are not responsive to parenteral antibiotics

Aerosolization treatment should be preceded by administration of bronchodilators

Corticosteroids may help alleviate clinical signs but should be used concurrently with an antibacterial agent; they are contraindicated in severely ill, coughing dogs.

Prevention:

When the risk of *B bronchiseptica* infection is significant, a live, avirulent, intranasal vaccine or parenteral products containing subunit bacterial extracts should be used. A combination of an avirulent *B bronchiseptica* and a modified-live parainfluenza vaccine is available for intranasal use. One inoculation is administered to puppies >3 wk old.

Avian Bordetellosis

Avian bordetellosis is a highly infectious, acute disease of the upper respiratory tract of young turkeys.

Etiology

Bordetella avium and *B hinzii*

It is a gram-negative, nonfermentative, motile, aerobic bacilli that grow on a variety of media, including MacConkey agar, Bordet-Gengou agar, blood agar, veal infusion broth, trypticase soy broth, and brain-heart infusion broth.

B hinzii, but not *B avium*, grows on minimal essential medium

Epidemiology

Bordetellosis has been **identified in almost every area of the world**, where turkeys are intensively reared.

It appears to be rare or inapparent in some locations, whereas severe outbreaks occur in other geographic regions.

Morbidity in young turkeys with bordetellosis is usually 80%– 100% and mortality rate is low

Transmission

B avium is **highly contagious** and easily transmitted from infected turkeys to susceptible birds by direct contact.

It can also be spread through contaminated drinking water, feed, and litter, which can remain infectious for as long as 6 months.

Other domesticated and wild birds from which *B avium* has been isolated should be considered possible reservoirs of infection.

Clinical manifestation

Incubation period: 7– 10 days

Clinical signs include **sinusitis**, sneezing (snick), watery or foamy eyes, clear nasal discharge following gentle pressure to the nares, mouth breathing, dyspnea, tracheal rales, and altered vocalization.

Older turkeys may develop a dry cough.

During the first 2 weeks of disease, the nares and feathers of the head and wings often appear crusted with wet, sticky exudate. By 1 week after disease onset, tracheal softening can sometimes be palpated through the skin of the neck.

Copious production of mucus in the trachea and tracheal collapse can result in mortality due to suffocation.

Complicated disease often triggers more exaggerated signs, including airsacculitis.

In cockatiels, *B. avium* is known to cause temporomandibular rigidity (lockjaw syndrome).

Diagnosis

It is based on clinical signs and lesions and isolation of *B. avium* or *B. hinzii* from the respiratory tract.

Both *B. avium* and *B. hinzii* are MacConkey agar is recommended for primary culture because it will differentiate nonfermentative *Bordetella* from fermentative opportunists, including *E. coli*.

Treatment and control

Antimicrobial therapy is rarely effective but sound husbandry practices may reduce the impact of an outbreak.

Rigorous cleanup and disinfection after field outbreaks is essential. Most of the commonly used disinfectants are effective against *B. avium* when applied as directed

Glanders

(Syn: Farcy)

It is a highly contagious and chronic disease of equine characterized by the development of nodules, ulcers in upper respiratory tract, lungs, and cutaneous tissues.

Etiology

Glanders is caused by *Burkholderia mallei*, previously it is called as *Pseudomonas mallei*

Burkholderia pseudomallei causes Melioidosis

Pseudomonas is a medium sized 0.5-1.0 x 1.5-5.0 µm in length. It is a gram negative, rod shaped, strict aerobic organism. Motile by one or more polar flagella

B. mallei is a non motile organism

Some species produce soluble pigments, pyocyanin

Most will grow on MacConkey Agar

Epidemiology

Prevalence of infection

The disease is prevalent in horse rearing countries. Glanders has been reported from Eastern Europe, India and Africa

Host affected

Horse, Mule and Donkeys

All age group are susceptible

Acute form is common in mules and donkeys

Chronic form is common in horse

Sources of infection

Infected and carrier animals

Chronic nodular lung lesion, which have ruptured into the bronchi, infect upper airway and excreted through nasal and oral secretion

Transmission

Ingestion of contaminated fodder with nasal discharge and sputum of infected animals

Use of communal watering trough

Contamination of skin abrasions with direct contact or from harness or grooming tools

Inhalation -rare

Man get infection via broken skin, contact with exudates

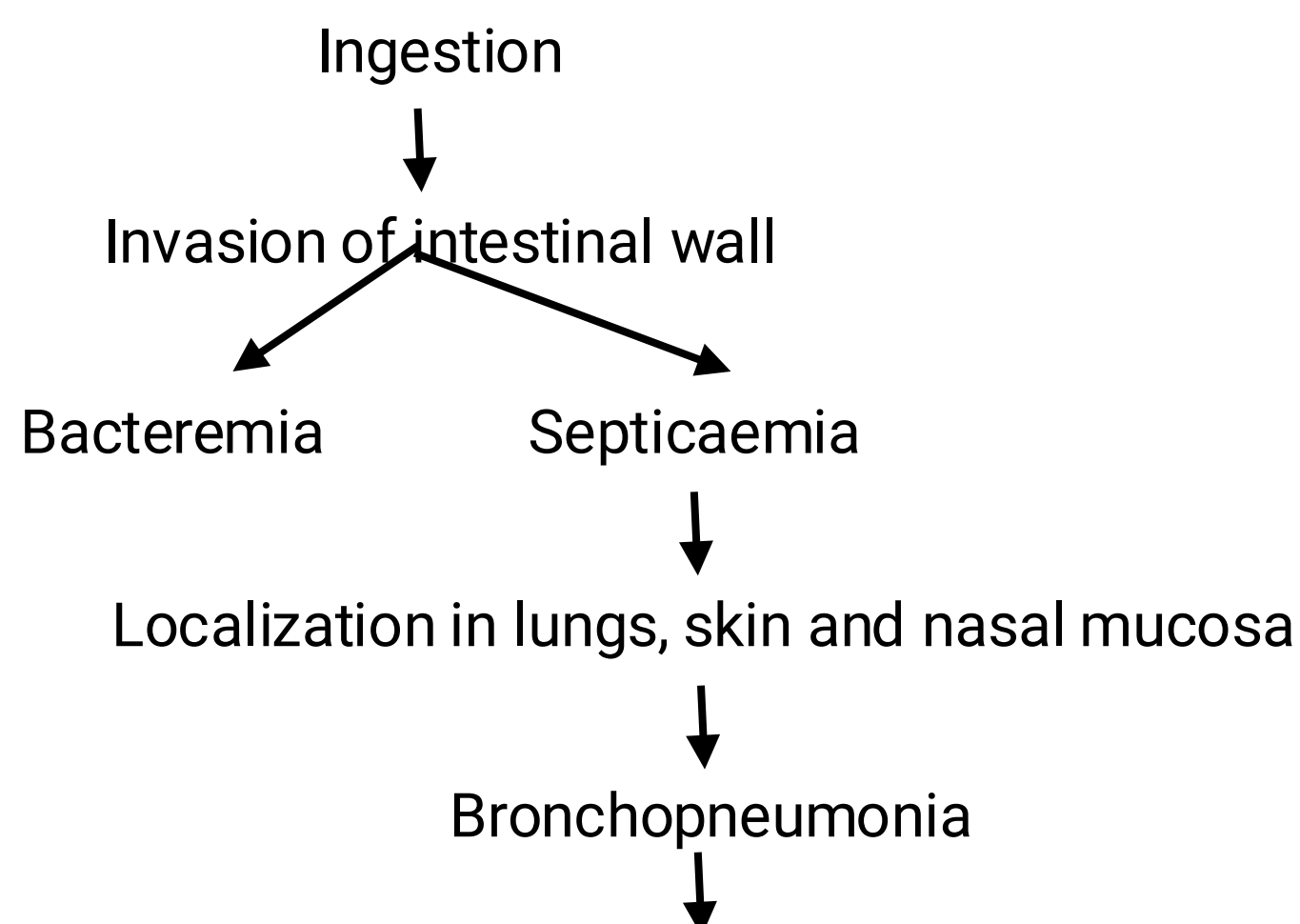
Pathogenesis

The organism gains access through intestinal mucosa leads to septicemia in acute and bacteremia in chronic phase

It invades the regional lymphnodes through pharyngeal mucosa and undergo proliferation

From the skin wound the organism reaches the blood stream and localized in various organs and forms nodules and ulcers in the skin

Involvement of lungs via inhalation leads to severe broncho-pneumonia



Anoxic anoxia



Death

Clinical signs

Incubation period varies few days to several months

Acute disease

High fever, cough and nasal discharge

Ulcer in the nasal mucosa

Nodules on the skin of the lower limbs or abdomen

Death is due to septicaemia

Chronic form

It occur as pulmonary, skin and nasal form

Pulmonary form

Chronic pneumonia with cough

Frequent epistaxis

Labored pneumonia

Nasal form

Lesion appear on lower part of turbinate bone and cartilaginous nasal septum

The lesion commence as nodule and ulcerate and become confluent

On healing the ulcer are replaced by stellate scar

In early stage there is serous nasal discharge, which may be unilateral later become purulent and blood stained

Enlargement of submaxillary lymph node is common

Skin form (Farcy)

Subcutaneous nodules (1-2cm dia), which soon ulcerate and discharge dark honey color pus

In some cases the lesions are deeply situated and discharge a pus through fistulous tract

Nodules and ulcers formed along the lymphatic channels of the skin.

Thickened fibrous lymph vessels radiate from the lesion and connect one to the other

The predilection site for cutaneous lesion is medial aspect of the hock, but they can occur in any part of the part

Zoonosis

Human-High temperature, swelling and pain usually on the hands, lips, eyes, accompanies with swelling of neighbouring lymphnodes

Ulcers develop on nose, mouth in some cases

Abscess development and pustules in the skin are the important features of the disease

Necropsy finding

Acute form-Multiple petechial haemorrhage throughout the body

Chronic form- Millitary nodules in the lung

Diagnosis

Based on clinical signs and necropsy findings.

Clinical pathology- low haemoglobin, low RBC and PCV

Mallein test-Intrapalberal test

0.1ml of mallein injected intradermally into the lower eyelid causes marked edematous swelling of the eyelid with blepharospasm and severe conjunctivitis after 48 hrs of inoculation in positive cases

Strauss reaction-Intra peritoneal inoculation of pus (about 0.5 ml) in male guinea pig causes severe orchitis and inflammation of the scrotal sac

Isolation of organism from pus by culturing in blood agar

Serological test- **complement fixation test is most accurate test**

Indirect haemagglutination test and ELISA

Sample collection

Pus material, lymphnode discharge and swabs

Differential diagnosis

Epizootic lymphangitis

Ulcerative lymphangitis

Sporotrichosis

Melioidosis

Others cases of pneumonia

Treatment

However, treatment with Sodium sulphadiazine has been found to be effective in experimental cases

Combination of formalized preparation of mallein and sulfadiazine or mallein and sulfadimidine can be given

Prevention

Glander is a notifiable disease under the glander or farcy act. As per the act animals declared positive must be destroyed and safely disposed

Confirmed infection may be indicated by restrictions on international trading and local movement of animals

The appropriate Ministry of agriculture must be informed of all suspected or confirmed cases as it is a biological weapon in the warfare

Control

All infected carcasses are buried and incinerated

All in contact, suspected, imported animals must be isolated and tested

Positive reactors must be slaughtered

Manure, bedding materials and residuals must be burnt and buried

Complete quarantine of affected animals

Remaining animals must be subjected to mallein test at interval of 3 weeks

Disinfection of premises

Readily destroyed by direct sunlight and by common disinfectants

The organism survive in contaminated stable for 6 weeks

Strangles

(Syn: Equine Distemper, Infectious adenitis)

It is an extremely contagious disease of horses, most commonly affecting foals, clinically characterized by marked swelling of the retropharyngeal lymphnodes in throat causes "strangled breathe sounds" the disease is named "Strangles".

Etiology

It is caused by *Streptococcus equi subspecies equi*

It is an obligate, gram positive and cocco-bacillus organism

The organism expresses beta haemolysin around the colonies in blood agar

High virulent nature of the organism is due to the presence of M protein, hyaluronic acid capsule and leukocidin toxin

The mild atypical form of the disease is associated with non-capsulated variant of *S.equi*

Epidemiology

Prevalence of infection

The disease has been widely distributed around the world

It is also recorded from Punjab and other places in India and Pakistan

The occurrence is highest in foals during February to May compared to winter and summer seasons

Predisposing factors

Sudden exposure to cold, high humidity, poor ventilation and severe stress are considered most important

Source of infection

10-40% of recovered horses is carriers and holds the organism in pharynx, guttural pouches for a long time

Discharges from nostrils, abscesses, contaminated grazing land, tracks, stalls, feed and water troughs, grooming kits, clothes of horses are sources of infection

Horses with guttural pouch empyema may excrete the organism for more than 3 years

About 25% of the recovered horses not immune and are susceptible to second cycle of infection

Over populated farms are at high risk

Economic impact

Death of infected horses, time required and expenses incurred for treatment, aesthetic unpleasantness due to running noses and draining abscesses are responsible for economic losses

Transmission

Direct or indirect contact with fomites

Inhalation also possible

Through skin wound from external environment

Host affected

Horses, donkeys and mules around the world are susceptible

Morbidity rate is (20-100%) usually more in foals and yearlings than in adults

Age specific attack rates in foals are 48% and in brood mares about 18%

Case fatality rate is 10% in untreated and 1-2% in treated horses have been recorded

Outbreaks occurs frequently in breeding farms, polo and race horses

Resistance to the disease is associated with the production of **serum and mucosal IgG antibodies** to the streptococcal **M protein**.

Pathogenesis

Following exposure to oral and nasopharyngeal mucosa

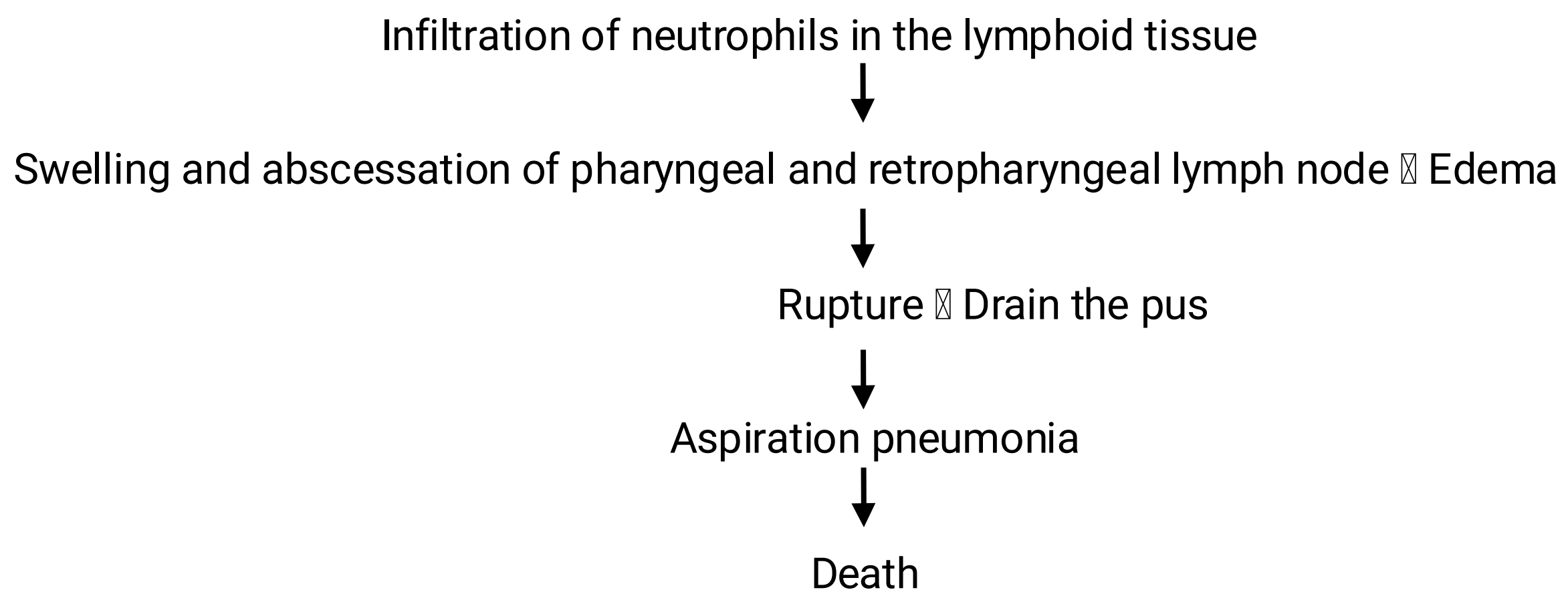


Localization of organism in pharyngeal and tonsillar lymph node



Replication and production of Streptolysin O & streptokinase enzyme





Clinical signs

Acute form

Incubation period is 1-3 weeks

Fever 103-105 ° F, sudden onset of anorexia, **serous, copious and mucopurulent nasal discharge, abscessation of sub-mandibular, retropharyngeal lymphnode** and guttural pouch

Severe pharyngitis, laryngitis and mild conjunctivitis

Lymphadenopathy

Distinct enlargement of sub-mandibular lymphnode

Pain in throat due to abscess, dysphagia, soft moist cough and extension of head

Hot, painful swelling of retropharyngeal lymphnode causes obstruction of oro-nasopharynx with respiratory distress which leads to death due to asphyxiation

Affected nodes rupture and discharge thick creamy, white to yellow pus in 10 days

The course of the disease may be from 3 weeks to 3 months in severe cases

Retropharyngeal lymphnode rupture and draining of pus into guttural pouches leads to empyema and chondroids formation

In severe cases, pharyngeal, sub-maxillary, parotid and retrobulbar nodes have abscess and spread to local lymphatic vessels develops obstructive edema

Rarely death occurs due to purpura hemorrhagica in horses as sequelae to *S.equi* and is associated with high serum antibody titre to M protein

Atypical form

It is characterized by widespread sub-clinical infection may have transient fever, profuse nasal discharge

Strangles in Burros

Slow developing debilitated disease and have caseation and calcification of abdominal lymphnode

Chronic form (Metastatic infection or Bastard strangles)

It is a chronic illness eventually leading to death

Arise due to bacteraemia or extension of infection in lymphnodes

Heart valves, brain, eyes, joints, tendons, sheath are affected

Abscess in brain leads to depression, meningitis, head pressing, abnormal gait, circling and seizures

Complications

Suppurative necrotic bronchopneumonia, due to aspiration of pus from ruptured abscess in the upper airways / metastatic infection of lungs

Spreading of infection from RLN into guttural pouches causes **guttural pouch empyema and conchoid formation**

Necropsy finding

Suppuration of liver, spleen, lungs, pleura and peritoneum

Meningitis

Lesions in guttural pouch and in various lymphnodes

Diagnosis

Based on clinical signs

Isolation and identification organism

The smear prepared from nasal swab is stained with gram stain. The *S.equi* appear as fluorescent purple color on microscopic examination

A commercial test, measures the serum IgM antibody titer to SeM (M protein) used to identify vaccination immunity

The test is not suitable in diagnosis of acute diseases

Serum antibody titre to SeM are very high (>1:12800) in horse with metastatic infection or purpura haemorrhagica

Detection of shedding *S.equi* by PCR. It is more specific than culture

Acute form: Leucocytosis with neutrophilia reaches the peak

Hyper-fibrinogenaemia is a characteristic of both acute and chronic disease.

Leucocytosis with a hyperproteinemia: Polyclonal gammaglobinemia is an important change of metastatic and chronic abscessation

Hypoalbuminaemia

Affected animals may or may not have anaemia, it depends on haemolytic effect of streptolysin O and immune mediated hemolysis

Sample collection

Nasal, pharyngeal, guttural pouch discharge and abscess, swabs from animals with active diseases and carrier state

Differential diagnosis

Glander

Equine viral arteritis

Equine viral rhinopneumonitis

Equine influenza

Treatment

Provide horses more comfort by feeding and watering

Perfect monitoring in order to avoid the development of secondary bacterial infection

Procaine penicillin G 22,000IU/kg bw by I/m for every 12 hours for 5 days is given

Tetracycline 6-7.5 mg/kg bw i/v for every 12-24hours

Sulphonamide-trimethoprim combination 15-30mg/kg orally or i/v every 12 hours may be effective but only be used if procaine penicillin administration is not possible

Guttural pouch empyema is flushed with sterile isotonic electrolyte solution (0.9% NaCl)

Non-steroidal, anti-inflammatory drugs (NSAIDS) and hot poultices to reduce swelling relieve pain and rupture of abscess

Prevention

Serum IgGb developed against SeM protein can be used to prevent infection

Vaccination of mares 4-6weeks of gestation and foaling period by i/m

Colostrum fed foals are passively immune for 4 months

Majority of the horses develop solid immunity for 5 years

Genes HasA and HasB that are responsible for capsule formation and its removal may increase the genetic stability of live vaccine strain (707-27) because they lack capsule and become avirulent. It is only for intra nasal route for healthy horses not for infected horses and outbreak

Control

Isolation of infected horses

Stop the movement of animals until outbreak is controlled

Maintain strict hygiene

Identification of carrier status from clinically recovered horse by collection of three nasopharyngeal swab or washing at weekly interval by culture and to prove negative

Endoscopic examination of upper airways and guttural pouch for identification of lesion and treatment animals

Topical application is effective for open infection

Tracheotomy for upper air ways obstruction

Ulcerative lymphangitis

(Syn: Pigeon fever, Dry land distemper)

It is characterized by the development of pyo-granulomatous abscess in sub-cutis and internal organs, a necrotic and ulcerative dermatitis on the heel of the foot.

Etiology

Corynebacterium pseudotuberculosis biotype-2

It is a saprophytic and intracellular organism

They are gram positive, non-spore forming, non-motile, pleomorphic, rod-shaped bacteria that are straight or slightly curved

The bacteria group together in a characteristic way, which has been described as the form of a "V", "palisades", or "Chinese letters"

Epidemiology

Prevalence of infection

It is the most important disease in USA, North Africa, Israel and Asia

Incidence of the disease is more during summer months

Morbidity rate: 5%

Internal infection – Mortality rate is 30%– 40%

Culling rate: 16%

Horse and cattle are mostly affected

Predisposing factor-Unhygienic and wet conditions

Source of infection-arthropods and contact with fomites such as contaminated soil and grooming equipment

Transmission-, spread through abrasions, contact with fomites (contaminated soil and grooming equipment)

Pathogenesis

The organism enters through skin abrasions

Invasion of lymphatic vessels

Produce phospholipase and exotoxin- cytotoxic in nature

Formation of abscess

Mature animals-Ulcerative lymphangitis, and skin abscessation

Heifer - ulcerative dermatitis in heel of foot

Clinical signs

Horse

Painful inflammatory swelling, nodules and ulcers on the lower limbs usually on the pastern region, severe lameness, nodules in the subcutaneous tissue, specially around the fetlock and all parts of the body

Lesions are restricted to limbs

Edematous swelling can extend up the entire limb

The exudate is odorless, thick, tan, and blood tinged

Usually, only one leg is involved

Abscessation of the pectoral or ventral abdominal region with secondary dissemination to internal organs

Ventral pitting edema, ventral midline dermatitis

Abscess or Nodules rupture and discharge a **creamy green pus**

Dermatitis lesions are **painful and mildly pruritic with alopecia, exudation, crusting, and ulceration**

Healing may take place in a week or two but new abscess leads to persistence of disease for up to one year

Cattle

The symptoms and lesions are similar as in horses

Enlargement of draining lymph nodes and discharge of gelatinous clear exudate from ulcers

Sample collection

Aspirates from abscesses

swabs of purulent exudate beneath crusts associated with folliculitis

punch biopsies

Diagnosis

Based on the clinical signs

Clinical pathology- leukocytosis and neutrophilia

Internal abscessation by ultrasound

Isolation and identification of the organism from infected tissues and discharging pus.

No serological tests are available for screening infection in cattle or horses.

Differential diagnosis

Epizootic lymphangitis

Glanders

Sporotrichosis
Staphylococcosis
Pseudomonosis
Equine cryptococcosis
Onchocerciasis

Treatment

Lymphangitis and internal infection should be treated with longterm antimicrobials

(1month duration)

The organism is sensitive to all common antibiotics

Parenteral injection of penicillin or tetracycline

Early stages of infection can be treated with autogenous bacterin

Lymphangitis and early abscess swellings are treated with **hot packs, poultices, or hydrotherapy until rupture**

Phenylbutazone is used to relieve pain and swelling

Prevention and control

- Clean the wound with antiseptics or antibiotic solution
- Disinfection of stables
- Control of arthropods by spraying of insecticides (synthetic pyrethroids)

Brucellosis

(Syn: Bang' s disease, Contagious abortion, Undulant fever and Malta fever)

Brucellosis is an important Zoonotic disease, causes significant reproductive losses through abortions, retained placenta, epididymitis and orchitis in infected animals.

Etiology

Brucella abortus -Cattle, buffalo, bison and human

Brucella melitensis, *B.ovis* - Sheep and goats

B.suis -Pigs

B.canis -Dogs

There are five biovars in *B.suis*, three in *B.melitensis* and nine in *B.abortus*

B. suis contains more diverse isolates and broader host range as compared to any other species of *Brucella*

B.abortus, *B.melitensis*, *B.suis* and *B. canis* causes *brucellosis* in human

It is a Gram negative, non-motile, facultative intracellular, coccobacillary or short, rod shaped organism

Epidemiology

Prevalence of infection

It is worldwide in distribution

Brucellosis is endemic in India

Source of infection

The large no. of organism present in aborted fetuses, fetal membrane and uterine discharge, licking of contaminated genitals of other animals

Brucella species also been found in secretions and excretions, which include milk, urine, feces, hygroma fluids, saliva, nasal and ocular discharge

The animal recovered from infection become asymptomatic carrier

Transmission

Ingestion, inhalation, skin injury and conjunctival route

Mostly, *Brucella abortus*, *B. melitensis* are shed through bull semen for a long period but venereal spread is seems to be uncommon

The **venereal route is the primary route of transmission for *B. ovis*, *B. suis* and *B. canis* and occurs frequently**

The infection spread also from milkers and farmers hands but **long-term shedding not occurs normally in ruminant's milk.**

Host affected

Cattle, bison, buffaloes, sheep, goats, pigs, dogs, marine mammals and rodents
Sheep and goats acts as reservoir for *B. melitensis* and deer as a maintenance host for *B. ovis*

Sexually mature first calf heifer are highly susceptible

B. abortus, *B. melitensis* and *B. suis* affects horses, camels, moose, chamois, alpine ibex, raccoons, opossums, dogs, coyotes, foxes and wolves (accidental hosts)

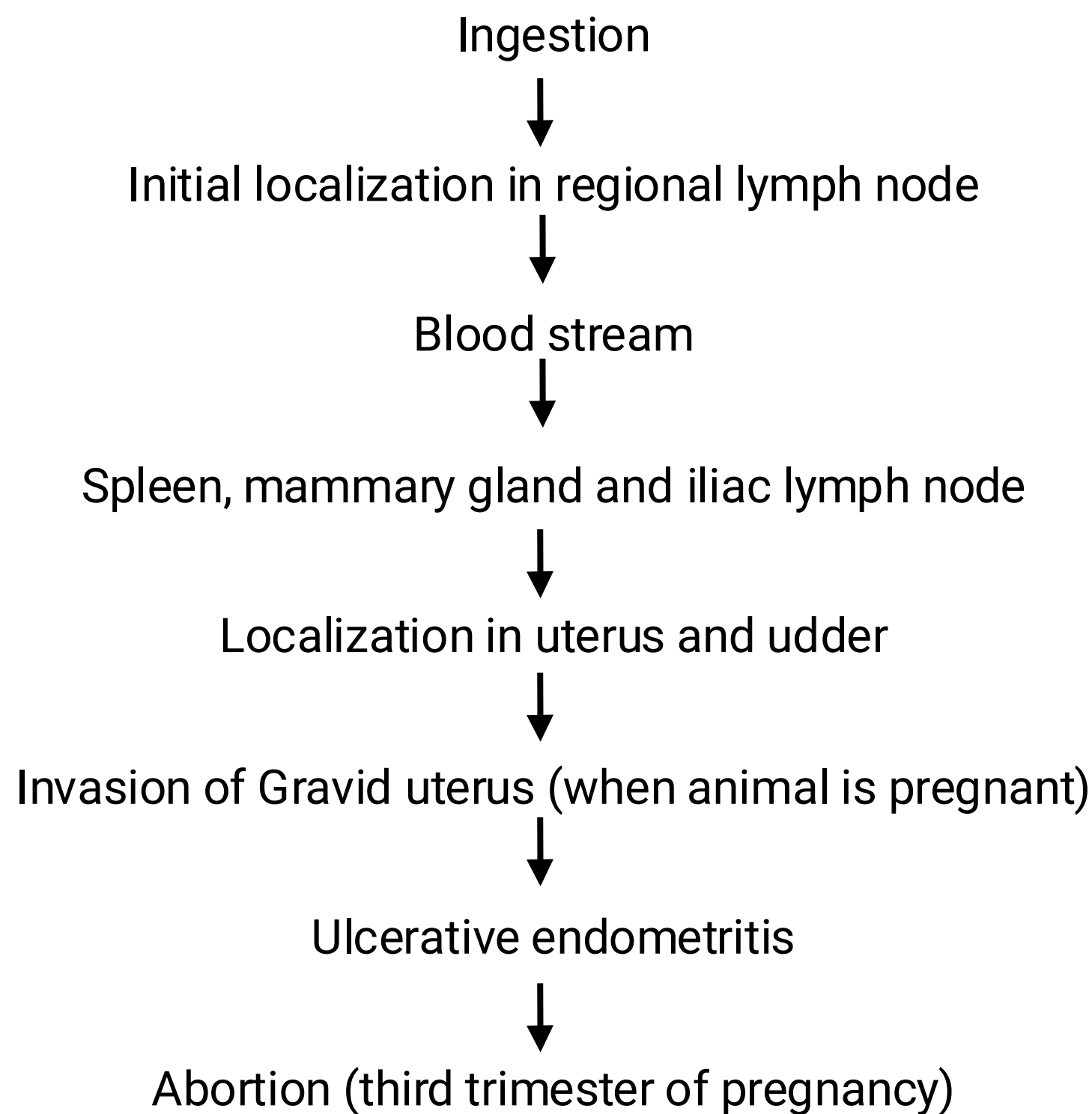
Pathogenesis

Following entry, the bacteria are swallowed by neutrophils, monocytes and macrophages. Within the phagocytic cells they proliferate intracellularly and remain in lymphnodes if not killed by the host immune cells and get localised in regional lymphnodes.

Brucella primarily localised in the organs rich in sugar, erythritol ie; breast, uterus, epididymis and testes etc.

Within the phago-lysosome *B. abortus* releases 5'-guanosine and adenine which inhibits the de-granulation of peroxidase-containing granules and thus inhibit the myeloperoxidase-peroxide-halide system of bacterial killing.

The intracellular persistence of the brucella leads to granuloma formation in organs and tissues are due to hypersensitivity reactions, mostly type-IV
The organism is distributed from lymph nodes to lympho-reticular organs such as spleen, bone marrow, liver and testes where they develop granulomas or microabscesses which resulted in abortion in the third term pregnancy.



Clinical signs

Cattle: *B.abortus*

Abortion in second half to full term pregnancy, still births, weak calves, retained fetal membrane and reduced milk production

Once aborted animals become symptomatically normal for the further pregnancy, but actively shed the organism in uterine discharge and in milk in the subsequent parturition

In bulls, epididymitis, seminal vesiculitis, orchitis, testicular abscess, infertility, in sexes, unilateral or bilateral hygroma and arthritis following long term infection

Rarely causing death of fetus

Sheep and Goats: *B.melitensis*

Abortion at 4th month of pregnancy, still birth, weak offspring, retained fetal membrane, re-invasion during subsequent pregnancies and reduction in milk yield

Absence of mastitis

In acute cases, orchitis, epididymitis in males which results in infertility
Arthritis develops in both males and females.

Sheep: *B. ovis*

It affects only sheep not goats

Epididymitis (unilateral, occasionally bilateral), orchitis, impaired fertility in rams, poor semen quality and atrophied testes etc.

Ram shed *B. ovis* for long periods while it is asymptomatic

In ewe it causes abortion and placentitis

Mortality in ewes is not common

Pigs: *B. suis*: (Porcine and rangiferine Brucellosis)

It causes abortion in pig at any time, weak and still birth piglets, vaginal discharge, infertility and metritis is occasional

Orchitis, temporary or permanent sterility in boars are common

Posterior paralysis and spondylitis

Infected boars are persistent carriers

Biovar-2 in hares produces nodules in internal organs, reproductive organs, subcutaneous tissues, muscles, and that may be purulent in nature.

Biovar-4 causes abortion, retained placenta, metritis, mastitis and orchitis in male, lameness, arthritis, bursitis, tenosynovitis or hygromas and subcutaneous abscess in caribou and reindeer.

Dogs: *B. canis*

It causes abortion at 7-9 weeks of gestation, followed by a mucoid, serosanguineous / gray green vaginal discharge that persists for up to 6 weeks

Early embryonic mortality

Pups born from infected dam are alive, but weak

Epididymitis, scrotal edema, orchitis, poor sperm quality, unilateral or bilateral testicular atrophy in chronic infection and lymphadenitis is common.

Lethargy, decreased appetites, poor performance, loss of alertness, disc spondylitis causes spinal pain, paresis and ataxia if spinal cord compression develops, osteomyelitis or polyarthritis causes lameness, uveitis, endophthalmitis, polygranulomatous dermatitis, endocarditis, and meningoencephalitis without fever

Horses: *B. abortus*, and *B. suis*

It causes inflammation of supra spinous or supra atlantal bursa and leads to “fistulous withers” or “poll evil”. The bursal sacs become distended and rupture leading to secondary inflammation.

In chronic cases, nearby ligaments and dorsal vertebral spines may become necrotic.

Zoonosis

Brucellosis is an occupational hazard. Veterinarian, butchers, farmers and some animal handlers dealing with shandy and exhibition animals, exposed to infection while contact with infected animals

Ingestion of un-pasteurized milk and other dairy products

Accidental injection of live brucella vaccines spread the disease, blood transfusion and bone marrow transplantation

It is a serious, debilitating and chronic disease develops a systemic infection in human.

In clinical cases, acute febrile illness with non-specific flu like signs such as fever, head ache, malaise, back pain and muscular pain noticed.

Drenching sweats particularly at night, splenomegaly, hepatomegaly, cough and pleuritic chest pain are sometimes seen

Necropsy finding

B.abortus, *B.melitensis* and *B.suis* develops autolysis of fetus, subcutaneous edema, blood stained fluid in body cavity in ruminants, enlargement of spleen, liver, lung, pneumonia and fibrinous pleuritis, placentitis. Cotyledons becomes red, yellow, normal or necrotic.

In cattle, small ruminants **inter-cotyledonary parts typically leathery** with focal thickening and exudate

In adult granulomatous to purulent lesion in mammary gland, superficial lymphnodes, lymphoid tissues, bones, joints and other tissues and organs lesions are found

Diagnosis

Based on clinical signs and necropsy findings

Identification of organism by Stamp's modification Ziehl Neelsen method. Stained smears reveals coccobacilli or short rods, arranged in a single chain, but sometimes in pairs or in small groups.

Isolation of organism by culture

Immunostaining demonstrate the presence of *Brucella* spp. in a clinical specimen

PCR techniques can also be used for diagnosis

Intradermal Brucellin Skin test

Serological test used to identify antibodies are

- ☒ **Rose Bengal Plate agglutination test-** very sensitive, antibody screening test
- ☒ Serum tube agglutination test: The **antibody titre of 1:40 is considered as positive** in animals infected for the **first time in uninfected area** and 1:80 is considered positive for animals maintained in an endemic areas
- ☒ Milk Ring test: It is a simple, qualitative, antibody detection test in which 1 or 2 ml of milk from either bulk milk take for herd screening or individual animal disease
- ☒ **Complement fixation test- confirmatory test for brucellosis**
- ☒ *Indirect Enzyme Linked Immuno Sorbent Assay:* It is a test of international trade
- ☒ *Fluorescent polarization assay:* It is a simple and international trade test. Its sensitivity and specificity (99%) is similar to the C-ELISA
- ☒ Gamma interferon test : It is a test of choice for detection of cell mediated immune response
- ☒ Most human infections are diagnosed by serology. Tests used include serum agglutination, a modified Coombs' (antiglobulin) technique, ELISAs and Western blot

Sample collection

Blood, uterine discharge, foetal stomach content, hygroma fluid and milk

Differential diagnosis

Trichomoniasis

Leptospirosis

Campylobacteriosis

IBRT

Treatment

No treatment is advised in animals

Long term antibiotics treatment is advised in human with combination of antibiotics

Prevention

Brucella abortus strain 19 (cotton strain) vaccine

- o This vaccine is most widely used in the prevention of brucellosis. It is a reference live vaccine normally given to female calves of 3-6 months in subcutaneous route with dose rate of $5-8 \times 10^{10}$ viable organisms, where as in adult cattle 3×10^8 organism is given s/c route but some animals develop persistent antibody titers and may abort and excrete the vaccine strain in milk.

Brucella abortus strain RB51 vaccine

- o It is an official vaccine being used since 1996. This strain was produced from a laboratory derived **rough mutant of smooth *B.abortus* strain 2308**. Calves of 4-12 months old are vaccinated with $1-3.4 \times 10^{10}$ viable strain RB51 organisms.

Brucella Melitensis strain Rev.1 vaccine

- o This vaccine may be more effective in countries with a high prevalence of brucellosis in small ruminant and this vaccine should not be used in cattle because the safety of the vaccine in cattle is not studied yet.

Control

All positive animals must be separated and disposed

Vaccination of all susceptible in contact animals with either *B.abortus* S19 vaccine or *B.abortus* RB51 vaccine. *B.abortus* 45/20 vaccine is also used

Stamping out policy by quarantine, vaccination, test and slaughter, depopulation, cleaning, disinfection, surveillance, trace back of animals, eradication and neutering

In endemic area pasteurization of dairy products is an important safety measure

Good hygiene and protective clothing/equipment are very important in preventing occupational exposure

Measures should be taken to exclude the entry of organism through skin, inhalation, and ingestion, while performing a necropsy, butchering and parturition or while removing fetal membranes and fluids

The Strain 19 *B. abortus* vaccine and *B. melitensis* Rev-1 vaccine must be handled with precaution to avoid accidental injection or exposure. *Brucella organism* should be handled under bio-safety containment level 3 or higher.

2.5% sodium hypochlorite, 2-3% caustic soda, 20% fresh slaked lime suspension, 2% formaldehyde solution (all tested for one hour) can be sprayed for inactivating the organism in contaminated surfaces

Xylene (1ml/liter) and calcium cyanamide (20 kg/m³) are used to decontaminate liquid manure in 2 to 4 weeks if its use.

Gamma irradiation (e.g. in colostrum) and pasteurization can inactivate the organism.

Screening, culling or isolation or depopulation of infected animals

Proper disposal of infected carcass, aborted fetus and placental membrane

Tuberculosis

(Syn: Consumption, Pthisis, Pearl' s disease and Scrofula)

Tuberculosis is a chronic contagious g ranulomatous disease of animals, characterized by development of tubercle nodules followed by caseation and calcification, debility and muscle wasting. The disease is transmissible between animal and human beings.

Etiology

Cattle: *M.bovis* and *M.tuberculosis*

Human – *M.tuberculosis* and *M.bovis*

Horses: *M. bovis* (majorly), *M.tuberculosis* and *M.avium*

Dogs: *M. bovis*, *M. tuberculosis* and *M. avium*

Pigs: *M. bovis*, *M. tuberculosis* and *M. avium*

Birds: *M.avium*

Mycobacterium is a slender, rod shaped acid fast organism, appears as single or pair or in group, resembles bundle of faggots. The cell wall of the organism is rich in mycolic acid (waxy coat) and sulphur rich glycolipids, which facilitate the bacteria to take up certain dyes such as Aniline and Ziehl Nielsen stain

Physical and Chemical features

The organism is a slow grower in laboratory media

It is sensitive to heat

The organism is killed by creosote and sodium orthophenoate

Resistant to chlorine and 6% NaOH

Epidemiology

Prevalence of infection

Tuberculosis is prevalent worldwide, both in animals and human beings

TB is endemic in India

Economic impact

Apart from the mortality and productive loss

Disease is relatively benign in pigs

Discarding and condemning infected carcasses at slaughter

Predisposing factors

Overpopulation in small area

Purely intensive rearing

Intercurrent infection

Poor sanitation

Inadequate ventilation

Source of infection

Exhaled air, feces, milk, nasal discharge, uterine discharge from infected animal and human being, contaminated feed, water and soil

Transmission

Inhalation of droplet nuclei from aerosol and ingestion through common use of feed and water troughs

Human get infection by ingestion of unpasteurized milk and milk products from tuberculosis infected animals

Direct contact with infected animals.

Congenital infection by vertical transmission

Host affected

Cattle are the primary host highly susceptible for the infection.

Pure bred and cross bred are highly affected as compared to zebu cattle.

Sheep and goats more or less resistant to infection.

Badgers, possums, pigs, dogs, primates, large cats and elephants other than cattle are affected and act as maintenance host

Pathogenesis

Mechanism of development of infection -Primary Tubercle Complex or Ghon' s focus

The mycobacteria which gain entry in the host by inhalation get lodged in the alveolar surface of the lung and its bronchial and thoracic lymph nodes

By ingestion the organism reaches pharyngeal and mesenteric lymph nodes get locked up there

Within the corresponding lymphnodes the mycobacteria undergoes multiplication and develop small tubercles. The focus of such infection in the lymphnodes given the condition “ Primary complex of infection” or “ Ghon' s focus”

Post Primary dissemination and Stage of late generalization

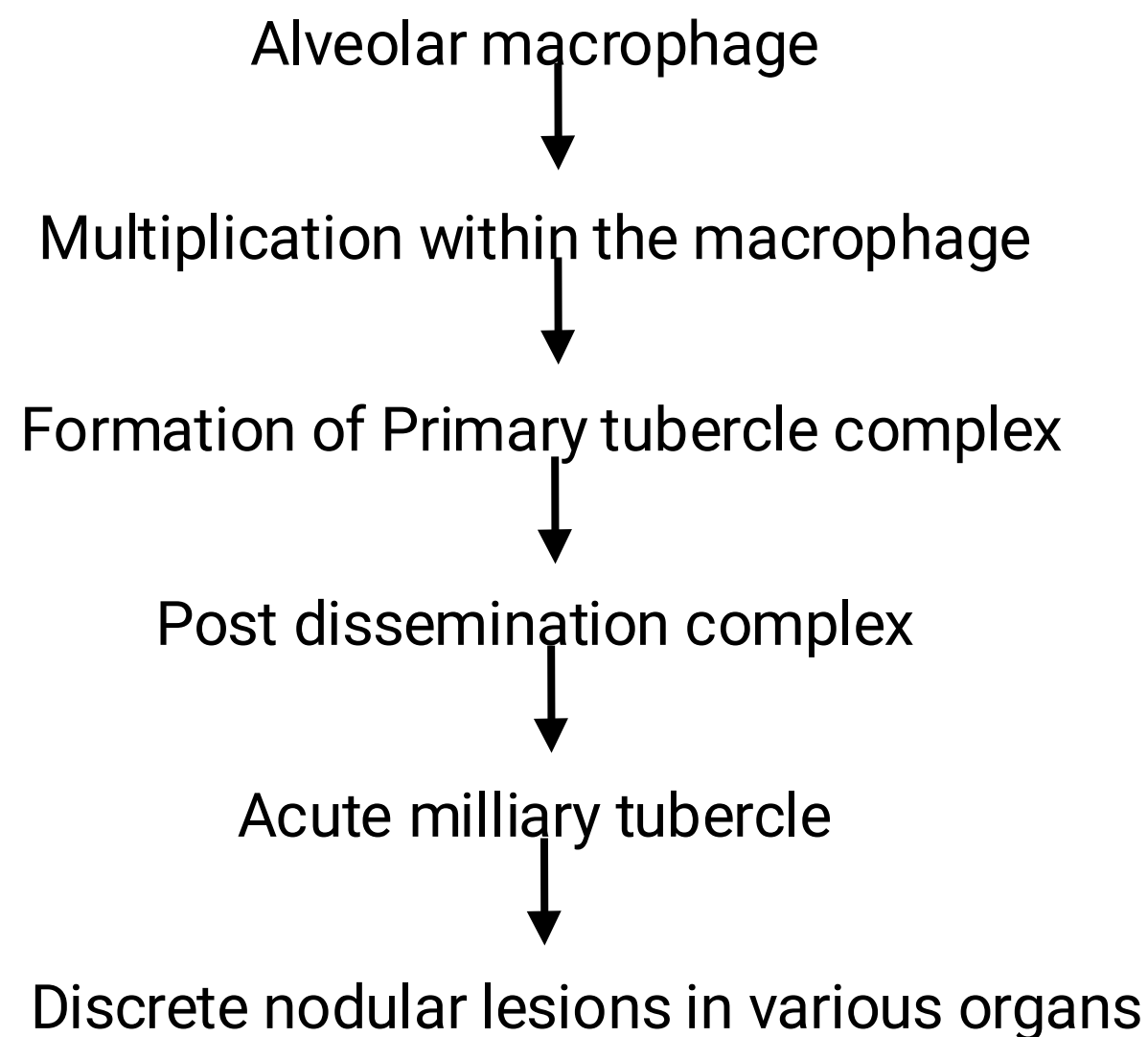
Through draining of lymph nodes, bacteria reach the blood circulation and spread into other body cavities due to drop in immune response

As a result the bacteria overcome the killing effect by host immune mechanism and causes infiltration and necrosis of neutrophils

Together, the bacteria and the dead immune cells are surrounded by multi-epitheloid cell layer which is called as Langhan' s giant cells or granulomas in various organ (miliary tubercle) and further this is surrounded by lymphocytes and fibrous connective tissues

Inhalation





Clinical signs

General form

Affected animals become docile, lethargic but seems to be bright and alert

Cows with prominent miliary tubercle lesions are clinically normal

Progressive emaciation

Capricious appetite, fluctuating body temperature and rough / sleek hair coat

In spite of good appetite animal does not put up weight

All these general signs are pronounced following calving

Mastitis with induration of udder and enlargement of supramammary lymph node

Early stage of mastitis- fine floccules settle at the bottom and later the secretion become amber coloured

Respiratory form

Silent or **paroxysmal cough** especially during early morning and chilled weather

Chronic cough due to bronchial pneumonia

Rapid respiration with dyspnoea revealed by auscultation and percussion of the chest due to enlargement of bronchial lymphnode and obstruction of air ways

Pleurisy without effusion

Squeaking crackles are audible but noise audible over the caudal lobes

Enlargement of retropharyngeal lymphnode causes dysphagia and noisy breathing due to pharyngeal obstruction.

Reproductive form

Metritis and inflammation of placenta leads to infertility, abortion and failure in conception

Nervous signs

In horse painful cervical osteomyelitis causes stiffness of neck and difficulty to eat.

Sheep and goats

Bronchopneumonia and terminal dyspnea

Intestinal ulceration with diarrhea which is rare in cattle

Enlarged lymphnode of alimentary tract

Kids are more progressive to early death

Pigs

TB lesion is cervical lymphnodes occur externally

Tuberculus involvement of meninges and joints are most common

Horses

Cervical vertebrae show osteomyelitis, and causes stiffness of neck

Zoonosis

It is an important zoonotic disease. There is an increased incidence with an increased spread from animal to human

M. bovis is responsible for 5-10% of human tuberculosis. Children getting infected via drinking of infected breast milk

Necropsy finding

Cattle, Sheep and Goats

Granulomatous lesion in bronchial, retropharyngeal and mediastinal lymph nodes

Enlargement of pharyngeal and supramammary lymphnode

Miliary abscesses in lung

Pus is characteristic creamy and cheesy

Lesions in placenta with chronic purulent material

Orchitis

Chronic lesions

Discrete, nodular, thick yellow to orange caseous material followed by calcification covered by a thick fibrous capsule

Pigs

Enlarged miliary TB localized in tonsils, submaxillary, cervical, hepatic, bronchial, mediastinal and mesenteric lymph node with white caseous calcified material surrounded by a strong fibrous capsule

Horses

TB lesion in intestinal wall, mesenteric lymphnode and spleen

Cut surface of the LNs appears fleshy and resemble neoplastic tissue

Mineralisation and tissue necrosis noticed

Sample collection

Sputum, feces, milk, blood and tracheal secretions must be collected for the diagnosis

Diagnosis

Based on clinical signs and necropsy findings

Palpation of supramammary lymph node essential for suspected TB mastitis cases

Isolation and identification of organism by culture or acid fast staining

Intra dermal test is used to detect cell mediated immunity

Gamma Interferon Assay

ELISA for detection of antibody

Molecular diagnosis by PCR

Intra dermal test**Single intradermal test**

0.1 ml of PPD is injected intra dermally in neck region of cattle and caudal fold region usually at the base of the tail in sheep and goat

0.2 ml herds with known infected herds

Skin thickness is measured after 72 hours post injection

Increasing skin thickness more than 4mm is taken as positive

Short thermal test

Tuberculin PPD 4ml injected s/c in the mid neck of cattle

Initial rectal temperature 39 ° C (102 ° F) at the time of injection

If the temperature increases in 4, 6, 8hrs to above 40 ° C animal is classed as positive reactor

Stormont test

First intradermal injection of 0.1 PPD at middle of the neck followed second injection should be given 0.1ml after 7 days at the same site

Increase in skin thickness of 5mm or more at 24 hrs after this second injection is considered positive

Cattle with *M. avium* does not give positive reaction

It is more accurate than SID

Comparative test

For testing the animals for TB, along the side of PPD tuberculin injection, 12cm apart, *M. avium* PPD should be given and the test result should be compared

Test results read after 72 hrs.

This test is not used for initial screening

But can be used as a primary test when there is a high incidence of Avian TB and JD

Differential diagnosis

Lung abscess
Pleurisy
Pericarditis
Traumatic reticulitis
Chronic contagious bovine pleuropneumonia
Bovine leukosis
Lymphadenopathy
Mastitis

Treatment

There is no successful antibiotic treatment for control of tuberculosis
Supportive treatment to enhance the immune response can be done

Prevention

All animals over six months and above should be tested by tuberculin test
Positive reactors disposed of according to local legislation
Suspicious cases must be retested by comparative test
Retesting: If incidence is higher, retesting should be done in 45-60 days after desensitization of intradermal injection
Annual testing of all cattle and slaughter of positive reactors
BCG vaccination is available for calf hood vaccination. But vaccinated animals react positive to the skin test. Immunity is not strong and long lasting
Establishing **Accredited area**: All cattle in this area are tested and reactors removed and monitored until next generation off-springs become free from infection by tests

Control

Disinfection of utensils with 5% hot phenol and cresol
Feeding of calves with milk free from infection
Test and introduce new stock
Educate public about the significance of the disease

Paratuberculosis

(Syn: Johnes disease)

Paratuberculosis is chronic contagious granulomatous enteritis of ruminants caused by *Mycobacterium avium subsp. paratuberculosis* (M. paratuberculosis) and

clinically characterized by persistent diarrhoea, progressive weight loss, debilitation and eventually death. Infection generally occurs early in life, and many infected animals become chronic carriers.

Etiology

Mycobacterium avium subsp. paratuberculosis, an acid-fast rod previously known as *Mycobacterium paratuberculosis* and *M. johnei*. *M. avium subsp. paratuberculosis* is a member of the *M. avium-Mycobacterium intracellulare* (MAI) complex of organisms

Some strains of *M. avium subsp. paratuberculosis* preferentially infect specific hosts. The two main types, which can be distinguished by restriction fragment length polymorphisms, are the Type II or C strains, found in cattle, and type I or S strains, found in sheep. The type C strain has broad host range, whereas S strain infects mainly sheep.

M. avium subsp. paratuberculosis is resistant to environmental conditions and can survive on pastures for more than a year

High calcium content and pH of the soil influence the persistence of organism

Epidemiology

The disease occurs worldwide most commonly in cattle and to a lesser extent in sheep and goat.

Species affected

Paratuberculosis affects domesticated and wild ruminants including cattle, sheep, goats, llamas, alpaca, camels, moose, elk, bighorn sheep, buffalo, deer and reindeer. Disease has been reported in wild rabbits, nonhuman primates and pigs. Horses and dogs can be infected experimentally.

Source of infection

Large number of organism excreted through faeces of infected animals and lower numbers in the milk or colostrum

Soiled feed, licking and grooming behaviour in contaminated environment

Introduction of infection in non infected herd through purchase of replacement stock

Transmission

Faeco-oral route is the main mode of transmission

The infection can spread vertically to the foetus

The disease also transmitted through semen

The primary source of infection in calves is milk from infected cows or milk that is contaminated with the faeces of diseased cattle

Animals usually become infected when they nurse from an udder soiled with faeces or housed in contaminated pens

Transmission can occur on fomites, and insects may act as mechanical vectors
Both horizontal and vertical transmission seems to take place in wild rabbit populations

Predation might be a route of transmission to carnivores or omnivores

Zoonosis

In human *M.paratuberculosis* causes crohn' s disease

Pathogenesis

Following oral ingestion, the organism localize in the mucosa of the small intestine, its associated lymph node and to a lesser extent, in the tonsil and supra pharyngeal lymph nodes. The primary site of bacterial multiplication is the **terminal part of the small intestine and large intestine.**

Depending on the host parasite relationship the animals are grouped into three groups.

1. **Infected resistant-** the animal quickly develop resistance , control the infection and do not become shedders
2. **Intermediate** – infection is not completely controlled, some animals will partially control the infection, and will shed the organism intermittently
3. **Clinical case-**the organism persists in intestinal mucosa

Clinical signs

The incubation period is usually months or years; periods ranging from 4 months to 15 years have been reported

It occur in four stages

Stage I-Silent infection

Calves generally become infected soon after birth but rarely show clinical signs before they are two years old

It occur in animals< 2 yrs of age

There is no clinical signs and no effect on body weight gain

Stage II-Subclinical disease

No clinical signs, but may be affected by other abnormalities such as mastitis or infertility

Stage III- Clinical disease

Most cases are seen in 2 to 6 year old animals. The initial symptoms can be subtle and may be limited to weight loss, decreased milk production or roughening of the hair coat

The animal is eating well throughout the period, but thirst is excessive. The faeces are soft and thin like thick pea soup, homogenous and without offensive odour. This may be continuous or intermittent. Tenesmus is not seen

Stage four- advanced clinical disease

Emaciation, **inermamandibular edema**, or ventral edema, the **diarrhoea is characterized by fluid waterhose or pipestream passage of faeces**, and death is due to severe dehydration and cachexia dehydration

Sheep and goat

Emaciation is more prominent. Diarrhoea is less common, wool is often damaged and shed easily

Necropsy findings

The intestines should be opened from the duodenum to the rectum to expose the mucosa

The mucosa, especially of the terminal ileum, is inspected for pathognomonic **thickening and corrugation**

The **mesenteric lymph nodes are usually enlarged and oedematous**. Smears from the affected mucosa and cut surfaces of lymph nodes should be stained by Ziehl– Neelsen’ s method and examined microscopically for acid-fast organisms

The typical lesions of paratuberculosis consist of infiltration of the intestinal mucosa, submucosa, Peyer’ s patches and the cortex of the mesenteric lymph nodes with large macrophages, also known as epithelioid cells, and multinucleate giant cells, in both of which clumps or singly disposed acid-fast bacilli are usually, but not invariably found. Discrete plaques may be seen early in the disease

Sheep and goat

The mucosa is often only slightly thickened in these species, but **caseated or calcified nodules** are sometimes found in the intestines and associated lymph nodes. Some (S) strains of *M. avium* subsp. *paratuberculosis* produce a pigment that stains the intestinal lesions brownish– yellow.

The lesions in other ruminants resemble those found in cattle

Sample collection

Rectal washing, rectal pinch, faeces and serum

Diagnosis

Based on clinical signs

Delayed-type hypersensitivity (DTH) is detectable early in the infection and remains present in a proportion of the subclinically infected carriers

DTH wanes and may be absent in clinical cases

Intradermal johnin test

Serum antibodies are detectable later than DTH

Serum antibodies are present more constantly and are of higher titre as lesions become more extensive, reflecting the amount of antigen present

Confirmation depends on typical gross lesions and demonstration of acid-fast organisms in impression smears

Isolation of MAP in faecal culture- Suitable media include Herrold's egg yolk medium, modified Dubos' medium and Middlebrook 7H9, 7H10 and 7H11 media; mycobactin is necessary for bacterial growth.

Polymerase chain reaction

Differential diagnosis

Gastrointestinal parasitism

Coccidiosis

Copper poisoning

Starvation

Salmonellosis

Sheep and goat

Caseous lymphadenitis

Internal abscess

Gastrointestinal parasitism

Ovine progressive pneumonia

Dietary deficiencies

Treatment

Streptomycin 50 mg/ kg b.wt

Isoniazid 20mg/kg b.wt

Clofazimine 600 mg daily for 10 months in sheep and goat

Prevention and control

Identification and elimination of infected animals by test (**ELISA and faecal culture**)

Prevention of introduction of new infection

Prevent exposure of young animals to infectious manure, colostrums and milk

Calves from infected dams should also be considered for removal

Hygienic practices (common practice of using same loader for feeding and manure handling

Isolated rearing of calves

Vaccination

Vallee' s vaccine- Live vaccine in paraffin oil and pumice stone vehicle

Sigurds son vaccine-killed vaccine

Vaccination is done only in calves, less than 1 months of age

Haemorrhagic Septicaemia

(**Syn:** Septicaemic Pasteurellosis, Barbone)

Haemorrhagic septicaemia is an acute potentially fatal septicaemia mainly affecting cattle and buffaloes.

Etiology

It is caused by *Pasteurella multocida* serotype type B and E

It is a gram negative, coccoid and short rod shaped organism

The organism stains at either pole, so it is called as **Bi-polar**

Carter classified the serotypes as A- E based on the analysis of capsular antigen

Epidemiology

Prevalence of infection

The disease is enzootic in tropical and sub tropical countries

It is an economically important disease in India

Out breaks usually occurs in wet and humid weather during rainy season

Morbidity rates 50% and case fatality rate 100% but it is three times higher in buffaloes

Adult animals in endemic area acquire a natural immunity for infection

Source of infection

Recovered carrier and clinical animals carries the organism in tonsillar and nasopharyngeal mucosa in 45% of healthy and 3-5% diseased cattle

The organism is abundant in saliva and blood

Predisposing factors

Stress due to exercise, mal-nutrition, adverse climate, health related fatigue, over exertion and starvation

Wet humid climate during rainfall

Overcrowding Damp or humid weather

Transportation of feed lot cattle

Excessive worm burden

Transmission

Ingestion of contaminated feed and water

Ticks and insect bites.

Host affected

Buffaloes are more susceptible than cattle

All ages can be affected, but more common between 6 months to 2 years of age

Cattle, sheep, goats, yak and water buffaloes are highly susceptible.

Horses and pigs are less susceptible

Pathogenesis

Organism by ingestion enter the tonsils, GI tract and respiratory tract, via the blood circulation reaches heart, liver and spleen followed by development of severe septicaemia which ensues death in 24 hours

Animals recovered from natural infection are immune

Clinical signs

Incubation period 2-4 days

Death within 24 hrs

Sudden onset of high fever

Profuse salivation

Sub-mucosal petichiation, severe depression and death in 24 hours

Warm, painful swelling at throat, dewlap, brisket and perineum

Severe dyspnoea and respiratory distress

Severe toxemia and enteritis and dehydration in gastrointestinal form

Necropsy finding

Generalized haemorrhages in serosa

Oedema, congestion to patchy or extensive consolidation of lungs and lymphnodes

Infiltration of gelatinous fluid in subcutaneous area

Haemorrhagic gastroenteritis

Prominent thickening of interlobular septa

Diagnosis

Based on clinical signs and necropsy findings.

Demonstration of the bipolar organism in the blood smear at the height of temperature

Isolation of organism in blood agar

Animal inoculation: Intra peritoneal or subcutaneous inoculation of blood into mice or rabbit causes death in 24- 48hrs

ELISA is used to detect antibody

Sample collection

Blood smear, heart blood smear, heart blood swab, bone marrow, blood, heart and spleen immediately few hours following death

Differential diagnosis

Mycoplasmosis
Chronic respiratory diseases
Parasitic infection
Pneumonic pasteurellosis

Treatment

Cattle – Inj. Sulphadimidine @150 mg /kg bw, i/v for 3 days.-drug of choice
Pig – Inj. Oxytetracycline @10mg / kg bw i/v for 3-5 days is highly effective

Prevention

Cattle & Buffalo-Formalin inactivated alum precipitated haemorrhagic septicaemia vaccine 5-10 ml S/c
Immunity develop in 10 days and last for about 6 months

Control

The organism is killed by 0.5% Phenol for 15 minutes
Vaccinate all susceptible population before the onset of monsoon
Avoid all types of stress

Contagious Bovine Pleuropneumonia (Syn: CBPP, Lung Sickness)

Contagious bovine pleuropneumonia (CBPP) is a severe contagious disease of cattle, characterized by high fever, anorexia, coughing, dyspnoea, thoracic pain, arched back, expiratory grunt, and pleuritic friction rubs, edema of throat and dewlap.

Etiology

CBPP is caused by *Mycoplasma mycoides subspecies mycoides* small colony type
The organism is pleomorphic in nature. It can be grown in a special media and embryonated hens eggs
The organism is highly susceptible to environmental influences including disinfectant, heat and drying

Epidemiology

CBPP is considered one of the economically important cattle plague affecting Africa with relaxation of import control and international trade.
OIE categorized CBPP as List A disease, transmissible disease that have potential for very serious and rapid spread irrespective of national border

The disease was eradicated from many countries during 19th and 20th century through strict movement control, culling of infected cattle, it has **re-emerged due to change of ecological and environmental factors such as drought, floods and increased movement of livestock for long distance, reduced serological screening, abattoir studies and limited potency of CBPP Vaccine**

Prevalence of infection

It is reported to be endemic in most parts of Africa and it has been reported in Europe, USA, Asia, Australia and Japan

In Asia, CBPP has been reported from Assam in India, Bangladesh and Myanmar

Economic impacts

Direct loss in CBPP is due to mortality, decreased milk yield, vaccination, disease surveillance and research program

Indirect loss occurs through loss of weight, reduced working ability, decreased fertility, loss of trade and fatal nature of the disease

Predisposing factors

Stress due to starvation, exhaustion, intercurrent disease can cause the **sequestrum to break down** and convert the animals into an active cases

Sources of infection

The infected animals carry potential organism for at least 3 years period

Large number of viable organism are passed through urine

The organism is isolated from semen, preputial washings, urine and frozen embryos.

Transmission

Inhalation of infective droplets from active or carrier cases

Direct and repeated contacts between sick and healthy animals

Host Susceptible

CCBP occur only in cattle, rare natural cases have been recorded in buffalo, yak, bison, reindeer, and antelopes

Pathogenesis

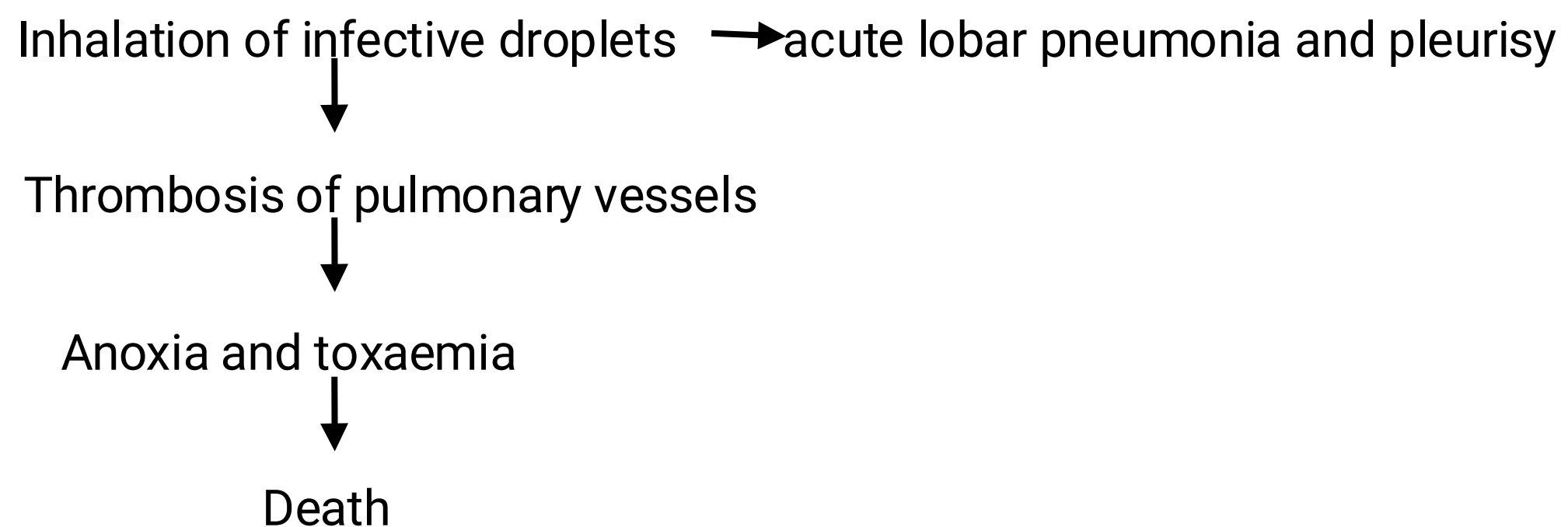
The disease spread principally through **inhalation of infected droplets**

The organism possess carbohydrate cell capsule and produce hydrogen peroxide.

It develops an **acute lobar pneumonia, pleurisy** and invades the lungs of cattle to cause mycoplasmaemia resulting in localization in numerous other sites including kidneys, joints and brain, resulting in high morbidity and mortality

Thrombosis in pulmonary vessels prior to the development of pneumonic lesions and there is no general increase in blood coagulability and no generalized tendency to spontaneous thrombosis

Death results from **anoxia and toxaemia**



Clinical signs

Per acute form

Affected cattle may die in a week after the onset of respiratory distress

Acute form

Incubation period 3-6 weeks and may exceeds 6 months sometimes

Sudden onset of high fever 105⁰F, fall in milk yield, anorexia and cessation of rumination

Severe depression, **animals stand apart or lag behind other group**

Coughing at first only **on exercise** and **thoracic pain** are evident, disinclined to move, **standing with the elbow out, arched back**, extended head, shallow and rapid respiration with **expiratory grunting**

Pain evinced on percussion of chest

Auscultation reveals a **pleuritic friction sounds** in early stages, dullness, fluid sounds, moist gurgling, and noise in later stages of infection

Dullness on areas of lung may detectable on percussion

Edematous swelling of throat, and dewlap, swelling of the large movable joints may be present

In calves valvular endocariditis and myocarditis may occur

In fatal cases death occurs at a variable course from several days to 3 weeks.

Chronic and sub acute form

Recovered animals may be clinically normal but in some animals inactive sequestrum forms in the lung with a necrotic centre of sufficient size to produce toxaemia causing unthriftiness.

A chronic cough, mild respiratory distress on exercise

During exercise this sequestra commonly break down when the animal is exposed to environmental stress and cause an acute attack of disease

Necropsy finding

Lesions limited to thoracic cavity, lungs and the lesion is **unilateral**

The **pleural cavity** contains large quantity of clear, **yellow brown fluid containing pieces of fibrin**. This fluid is ideal for culture of the organism

Caseous fibrinous deposits are present on the parietal and visceral surfaces of the pleura

Interlobular septa are predominantly distended with amber colour fluid surrounding distended lymphatics

Consolidation of lungs with a typically marbled appearance is characteristic

In chronic or advance case a **sequestrum of necrotic lung** varying size from 1-10 cm in dia, which may be surrounded by a fibrinous capsule

The sequestrum rupture and are drained by a bronchus

In **calves'** **exudative peritonitis, arthritis, bursitis and fibrinous anrthritis** of carpal and tarsal joints are present

Perivascular organization foci or organization in the interlobular septa are considered typical of CCBP

Sample collection

Nasal swab, bronchioalveolar lavage or transtracheal wash fluid, Pleural fluid, urine, blood, serum and pulmonary tissue

Diagnosis

Based on clinical signs and necropsy findings

Isolation and identification of organism from pleural fluid and sequestrum

Molecular diagnosis by PCR

Loop mediated isothermal amplification (LAMP) detect DNA of Mycoplasma from pleural fluid and plasma within one hour

Immunological tests: Indirect Fluorescent antibody test (IFA) and Fluorescent antibody technique (FAT)

Serological test- Complement Fixation Test (CFT) and Competitive enzyme linked immunosorbent assay (C-ELISA)-These are OIE prescribed test for testing of antibodies against Mmmsc

C-ELISA is superior in detection of antibodies in chronic cases

Immunoblotting test (IB) is useful at herd level diagnosis

Slide flocculation and rapid slide agglutination test

Differential diagnosis

Haemorrhagic septicaemia
Pneumonic pasteurellosis
Theileriosis (East coast fever)
Ephemeral fever
Tuberculosis
Pulmonary abscesses
Actinobacillosis
Echinococcosis (Hydatid cyst)

Treatment

The use of antimicrobials is legally prohibited in many countries affected by CBPP, because it gives rise to persistent infection

There is no specific treatment available

However, use of antibiotics have been found effective for controlling the secondary bacterial infection that may aggravate the infection

Tulathromycin @ 2.5 mg/Kg, SC, single dose

Tilmicosin @ 10mg/Kg, SC single dose

Gamithromycin @ 6mg/Kg SC, single dose

Danofloxacin @ 2.5 mg/Kg SC

Oxytetracycline @ 10 mg/Kg for 3 days

Control

Tulathromycin @ 2.5 mg/Kg, SC, single dose

Tilmicosin @ 10mg/Kg, SC single dose

Gamithromycin @ 6mg/Kg SC, single dose

Danofloxacin @ 2.5 mg/Kg SC

Long acting Oxytetracycline @ 20 mg/Kg IM

Prevention

Vaccination of healthy cattle

Currently vaccine in use live vaccine derived from T1 strain (T1/44 and T1SR) of live attenuated mycoplasma mycoides subsp mycoides attenuate through repeated passage in embryonated egg

Route of administration-Subcutaneous

T1/44 induce good immunity, when herds are revaccinated annually

Disposal of sick cattle during epidemics

Control

Disposal of all sick and contacted animals

At herd level, removal of sources of infection by identification of infected animals and their removal

Testing of all animals over 12 months of age

It is necessary to do at least three negative tests at interval of greater than 3 weeks apart before the herd is classified as free from infection and vaccination of all animals at risk

Maintenance of hygiene

Contagious Caprine Pleuropneumonia

(Syn: Pleuropneumonia contagiosa)

It is a chronic contagious disease of goats and sheep resembles the contagious bovine pleuropneumonia and clinically characterized by pneumonia, fibrinous pleurisy, profuse pleural exudates and death.

Etiology

It is caused by *M.capricolum subsp. capripneumoniae*. The causative agent is previously known as **Mycoplasma strain F38**.

It is difficult to differentiate from *M.mycooides subsp Capri*

Mycoplasma mycooides subspecies Caprae, *M.mycooides subspecies mycooides*, *M.galactiae*, *M.arginini* and *M.capricola* have been considered as the principal cause of pleuropneumonia in kids upto 6 months

Epidemiology

Prevalence of infection

It is widespread in tropical countries

In India the disease was reported from Chennai, Assam, Mumbai, Haryana, Uttarpradesh, Bihar and West Bengal

Morbidity rate -100% and Mortality rate is f 60-100%

Economic impact

Direct loss- high mortality, reduced milk and meat yield, cost of treatment, disease diagnosis, surveillance and control measures

Indirect loss- imposition of trade restrictions

Source of infection

Infected carrier animals

Organism does not live longer period outside of animal body

Transmission

The disease spread through inhalation of organism present in soil and environment

The bot fly have been found to spread the infections

Experimental transmission is possible

Host affected

Goats are highly susceptible when compared to sheep

Goat strain is not transmissible to cattle

Clinical signs

Incubation period- 6-10 days

Affected animals continue to graze for some time but eventually become anorectic

Fever (Temp. 104.5 to 106°F) cough, dyspnoea, lagging, lying down for lot of time, but the animal can stand and walk, abducted legs, the stiff neck and extended head with painful grunting

Open mouth breathing, protrusion of tongue with frothy salivation occur in terminal stage of illness and death in 2 or more days

In adverse climatic conditions disease occurs as septicemic with little signs or with evidence of pneumonia

In endemic areas subacute and chronic cases are common and the symptoms are milder, dominated by intermittent coughing

Necropsy finding

Usual findings are similar to bovine but in **caprine sequestra are not formed in the lungs**

Lesions are **restricted to lungs and pleura with hepatization** of part of the lung (marbled appearance on cut surface of lung) and increase in pleural fluid with a fibrinous pleuritis

Sample collection

Pleural fluid and affected lung tissue and serum

Diagnosis

Based on clinical signs and necropsy findings.

Isolation and identification of organism from pulmonary lesions with specific culture media

More effective detection is by **PCR, which is performed on samples of pleural fluid** dried on filter paper.

Detection of antigen from lung tissues.

Antibody detection by CFT, ELISA and Latex agglutination test.

A blocking ELISA using monoclonal antibody is highly specific for detection of CCPP

Differential diagnosis

Other pulmonary mycoplasmosis

Pneumonic pasteurellosis

Treatment

The organism is highly sensitive to Oxytetracycline @15mg/kg bw /day

Tylosin tartrate @ 10mg/kg bw i/m along with cotrimaxazole tablet for 5 days

Tiamutin 36mg/kg bw is satisfactory

The macrolides (erythromycin, spiramycin, and tylosin), tetracyclines and quinolones are active against *M. capricolum* subsp. *capripneumonia*

Prevention

Vaccine consists of live culture of *M. caprae*

Dose 0.2ml @ tip of the ear s/c

Saponin killed lyophilized vaccine can be used for revaccination @ 6 months interval

Control

Implementation of good biosecurity measures

Vaccination with an **inactivated mycoplasma F38 vaccine** induces an immune response which is effective in reducing morbidity and mortality rates

A booster dose /month after the first vaccination provides an additional protection

Immunity is generally short lived

Age of vaccination is >12 weeks

Avoid mixing of infected animals with healthy animals and maintenance of good hygiene

Movement restrictions and slaughtering of infected animals are recommended for countries that are newly infected

Actinobacillosis

(Syn: Wooden tongue, Timber tongue and Big head disease)

It is a chronic pyogranulomatous inflammation of soft tissue of the tongue, head, neck, oro-pharyngeal lymph nodes, oesophageal groove and nasal cavity and clinically characterized by induration of tongue.

Etiology

It is caused by *Actinobacillus lignieresii*

It is a gram negative motile rod shaped organism and appears as single or in short chain

The organism is susceptible to external environment and disinfectants.

Blood and serum is required for growth of the organism.

Epidemiology

Prevalence of infection

The disease is world wide in distribution

Sporadic occurrence in India

Predisposing factors

Eruption of permanent teeth in young animals favours the organism to penetrate sub-epithelial tissues

Grazing on abrasive pasture or spiny awns

Source of infection

Infection is spread through discharges from infected animals

Grazing on infected pastures

Transmission

Through injury and abrasion made by sharp objects permit the entry of organisms through oral mucosa.

External parasites help the development of cutaneous actinobacillosis.

Host affected

Cattle, buffalo and sheep are susceptible

Horses, pigs and human beings are also affected by the natural infection

Young animals of 2-5 years of age are found more affected

Pathogenesis

The organism is a normal inhabitant of digestive and respiratory tract, they invade the nasal and buccal mucosa and adjacent lymph node following trauma or injury of them

In localized infection causes an acute inflammatory reaction and subsequent development of granulomatous lesions in which necrosis and suppuration with discharge of pus to the exterior and spread into adjacent lymph nodes

Infection of tongue interfere in prehension and mastication

Theoretically, inflammatory changes occur at an earlier stages and deformity of the tongue occur at an later stages

Clinical signs

Cattle

Excessive salivation

Gentle chewing

On palpation tongue is swollen and hard particularly at the base of the tongue

Nodules and ulcers are present on the side of the tongue

Affected tongue appears like wood and hence the disease has got the name wooden or timber tongue.

Acute inflammation is replaced by fibrous tissue, the tongue becomes shrunk and immobile

Lymphadenitis is common

Enlargement of submaxillary, parotid and retropharyngeal lymph node

Difficulty in swallowing

Loud snoring respiration

The lesion also found in external nare

The organism also produce infection in liver, rumen, reticulum, peritoneum and lungs

In testis produces granulomatous lesion

Recurrence is common

Sheep

Tongue is usually not affected in this species

Lesion seen in cutaneous portion of lower jaw, face and extending into sternum

Thickening and scabbing of lips

Affected lymph nodes have yellowish greenish pus discharging from the openings

Bilateral nasal discharges

Cud dropping

Occasionally causes mastitis in sheep

Affected sheep may die due to starvation

Necropsy findings

The tongue becomes hard by proliferation of fibrous tissues

Lesions of soft tissue appeared as thick walled abscesses filled with thick tenacious pus

Firm, pale nodules containing in the center with minute yellow "sulfur granules".

Diagnosis

Based on clinical signs and lesions.

Identification of gram negative coccobacilli in stained pus

Examination of rosettes formation in granules (less than one mm in diameter).

Press cheesy mass on a glass, put a drop of 10% KOH solution, covered with cover slip and observe rosette mass under microscope without staining

Differential diagnosis

Actinomycosis

Rabies

Foreign bodies in the mouth

Tuberculosis

Treatment

X-Ray radiation for 10 consecutive weeks along with antibiotics

Standard treatment- Sodium Iodide 7-10g/ day for 7-10 days per os

Abortion occurs occasionally in pregnant animal following treatment with sodium iodide

Lugol's Iodine injected into the lesion may be useful

Oral feeding of cattle with Iodine 10 gm/1000 lb bw daily for 2-4 weeks until iodism signs appear (anorexia, salivation and depression)

Administration of Sodium Iodide @70 mg/kg bw (10-20%) solution in slow i/v and repeated after 7-10 days

Administration of streptomycin sulphate @5 g/day i/m for 3-5 days

Sheep- Sodium iodide 10% solution 20ml s/c route weekly once for 4-5 weeks

Control

Isolation of affected animals

Slaughter of affected animals

Changes in grazing management

Feed and water trough meant for affected animals should be disinfected

Actinobacillosis is also affecting human beings. Hence, due care in handling of diseased animals and processing of infected tissue samples in laboratory is very important

Actinomycosis

(Syn: Lumpy Jaw)

Actinomycosis is a common sporadic disease of cattle occasionally occurs in pig and horse and clinically characterized by rarefying osteomyelitis of the bone in skull, enlargement of mammary gland, inflammation of bursa of the ligamentum nuchae.

Etiology

It is caused by *Actinomyces bovis*, a normal inhabitant of oropharynx and digestive tract

It is a gram positive non spore forming filamentous organism

As cultured organism has both bacterial and fungus features it is ascribed as fungal like bacteria.

Epidemiology

Prevalence of infection

The disease is world wide in distribution

It is recorded in various regions of India

Incidence of the disease is higher in straw and silage fed cattle, because, these coarse materials injure the buccal mucosa and spread the infection further into the other parts of the body

Predisposing factors

Eruption of permanent teeth in young animals favors the organism to penetrate sub-epithelial tissues

Source of infection

Infection is spread through discharges from infected animals

Grazing on infected pastures

Transmission

Infection occurs through abraded injury or wounds induced by coarse feed stuff

Infection spread through dental alveoli at the time of eruption of teeth

Alimentary tract harbour *A. bovis* from where it invade subcutaneous tissues through injury made by sharp object

Animal to human transmission is rare

Host affected

Infection is sporadic and more common in cattle of 2-5 years old

The disease rarely affects pigs, horses and goats

Chronic actinomycosis prevails in horses and dogs

Pathogenesis

A. bovis causes **rarefying osteomyelitis of jaw bone, mandible or maxilla**

Involvement of jaw interfere with prehension and mastication

Infection of rumen leads to rumenitis. Which interfere with rumen motility and digestion results in partial starvation

Rarely localization occur in other organs by haematogenous spread

Clinical signs

Cattle

Initially **hard, painless bony swelling at the level of central molar teeth of the mandible or maxilla**. Which may be diffuse or discrete

Enlargement of mandible may appear only as a thickening of lower edge of the bone with most of the enlargement in the intermandibular space

The more common discrete lesions appear on the lateral surface of maxilla

The swelling are hard, immovable and become painful in the later stage, which break through the skin and discharge small amount of honey like pus through

one or multiple openings. The consistency of pus is honey like contain yellow white sulphur granules

There is a **tendency for sinus to heal and fresh one develops periodically**

Excessive swelling of maxilla may cause **dyspnoea**

Involvement of the local lymph node does not occur

Emaciation

The most common soft tissue involvement are oesophageal groove region, from which extension of infection to the reticulum cause impaired digestion, periodic diarrhoea with undigested food material and allotriphagia

The less common lesion of the soft tissue include orchitis in bull, partial obstruction of trachea and abscess in brain or lung

Horses

A. bovis in association with *Br. abortus* causes inflammation of bursa of ligamentum nuchae, which leads to suppuration and granulation of nuchal ligament is called as **poll evil and Fistulous withers**

Pigs

Extensive granulomatous lesions on the skin, particularly over the **udder** are more common

Emaciation is due to visceral involvement is rare

Necropsy findings

Lumpy jaw is an important manifestation of a **suppurative osteomyelitis of the mandible**

Affected bone is filled with distinct granulation tissues and nodules

Millary nodules present in udder and lungs

Oesophagus and anterior wall of the reticulum containing pockets of pus and spread from these may causes chronic local peritonitis

In sows infected udder is indurated and fibrosed with suppuration

Diagnosis

Based on clinical signs and necropsy finding

Isolation and identification of organism from pus

Gram staining of pus contained sulphur granule, which appear as club colonies with typical filamentous bacteria at centre

Sample collection

Pus from unopened lesion must be aspirated by a syringe and examined the visible sulphur granules under naked eye

Differential diagnosis

Foreign body problem

Chronic peritonitis
Cutaneous or mammary disease
Borreliosis

Treatment

Surgical debridement with antibacterial therapy is effective
Administration of **Sodium iodide 7-10 gm** mixed with water and should be given orally for 7-10 days
Isoniazid 20 ml /kg body weight for 10 days @ 10-20 mg /kg body weight daily orally for 30 days
Streptomycin is used successfully
Combination of Streptomycin with iodide and surgical drainage may provide good response
Cryotherapy with liquid nitrogen is found to be effective for treating abscess

Prevention

Vaccine is not available

Control

Isolation and disposal of affected animals
Feed and water trough meant for affected animals should be disinfected
Changes in grazing management

Dermatophilosis

(Mycotic dermatitis, cutaneous streptothricosis, Senkobo disease in cattle, Lumpy wool of sheep)

Dermatophilosis is an exudative, pustular dermatitis of cattle, sheep and goat, horses and also occur in dog cat, wild mammals, reptiles and occasionally humans. The severe disease in ruminants is promoted by immunomodulatory effects induced by infestation with the tick, *Amblyomma variegatum*.

Etiology

Dermatophilus congolensis

Gram positive, filamentous, branching actinobacterium

The organisms are dimorphic and grow as branching filamentous mycelia containing dormant zoospore. Which are transformed to active motile cocci under moist condition

Epidemiology

Prevalence of infection

Disease is prevalent tropical and temperate region of the world

Economic Impact

Damage to fleece causes severe losses upto 30%

Loss of skin value upto 40%

Predisposing factors

Prolonged wetting of fleece

Prolonged and heavy rain

Shearing cuts

Trauma to the face and limbs of animals grazing on thorny scrub

Intercurrent disease and stress

Moist and cold weather conducive for spread of the diseases and most of the outbreaks occur during rainfall.

Tick infestation (*Amblyomma variegatum*, *Boophilus microplus*, *Hyalomma* sp),
Sarcoptes, Demodex, mosquitoes, stinging and blood sucking flies favour the growth of *Dermatophilus* sp through bite wound

Source of infection

Healthy carrier animals

Scab of an infected animals

Transmission

Direct contact with active lesions of infected animals

Mechanical transmission through instruments, brushes, harnesses, bedding materials are possible

Transmission occurs through anthropod vectors like flies, *Musca domestica*, *Lucilia cuprina*, *Stomoxys calcitrans* and *Calliphora* spp.

Ticks and sheep keds acts as a mechanical vector

The disease also spread from one animal to other while dipping

Infection spread through abrasion

Host affected

Most commonly seen in cattle, sheep, goat and horse.

The disease was also reported in donkeys, dogs, fox, cat, deer, antelope, zebra, pig, monkey, raccoons and wide variety of wild animals

Man is readily infected with dermatophilosis therefore, may be considered as zoonotic disease

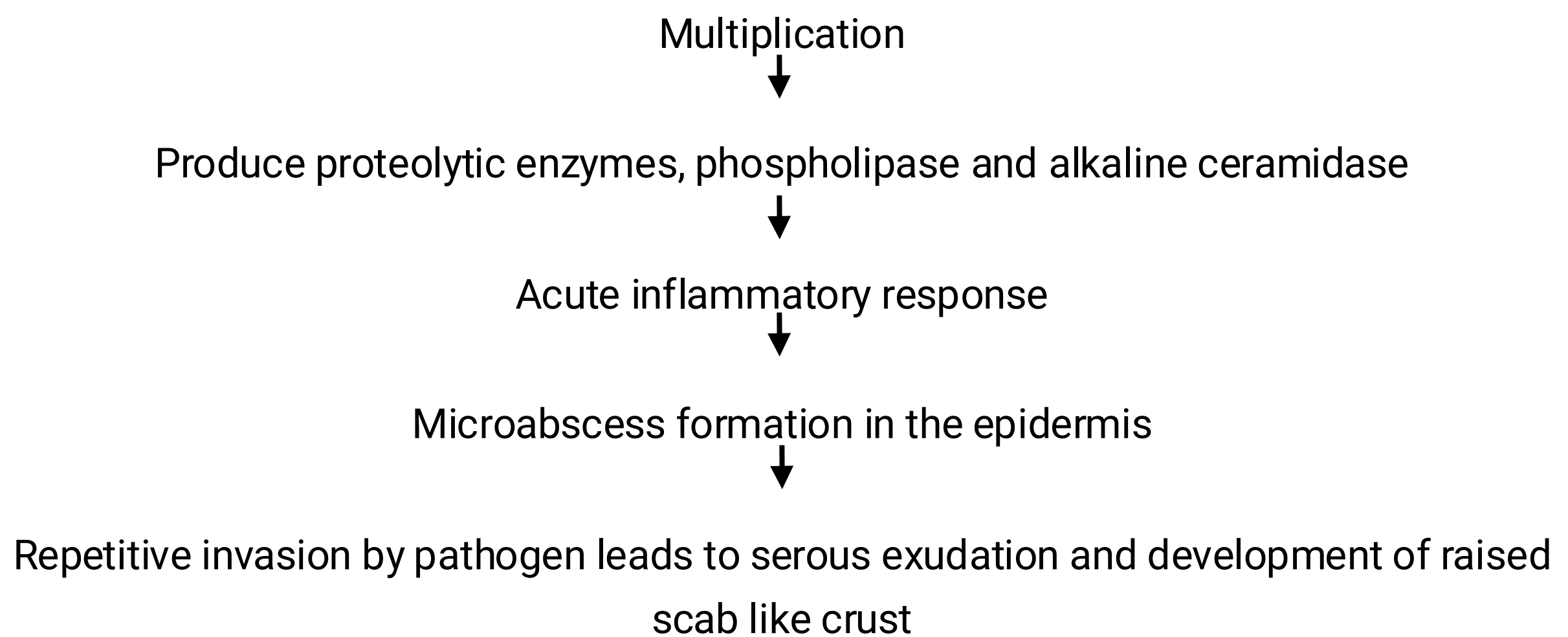
Pathogenesis

Trauma and prolonged wetting damage the cornified epithelium of skin



Invade the damaged skin





Clinical signs

Cattle

Early lesions may present as papules and are often detectable only on palpation

Serous exudates causes matting of hairs give them a tufted appearance (**Paint brush lesion**)

Lesions may coalesce to form irregular elevated crusty scab

Tuft of hair can be readily plucked from the lesion along with adherent scab material

The crust are hard to remove

Purulent exudates below the crust and removal of crust will reveal moist and inflamed areas of granulation tissues

Lesions commonly found on **withers to rump and extend halfway down the rib cages**, neck, body, udder and leg

Sheep

Lesions are commonly not visible in sheep because they are obscured by fleece

Crust can be palpated as hard mass at the surface of the skin (lumpy wool disease) typically distributed over the **dorsal midline with ribs spreading laterally and ventrally**

The crust are roughly circular, thick, upto 3 cm, often distinctly **pyramidal with concave base**

The crusts are easy to remove

Goats

Lesions appear on the lips and muzzle and spread to feet and goat possibly by biting

Sometimes lesions are found on all parts of the body, mostly on dorsal midline and inside the thigh

Heavy crust formation in ear

Horse

No pruritis

The affected skin is thickened and dried up exudates mat together form scab

The lesions on lower limbs are most common behind the pastern, around the coronet leads to lameness

Lesion can also be found on face, eye and muzzle

Removal of scab will exhibit a bleeding surface

Dogs and cats

Exhibits the feature of **exudative dermatitis** manifested as discrete, multiple focal raised lesions

The incidence is extremely rare in cat

Necropsy finding

In animals that die, there is extensive dermatitis, sometimes a secondary pneumonia, and often evidence of intercurrent disease

Sample collection

Scab, impression smears from concave surface of scab after soaking with distilled water for minimum 30 minutes or overnight

Diagnosis

Based on clinical signs

Gram staining of impression smear reveal filamentous form of bacteria

Gram staining does not give as good results as Giemsa because it may overstain the background and does not clearly show the characteristic laddering of the coccoid forms

Giemsa staining is commonly used to visualize the morphology of bacteria as rail road or tram track appearance

Isolation of organism in blood agar with 10% Carbon di-oxide produce haemolytic, small pin point greyish-yellow colonies

Specific fluorescent antibody test, ELISA and Counter immunoelectrophoresis for detection of antigen

Polymerase chain reaction

Differential diagnosis

Ring worm

Staphylococcal dermatitis

Scabies

Pediculosis
Fleece rot
Photosensitization

Treatment

Sheep

Combination of Procaine penicillin @ 70,000 IU/kg and streptomycin 70 mg/ kg for three consecutive days are effective (intramuscular route)
Long acting tetracycline @ 20 mg/kg b.wt are usually effective (intramuscular route)
Combination of lincomycin @ 5 mg/kg and spectinomycin @10 mg/ kg can be used
Bactericidal dips are used but have limited efficacy do not penetrate the scab to the active lesions

Cattle

Long acting tetracycline@ 20 mg/kg b.wt- single dose
Tetracycline@ 5 mg/kg b.wt and should be repeated at weekly interval whenever required
Parenteral procaine penicillin@ 22,000 IU/G for three consecutive days

Horse

Topical therapy is commonly used
Scab can be removed under sedation and lesions are treated topically with povidone iodine or chlorhexidine until heal
Topical application of 2.5% benzyl peroxide is found to be effective

Prevention and Control

Isolation of infected animals
Provision of shelter during prolonged rainfall
Spraying or dipping of sheep a in 0.5-1.0% Zn So₄ solution immediately after shearing
Spraying or dipping of sheep in a 1% solution of Alum (potassium aluminum sulfate) provide protection upto 70 days after shearing
Tick control (by dipping or spraying of acaricide at weekly interval)
Prophylactic use of long acting tetracycline in an endemic area
Control of intercurrent diseases

Mastitis **(Syn: Mammitis)**

Mastitis is an inflammation of the parenchyma of mammary gland regardless of cause. It is characterized by physical and chemical changes in milk and pathological changes in glandular tissue. The most important changes in milk include discolouration, presence of clots and large number of leukocytes. There is swelling, heat and induration of the mammary gland in many clinical cases.

Etiology

Based on infection, mastitis is divided into contagious and environmental mastitis

Contagious mastitis is caused by contagious pathogen, the infection spread from infected quarter to other quarter and cows

Environmental mastitis is caused by environmental pathogens, which is usually present in the cow's environment and reaches the teat from that source.

Contagious pathogens (Major pathogens causing clinical mastitis)

Streptococcus agalactiae, *Staphylococcus aureus* and *Mycoplasma bovis*

Environmental pathogens

- a) Environmental streptococci - *Streptococcus uberis* and *Streptococcus dysgalactiae*
- b) Environmental coliforms - *Escherichia coli*, *Klebsiella spp*, *Citrobacter sp*, *Enterobacter sp*, *Enterococcus faecalis* and *Enterococcus faecium* and other gram negative bacteria include *Serratia*, *Pseudomonas* and *Proteus*

Minor pathogens

Coagulase negative *staphylococcus sp*, *corynebacterium bovis*

Uncommon mastitis pathogens-Sporadically causes mastitis

Arcanobacterium pyogenes, *Nocardia sp*, *Pasteurella multocida*, *Pasteurella haemolytica*, *Campylobacter jejuni*, *Mycobacterium bovis*, *Bacillus cereus*, *leptospira interrogans*,
fungal infection- *Aspergillus fumigates*, *Candida sp* and *Cryptococcus neoformans*

Epidemiology

Prevalence of infection

Mastitis occurs in early lactation usually acquiring in first 50 days after calving

The risk of mastitis increases with increasing parity

Source of infection

S.agalactiae and *S.aureus* resides primarily in the udder of infected cows

Exposure to uninfected quarters is limited to milking process

Streptococcus uberis, *S.dysgalactiae* and coliforms are normal inhabitants of the cows environment

Risk factors

Age and parity- prevalence of infected quarters increases with age and peaking at 7 yrs

Most new infections occur during the early part of dry period and in the first 2 months of lactation

In heifers, the prevalence of infection is often high in the last trimester of pregnancy and several days before parturition, followed by marked decline after parturition

The incidence of mastitis is greater in Holstein Friesian than Jersey breed

Milking character and morphology of udder-high milking rate and large teat canal diameter has been associated with increased somatic cell count or risk of intra mammary infection

Physical condition of teat-the efficiency of teat defense mechanism depends on integrity of teat tissue

Nutritional status – Vitamin E and A and Selenium may be involved in certain types of mastitis

Genetic resistance to mastitis

The production of keratin in streak canal and physical and biochemical characteristic of keratin are important contributors to mastitis resistance. Many of the defense mechanism of udder including lysozyme, lactoferrin, immunoglobulins and leukocytes are direct product of gene and have a genetic basis

Other concurrent infections

Immunological functions of mammary gland is impaired during periparturient period

Previous history of mastitis cows are highly susceptible

Quality and management of housing

Milking practices

Transmission

Through milkers hands

Through ingestion of infected material from environment

Fly biting influences the entry of organism

Economic impact

It is one of the most economically important diseases of dairy cows next to foot and mouth disease.

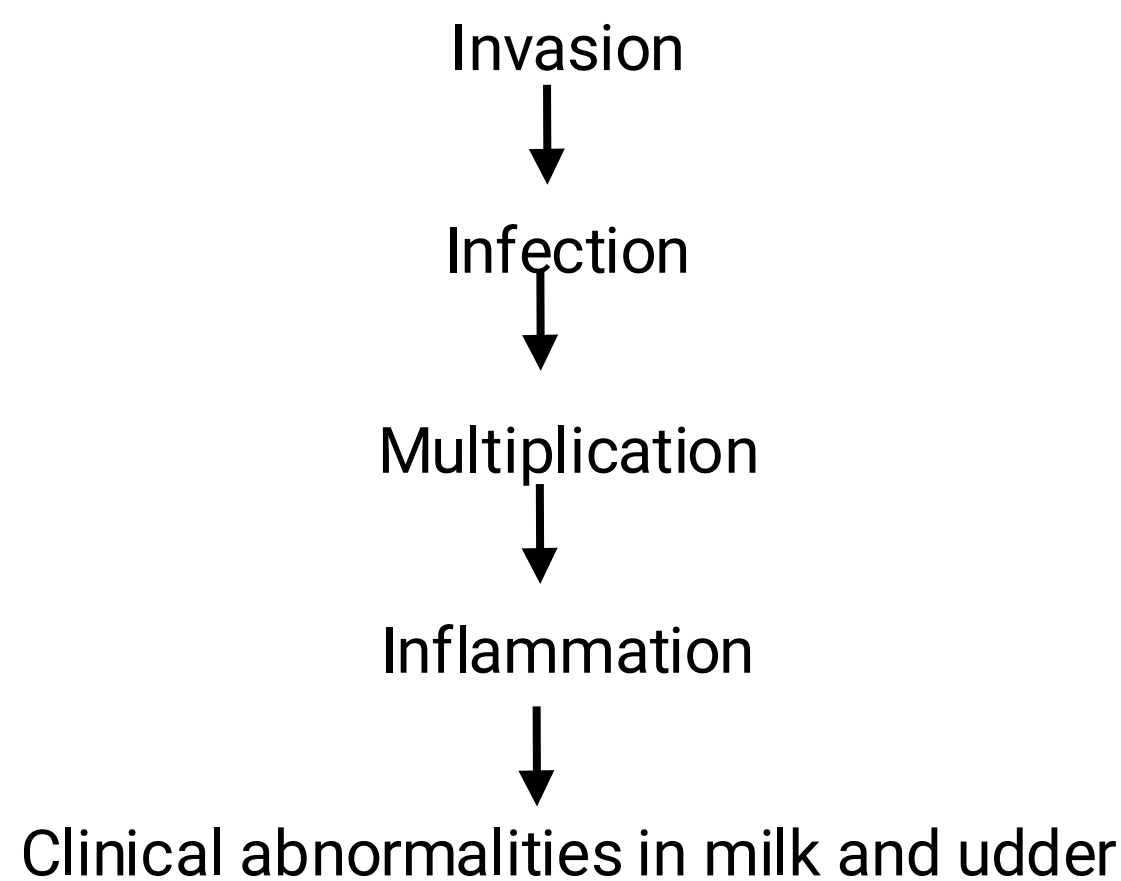
The economic loss occurs due to reduction in milk, changes in quality milk and milk products.

In addition, disqualification of dairy cow due to chronic mastitis.

Zoonotic potential

Tuberculosis, Streptococcal Sore throat and Brucellosis is transmitted to human through ingestion of unpasteurized milk

Pathogenesis



Clinical findings

It includes abnormalities of secretion, abnormalities of size, consistency and temperature of mammary gland and a systemic reaction. The clinical mastitis is usually classified according to severity and duration.

Based on severity

Peracute: Severe inflammation of udder with marked systemic reaction

Acute : Severe inflammation of udder without systemic reaction

Subacute: Mild inflammation with persistent abnormality of milk

Subclinical : high somatic cell count in the milk without visible abnormalities in milk and udder

Based on duration

Short term clinical or subclinical (eg. Coliform infection)

Recurrent clinical (eg. *Staphylococcus aureus*, *streptococcus dysgalactiae*)

Persistent clinical or subclinical (eg. *Streptococcus agalactiae*)

Gross and physical abnormalities of milk and udder

Milk is drawn onto the plate in pools and comparison made between milk of different quarters

Discoloration of milk (may be blood stained, wateriness, presence of clots or flakes)

Flakes at the end of milk indicate of tuberculosis mastitis in cow

Palpation and inspection of the udder are directed for detection of fibrosis, acute and diffuse inflammatory swelling and atrophy of mammary tissue

Diagnosis

Detection of subclinical mastitis by bulk tank milk somatic cell count and California mastitis test-indirect measure of detection of prevalence of mastitis within the herd

Physical examination of udder

Electrical conductivity meter test

Culture and sensitivity

Strip cup test

White side test

Hymast diagnostic test to differentiate gram positive and gram negative bacteria

Nagase test

Biopsy of mammary tissue

Other tests like Bromothymol Blue Test (BTB), Bromocresol purple test and Chloride test

Treatment

Parenteral antimicrobial therapy

Intrammary antimicrobial therapy with udder infusions

Selection of antimicrobials based on culture and sensitivity

Frequent stripping of the affected quarter

Anti-inflammatory agents-Ketoprofen 2g I/m and Flunixin meglumine 2mg/kg I/V twice 24 hrs apart

Supportive therapy- Intravenous administration of isotonic crystalloid fluid or hypertonic saline is indicated in cattle with systemic illness (Peracute coliform mastitis)

Dry cow therapy

The intra mammary use of antimicrobial therapy immediately after last milking of lactation to eliminate existing infection and prevent new infections during early weeks of **dry period** (eg. *Staphylococcus aureus* and *Streptococcus agalactiae*). The most dry cow preparations maintain an adequate minimum concentration in the quarter for about 4 weeks, but some persist for 6 weeks

Blitz therapy

The prevalence of subclinical mastitis due to *s.agalactiae* can be reduced more rapidly by treatment of infected cow during **lactation** and post milking teat dipping

Blanket versus selective dry cow therapy is the treatment of all four quarters at drying off compared to selective dry cow therapy

Sanitation versus Dry cow therapy

Antimicrobial residue test

Drying off chronically affected quarter or non responsive cases with infusion of 30-60ml of 3% silver nitrate solution, or 20 ml of 5% copper sulphate

Cytokines may be useful as adjunctive therapy with existing antimicrobials to improve efficacy

Prevention and control

Use of inactivated vaccine containing *Str.agalactiae*, *Str.uberis*, *Staph.pyogenes*, *Staph.aureus* and *E.coli* in some countries.

Reduction of time of infection

Treatment and drying off quarters

Immediate treatment of affected animals

Culling of non responding animals

Control of development of new infection

Dipping of quarter with disinfectant

Personal hygiene of the milk man before milking

Segregation of infected animals and proper treatment based on the antibiogram

Routine disposal of the soiled materials and thorough cleaning of shed using disinfectants.

Staphylococcal mastitis

Acute swelling of gland with fever, milk is abnormal with thick clots or pus, gangrene of gland and teat in peracute form. Systemic reaction with anorexia, toxemia, fever and ruminal stasis and severe lameness on the affected side

Differential diagnosis

Peracute coliform mastitis

Parturient paresis

Arcanobacter pyogenes mastitis

Streptococcal mastitis

The mammary gland is swollen, warm, and milk is watery and contains clots. Gradual induration of udder if not treated.

Blitz therapy

The prevalence of subclinical mastitis due to *s.agalactiae* can be reduced more rapidly by treatment of infected cow during **lactation** and post milking teat dipping

Mycoplasmal mastitis

Sudden onset of clinical mastitis in many cows, usually all four quarters, marked drop in milk production, swelling of udder and gross abnormality of milk- early stage

milk fairly normal at collection. On standing fine sandy material, flakes or floccules settle out leaving turbid whey like supernatant.

Treatment

Inj. Oxytetracycline 5g daily for 3 days

Mixture of tylosin 500 mg and tetracycline 450 mg used as infusion in some affected quarters

Coliform mastitis

Swelling of gland with watery milk and small flakes, mild systemic response

Peracute

Sudden onset of severe toxemia, fever, complete anorexia, shivering and trembling, tachycardia, polypnea, shock and cow may recumbent

Watery milk initially to thin yellow serous fluid containing small meal like flakes

Differential diagnosis

Parurient paresis and acute carbohydrate engorgement

Environmental Streptococcal mastitis

Swelling of gland. Abnormal milk without systemic signs and recovery occur in 2- 3milking

Pseudomonas aerogenosa

Acute swelling of gland, appearance of clotted or discoloured milk, functions of gland completely lost at one attack, but recurrent crisis may occur.

Arcanobacter pyogenes- summer mastitis

The teat is swollen and inflammed , affected quarter is very hard, swollen and secretion is watery with clots, later become purulent and putrid odour

Severe thelitis with obstruction of teat is a common sequel

Pasteurella sp

Mastitis caused by pasteurell sp is common in ewes and rare in cattle and ewes occurring in a peracute gangrenous forms

Fungal and yeast

Severe fall in milk yield and appearance of viscid mucoid, grey white secretion

Treatment : **Sodium iodide**

Colibacillosis

It is also called as Scour in ruminants, Colisepticaemia and Coligranuloma (Hjarre' s disease) in poultry

Colibacillosis occurs most commonly in newborn farm animals and poultry. Which occur as enteric form (Enteric colibacillosis) and septicaemic form (Colisepticaemia)

Enteric colibacillosis is characterized by varying degree of diarrhoea, dehydration, acidosis and death

Colisepticemia is a common disease of calves and lambs below one week old, which is clinically characterized by severe illness and rapid death

Etiology

It is caused by pathogenic serotypes of *Escherichia coli*

It is a gram negative, flagellated and rod shaped organism

Serotypes are mainly identified by their somatic antigen (O), capsular antigen (K) and flagellar antigen (H)

The capsular antigen can be identified by Rapid Slide Agglutination test

The most common enteropathogens which causes diarrhoea in newborn farm animals are

Enterotoxigenic *E.coli* (ETEC) most common enteropathogens-produce enterotoxin which causes hypersecretion of electrolytes and water into small intestine results in diarrhea

Enterotoxigenic colibacillosis is a major cause of diarrhea in calves less than 3 days of age and is not associated with outbreaks of diarrhea in calves older than 3 days.

Enteropathogenic *E.Coli*- Attachment and effacing *E.coli* (AEEC)- do not produce toxin

Enterohaemorrhagic *E.coli* (EHEC)-uncommon- capable of produce toxin similar to shiga toxin produced by *Shigella dysenteriae* type I- causes mild to severe haemorrhagic colitis

Attachment and effacing *E.coli* (AEEC)- haemorrhagic colitis

Necrotoxigenic *E.coli* (NTEC)-CNF1 or CNF2- causes diarrhoea and septicaemia
EHEC–O157:H7 highly prevalent in cattle and zoonotic to human causes highly fatal haemolytic uremic syndrome in children

Septicaemic *E.coli* strains of serogroup O78 are invasive and causes septicaemia in calves, piglets and lambs

Poultry-*E.coli* O1, O2 and O78 serogroup causes disease

Epidemiology

The disease is usually sporadic and is more common in dairy than beef calves

Morbidity rate: 75% in poorly managed condition

Case fatality rate: 10-50%

Cattle are important reservoir of Enterohaemorrhagic *E.coli* serotype O157H:7-
Haemorrhagic and hemolytic uremic syndrome in humans

Predisposing factor

Colostrum deprivation
Overcrowding, poor hygiene
Adverse climatic conditions-Cold, wet windy weather during winter
Insufficient or no colostral immunity
Inferior milk replacer
Colostrum deprived calves are highly susceptible

Pathogenesis

Enterotoxigenic *E.coli* (ETEC) K99 in calves & K88 in piglets are adhesin antigen is essential for colonization
Enterotoxigenic *E.coli* colonize in the small intestine, proliferative and produce heat stable enterotoxin, which results in net secretion of fluid and electrolyte from systemic circulation into gut lumen
The organism invade the blood stream following infection of the intestine, lungs or umbilical tissue causes colisepticaemia

Clinical signs

Colisepticaemia

It is frequent in calves, lambs and poultry
The invade the blood stream following infection of the intestine, lungs or umbilical tissue
Pyrexia, depression, weakness, tachycardia without diarrhoea
Hypothermia and death occur within 24 hrs
Watery mouth occur in lambs upto 3days of age, profuse salivation and abdominal distension and death is due to endotoxic shock
Meningitis and pneumonia are commonly encountered in affected calves and lambs
Post septicemic localization of organism in joints cause arthritis with swelling, pain, lameness and stiff gait

Enteric colibacillosis

Diarrhoea develops within the first days of life
Consistency of faeces is variable
Faeces are watery or pasty or white or yellowish with rancid odour
Rancid faecal material may accumulate on the tail and hind limb
Depression, dehydration and acidosis
Severely affected animals die within few days

Clinical signs-Poultry

Colisepticaemia

Very young birds are affected

Inappetance leads to listlessness, dejection, dry and ruffled feathers

Rapid laboured breathing, occasionally gasping

Morbidity >50% and Mortality <5%

Following recovery unsatisfactory performances of the birds are noticed

Yolk sac infection

Affected birds appears distressed with distended abdomen

Tendency to huddle, visibly thick navel which is prominent and necrotic

Egg peritonitis

Deposition of fibrin or albumin like material in the abdominal viscera causes peritonitis in adult birds

It is common cause of sporadic death in layers or breeders hen

Coligranuloma

This disease causes a sporadic death in adult

There is no specific signs

Clinical signs of oedema disease in pigs

It occur 1-2weeks after weaning in rapidly growing pig caused by haemolytic type of E.coli

Onset is sudden

Animal found dead without showing clinical signs

Posterior paralysis, muscular tremor, oedema of the eyelid and front face

Squeal may be hoarse due to laryngeal oedema

Faeces usually firm

Flaccid paralysis proceeds death within 36 hrs

Enteric colibacillosis in piglets

Piglet may succumb to enteric colibacillosis within 24 hrs of birth

piglets refuse to suck milk

Profuse watery diarrhoea

Death due to weakness and dehydration

Necropsy finding

Colisepticaemia- Calves, lambs and piglets

Submucosal haemorrhage

Occasionally fibrinous exudate

Enteric colibacillosis

Abomasum distended with milk clots

Poultry -Colisepticaemia

Air sacculitis, peritonitis, perihepatitis, pericarditis and fibrinous pericarditis seen

Pericardial sac appears to be thick with white membrane adhering to the surface of heart is a characteristic findings

liver, spleen, lungs and kidneys appears dark

Liver surface is covered by a thin fibrinous material

Air sacs are thickened, opaque and with white deposits adhere to this

Coligranuloma- Hard yellow, nodular granulomas in mesenteric wall of intestine and particularly in caeca

Sample collection

Colisepticaemia- blood

Enteric colibacillosis- faecal sample

Diagnosis

Hematology and biochemical profile

Increased PCV, decreased bicarbonate level and hyperkalemia

Culture of organism from faecal sample

Slide agglutination test- to identify O &H antigen

PCR

Differential diagnosis

Salmonellosis

Pullorum disease

Fowl typhoid

Fowl cholera

Treatment

Intensive critical care

Parenteral use of antibiotics

Fluid and electrolyte replacement therapy with isotonic NaHCO₃ (1.3%) and 5% dextrose

Prior to use sodium acetate (alkalizing agent)

Antimotility and intestinal protectant-kaolin

Oral electrolytes

Pig- introduction of new feed gradually

Control

Reduction of degree of exposure

Provision of adequate colostrums

Increasing specific resistance of newborn by vaccination of dam or new born

Provision of chlorinated water

Commercial bacterins administered to breeder hens or chicks have provided some protection against homologous *E coli* serogroups

Salmonellosis

(Syn: Paratyphoid)

It is an acute or chronic contagious disease of all mammals, birds and reptiles characterized by the development of severe dysentery and gastroenteritis. It causes the most important zoonotic problem in human beings.

Etiology

Caused by *Salmonella* sp

It is a gram negative, non-spore forming and rod shaped organism. It is susceptible to drying, sunlight and heat. But they may **remain alive in soil for 7 months in water, feces and pasture.**

In moist soil they may remain viable for 12 months or more.

The organism can be destroyed at 60° C for 20 minutes.

Some of the important salmonella species are as follows.

Cattle: *Salmonella typhimurium* , *Salmonella dublin* and *Salmonella newport*

Sheep and goat- *Salmonella typhimurium* , *S.aborus ovis*, *Salmonella dublin* and *Salmonella anatum*

Pigs- *Salmonella typhimurium* and *Salmonella choleraesuis*

Horse - *Salmonella typhimurium*, *Salmonella anatum*, *Salmonella newport*, *Salmonella enteritidis*, *Salmonella arizonae* and *Salmonella angona*

Virulence factors

Adhesion-pili, flagella, cytotoxin, enterotoxin and lipopolysaccharide (LPS)

Epidemiology

Prevalence of infection

The disease is world wide in distribution however; almost it has been reported from all the tropical countries

The prevalence and incidence of disease is varies with geographical area, depending on climate, population density, land use and farming practices

The morbidity and mortality are usually highest in calves under 12 weeks of age

About 40% of outbreak of salmonellosis in foals occur under 8 days of age

Risk factors

Intercurrent infection

Transportation

Overcrowding
Pregnancy
Extreme environmental temperature
Water deprivation
Sudden change in rations

Sources of infection

Infected animals excrete the organism through faeces
The organism may be present in water, soil, animal feeds, raw meat and offal and vegetable materials
Infected animals act as carrier
Carrier animals excrete the causative agents through their feces.
Contaminated water is an important source of transmission.

Transmission

Ingestion is main route of transmission
The disease can also be transmitted through the mucosae of the upper respiratory tract and conjunctiva
Cattle acquire infection from infected cattle or pigs or horse or sheep or dogs or poultry and pigeon

Host affected

Cattle, sheep, human, rodents, horse and dogs
Besides domestic animals, deer, elk, raccoons, possums may remain spread the infection to cattle

Pathogenesis

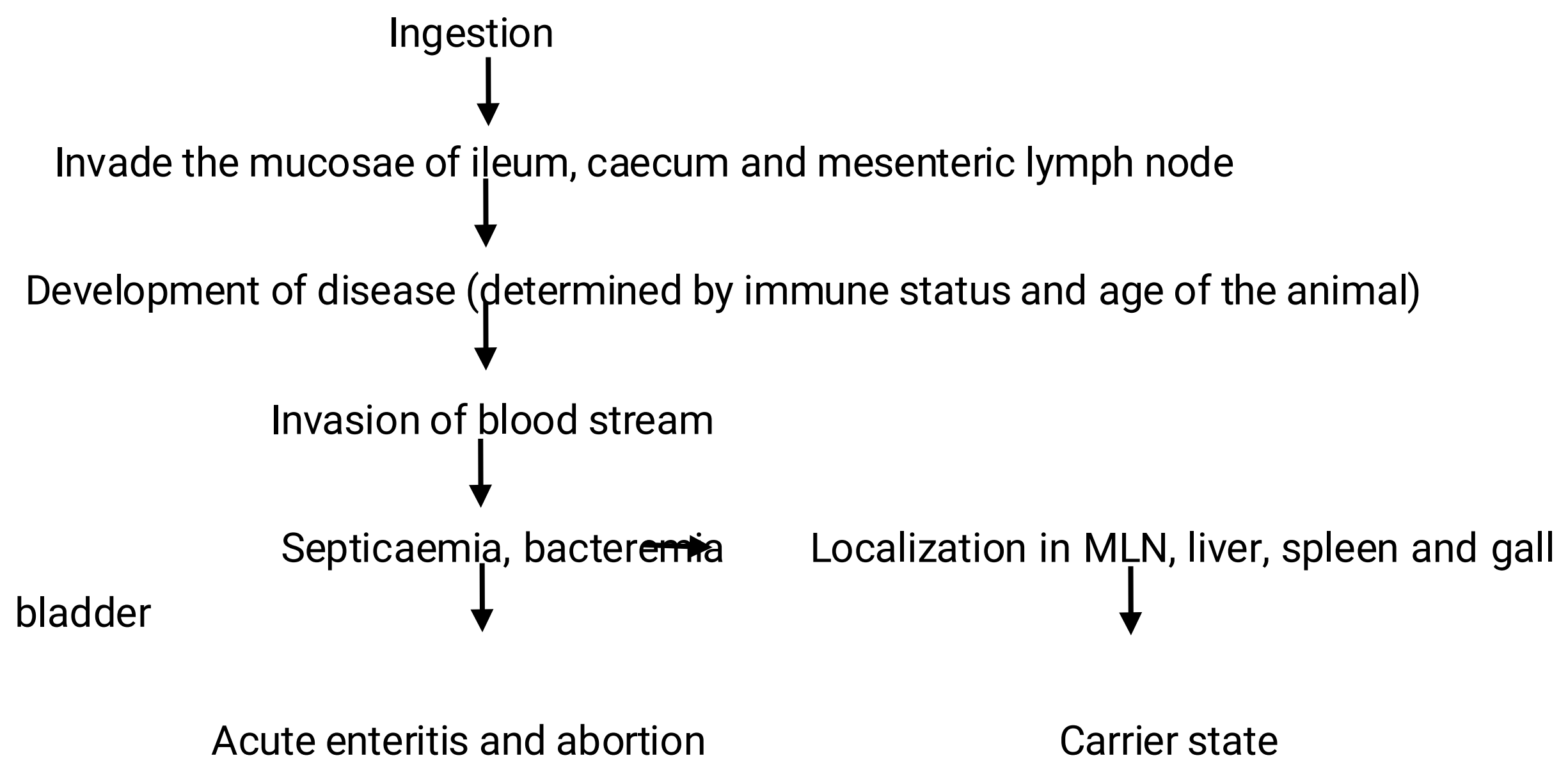
Infection is spread by ingestion through oral route followed by reaching the distal ileum, cecum and from there invade the mucosa, where they replicate in submucosa and peyers patches

In young animals and adults those have decreased immunity the organism spread beyond mesenteric lymphnode and establishes the infection in reticuloendothelium cells of liver followed by rapid entry into the blood circulation.

Invasion of the organism in the blood stream causes pyrexia in 24-48 hours and acute infection in 5-9days.

It is a rapidly fatal condition. If invasion is sufficient to cause only a bacteremia, acute enteritis may develop and abortion is a final sequelae in sheep and cattle. Many animals may survive out of infection and the organism localized in the mesenteric lymph node, liver, spleen, particularly in gall bladder.

In chronic cases, organism shed intermittently from gall bladder and causes the foci of infection in the intestinal wall into the feces and occasionally into milk.



Note

Cattle and sheep

Salmonella dublin causes **abortion** at mid- late gestation (124-270 days of pregnancy in cattle) and **terminal dry gangrene** due to endarteritis of the limbs, ears and tail, osteitis and polyarthritis in calves

Pig

Salmonella choleraesuis- causes pneumonia and enterocolitis

Clinical finding

Septicemic form

New born foals, calves, young pigs upto 4 months old are highly susceptible

Prominent depression

Dullness and prostration

High fever 40.5-42 ° C and death occurs in 24-48hours

Acute enteric form

Adult animals are highly affected

High fever, fluid diarrhea, sometimes dysentery and occasional tenesmus

Feces with putrid odor and contains mucus, blood, fibrinous casts, complete tubular casts of intestine

Intestinal mucosa in casts or sheets

Complete anorexia and increase thirst

Increase heart and respiration rate and congestion noticed

Pregnant animals abort

Severe dehydration, toxaemia, loss of weight, weakness, recumbent, and death in 2-5days

Chronic enteric form

Common in pig farm following a severe outbreak

Occasional in cattle and horses

Calves- intermittent or persistent diarrhoea

Occasional passage of spots of blood, mucus and firm fibrous casts present

Loss of weight and emaciation.

Cattle (*Salmonella dublin*, *Salmonella typhimurium*)

Septicemic form

Depression, toxaemia, fever, dyspnoea and weakness

Nervous signs: incoordination and nystagmus

Diarrhoea and dysentery

Acute enteric form

Abortion in pregnant cows

Polyarthritis in calves

Dysentery with large blood clots

Complete agalactiae, abdominal pain, with kicking of abdomen, rolling, crackling, groaning and looking at the flanks are more common features. Rectal examination at this stage creates severe distress.

Chronic enteric form

Inappetance, decrease weight gain, unthriftiness, acute enteritis

Terminal dry gangrene of the extremities in calves is characterized by lameness and swelling of the hindlimbs below the fetlock, separation of skin above the fetlock. The distal portion of limbs are cool, not painful, skin is dry and moist

The separation of metatarsus from phalanges

The tip of the ears may be indurated and deviated medially

Pigs

Dark purple discoloration of skin in abdomen and ear

Subcutaneous petechial hemorrhages

Nervous signs: Tremor, weakness, paralysis, convulsion and fatality 100%.

In four weeks old, meningitis, prostration, clonic convulsion and encephalitis.

Enteritis and pneumonia.

Horses

Sporadic occurrence.

Asymptomatic shedding of *S.typhimurium* in faeces intermittently or continuously for 4-6 days

Subacute enteric form in adult horses

The disease is endemic, with fever, depression, but without severe diarrhea

The consistency faeces is soft

Severe, acute fulminating enteritis with diarrhea, fever, dehydration and neutropenia

Abdominal pain

Necropsy finding

Lesions in the brain, neuro encephalitis, polyarthritis and fatal meningo encephalitis

Haemorrhage in gastric mucosa and edema

Necrosis and ulcers in intestinal mucosa

Caecum zebra markings produced

Liver is enlarged with necrotic foci

Spleen is enlarged and become pulpy

Mesenteric lymph glands are haemorrhagic

Adhesion and abscess in pleura and lungs

Pericardial and sub pericardial haemorrhage

Pig

Petechial haemorrhage in subserosa and submucosa are give the kidney turkey egg apperance

Diagnosis

Based on clinical signs and necropsy findings.

Blood culture

Isolation and identification of organism through culture, and FAT

Most of the serological tests are lacking both sensitivity and specificity.

Bacterial culture and detection of antigen

Antigen Capture ELISA

PCR and DNA probe test

Sample collection

Feces, blood, serum, milk, body fluid and tissues

Differential diagnosis

Coccidiosis

Pasteurellosis

Liver fluke infestation

Poisoning

Colibacillosis

Paratuberculosis

Treatment

Ruminants

Ceftiofur 5mg/kg 1/m 24 hours is effective. Concentration should be maintained above MICs

Amoxicillin, ampicillin, Trimethoprim, sulphadiazine, sulfadimidine and Framycetin and Nitrofurazone can be used

Horses

Gentamycin 3mg/kg. Combined with ampicillin 20mg/kg bwt I/V 8-12 hours intervals.

Trimethoprim and sulfonamide twice daily i/v combine dose 30mg/kg bwt
Combined sulfadoxin, sulfadiazine and sulfamethaxazole can be used in combined dose

Foals: Trimethoprim and sulfadoxine

Supportive therapy

5% Sodium bicarbonate solution 5-8 L/400 kg bwt I/V over a 2 hours period

Potassium chloride 30g in 8 L water twice daily

Prevention

To prevent the human infection the following measures may be adopted

Effective pasteurization of milk is to be made

Proper refrigeration of food is required

Sterilisation of animal food liable to contamination

High standards of abattoir hygiene required

Proper handling of fish has to be made

Control

Calving should be kept in a hygienic place

Calves should be allowed to have colostrum following their birth

Calves should not be exposed to adverse weather (too hot or too cold environment)

Overcrowding of animals should be avoided

Animals should be provided with pure wholesome water

Animal should not be allowed to drink water from sewage or stagnant water

Other susceptible animals like horse, sheep pigs, birds should be placed away from cattle shed

Affected animals should be brought under immediate treatment

Identify the carrier animals

Prophylactic use of antimicrobial- Oxytetracycline – feed/drinking water

Rigorous disinfection of building-5% formalin- 2 spraying one month apart

Heat treatment of feed
Proper disposal of dead animals
Restrict the movement of animals

Campylobacteriosis

It is a one of the important water borne and food borne disease of mammals and birds, characterized by acute gastroenteritis and abortion.

Etiology

Caused by *campylobacter* sp

It is a gram negative microaerobic, spiral or curved rods exhibit a characteristic corkscrew darting motility

It is an fastidious organism requires enriched media and special supplement for growth

The organism is susceptible to environmental influences like heat and drying

Campylobacter sp in different animal species

Livestock- *Campylobacter jejuni subsp jejuni* causes enteritis and abortion

Cattle and sheep- *Campylobacter fetus subsp fetus* causes sporadic abortion

Cattle - *Campylobacter fetus subsp venerealis*- causes early embryonic mortality and temporary infertility

Pig-*Campylobacter coli* and *Campylobacter mucosalis* causes enteritis

Dog – *Campylobacter upsaliensis* and *Campylobacter helveticus* causes enteritis

Poultry – *Campylobacter jejuni jejuni* causes enteritis

Epidemiology

Worldwide in distribution

Sources of infection

Feces, meat, milk, water, urine free living migratory birds

Transmission

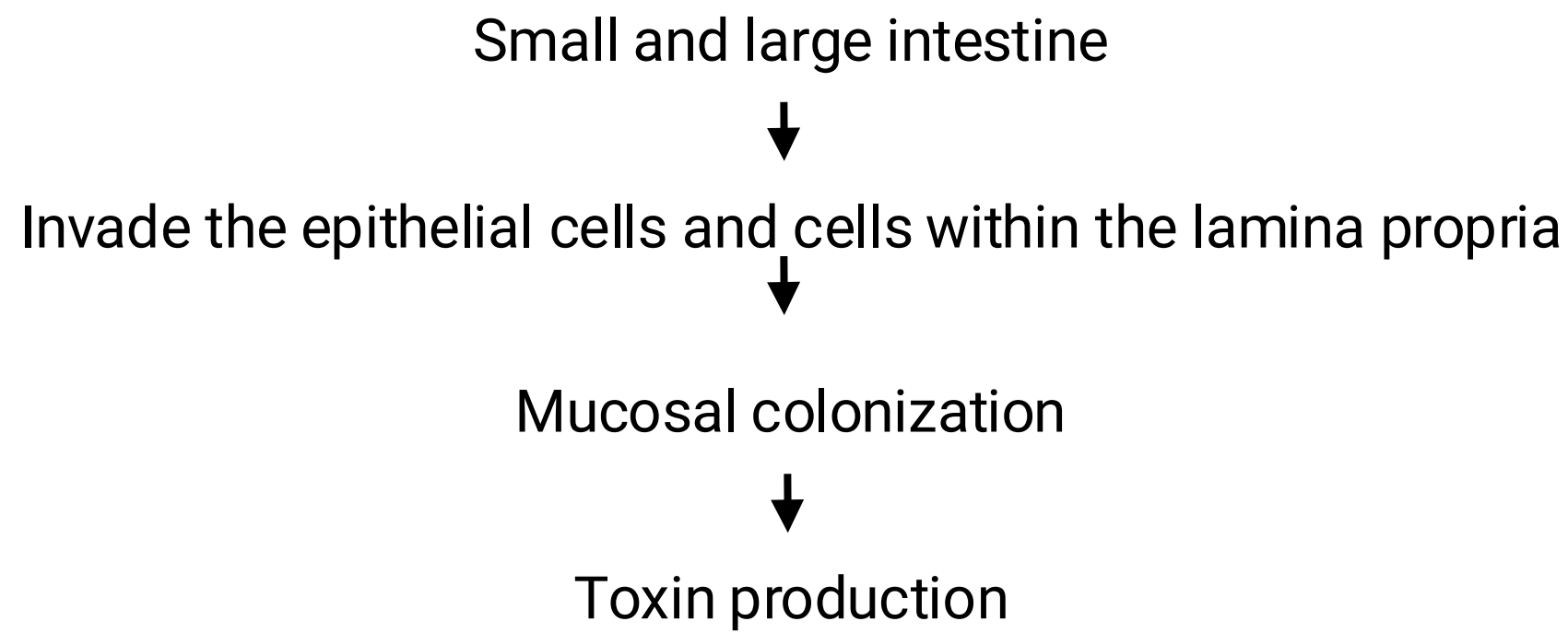
Cattle, sheep, pig and poultry- faeco oral route is the main mode of transmission

Dog, cat and human- ingestion of infected under cooked meat and unpasteurized milk

Pathogenesis

Ingestion of *campylobacter* sp in contaminated food and water





Clinical signs

Cattle and sheep

Enteritis and abortion

Localization of organism occur in gall bladder, large and small intestine and liver

Faecal shed in cattle leads to contamination of milk and beef

Pig

Colitis in weaning aged pigs

Anorexia, fever and diarrhea for 1-5 days

Dog and cat

Abdominal pain, fever, diarrhea, blood and inflammatory cells in faeces

Diarrhea is usually acute but can be recurrent

Diarrhea is watery to bloody with mucus, sometimes bile stained occur in dog and cats less than 6 months age

Campylobacter jejuni jejuni causes abortion in late pregnancy with odorless haemorrhagic vaginal discharge

Poultry

Hepatitis, loss of appetite, weight loss and yellow diarrhea

Necropsy finding

Typhlocolitis

Enlargement of ileoceccocolic and mesenteric lymph node

Abscess in intestinal crypt epithelium

Sample collection

Fresh faecal sample, meat and milk

Diagnosis

Based on clinical signs and postmortem findings

Isolation of organism in selective media under microaerophilic condition (Oxygen 6%, Carbon dioxide 10% and Nitrogen 84%)

Smear from culture or clinical specimen is stained with diluted carbal fuchsin for 4 minutes.

Polymerase chain reaction

Treatment

Gentamicin , tetracycline, erythromycin, cephalosporin and fluroquinolones are effective against *campylobacter* sp

Selection of antibiotics based on culture and sensitivity

Control

Maintenance of hygiene in slaughter house

Ingestion of properly cooked meat

Pasteurization of milk

Shielding of water tank to avoid contamination of faeces from migratory birds

Genital campylobacteriosis

Bovine genital campylobacteriosis is a venereal disease of cattle characterized by early embryonic death, infertility, a protracted calving season and occasionally abortion.

Etiology

Campylobacter fetus subsp venerealis and *campylobacter fetus subsp fetus*

It is a gram negative microaerobic, spiral or curved rods exhibit a characteristic corkscrew darting motility

It is an fastidious organism requires enriched media and special supplement for growth

The organism is susceptible to environmental influences like heat and drying

Epidemiology

Worldwide distribution

Outbreaks occur following introduction of infected bulls or cows into the breeding farms and the presence of increased number of estrus cows in the herd

Predisposing factors

Contaminated soil and housing environment

Introduction of infected carriers

Transport

Source of infection

Both young and older bulls have been identified as carriers for upto 18 weeks post infection

Older bulls are remaining as permanent carrier that has been attributed to the highest proliferation of *C. fetus* in the major site of epithelial crypts of the penis, in contrast, *C. fetus* concentration is very less in preputial fluid

Transmission

Venereal transmission-spread of infection occurs during coitus from infected cow to bull and vice versa.

The disease also transmitted through use of contaminated instruments, bedding, artificial insemination using contaminated semen

Spontaneous clearance occur in young bulls <3-4 years of age

In bulls >3-4 yr old, the deeper preputial crypt epithelium microaerophilic condition to establish chronic infection

Host affected

Cattle is highly susceptible for venereal campylobacteriosis as compared to sheep and goats

Sheep and goats affected by reproductive and gastrointestinal form of campylobacteriosis

Pathogenesis

Naturally infected bulls transmit *C.fetus subspecies venerealis* from one female to another, **without any quality changes in semen and breeding ability**

The organism establishes **infection in the prepuce**

In cows, in **cervicovaginal area**, during estrus, no uterine infection develop, due to presence of abundant neutrophils in the uterus at the time of estrus which prevent colonization of the organism within the uterus

But in the later stage of the estrus the organism invade and establish the infection in the uterus within 10 days following mating, the organism establishes infection in the mucous membrane of the vagina and cervix which leads to **mild vaginitis, cervicitis and endometritis**

The pathogenic nature of the organism increased because of a protective surface antigen or anti-phagocytic macrocapsule comprising a paracrystalline array of (surface array protein) protein which stimulates IgA, IgG, and IgM, following colonization of campylobacter

IgA found primarily in the cervico-vaginal mucus, immobilize *C. fetus subsp. venerealis* and **IgG in uterus** of convalescent animals which leads to carrier status

After conception if cows infected it may carry the organism. As there is insufficient stimulation of antigen the organism is restricted to the epithelial surface

Clinical signs

Bovine campylobacteriosis

Cows are systemically normal

Variable degree of mucopurulent discharge endometritis that causes early embryonic death, prolonged luteal phase, irregular estrous cycles, repeat breeding and protracted calving period

Reddening of cervix and extension of mild mucopurulent exudate from uterus into cervix and vagina occurs

In contrast, infected heifers which have natural immunity quickly conceive, but remain as carrier for months

Rarely causes abortion at mid gestation

Bulls are asymptomatic and produce normal semen

Sheep and goat

Abortion in late term pregnancy

Recovered ewes are immune for at least 3 years and flock fertility in subsequent breeding is good

Sample collection

Preputial washing and cervical mucus and serum

Diagnosis

Based on clinical signs

Isolation and identification of organism

Serum agglutination, cervical mucus agglutination, fluorescent antibody and culture tests have been used, however each test has certain limitations

Serum agglutination (SA) test

It is not a reliable test and antibodies are rarely found in the blood stream

Cervical mucus agglutination test: (CMAT)

It is a convenient and **accurate method for herd level bovine campylobacteriosis**, though this test is widely used there is a chance for false positive (Metestrual bleeding or rough manipulation of the vagina during collection) and false negative (when cervical mucus dilutes the agglutinins) results.

It takes **60 days following infection for the mucus to become positive on test.**

In individual animals the stage of estrus cycle and the time of exposure to the disease must be known to identify the infection. Around ten (selected) to twenty animals are enough to diagnose at farm level

Fluorescent antibody test is used to detect antibody

Differential diagnosis

Infectious bovine rhinotracheitis

Brucellosis

Trichomoniasis

Leptospirosis

Listeriosis

Treatment

Bull -Streptomycin @20mg/kg b.wt S/c route 1-2 dose, together with 5g streptomycin in an oil based suspension applied on the penis for 3 consecutive days

Cows are not generally treated for genital campylobacteriosis

Prevention

The use of artificial insemination is successful in dairy herds

Vaccination is important for prevention of the disease. Vaccination of breeding herd twice prior to the breeding season to increase the high serum IgG titre is imperative

Single vaccination at 2 months prior to breeding is a failure to mount good immune status. Because serum antibody titre decreased rapidly after vaccination and therefore annual booster vaccine is recommended

Both males and females require vaccination

Control

Proper disinfection of AI equipment with 1% sodium hypochlorite, 70% ethanol, 2% glutaraldehyde

Personnel hygiene

Pullorum disease

(**Syn:** Bacillary White Diarrhoea, Fatal septicaemia)

It is an acute septicaemic disease affecting primarily chicken and turkeys

Etiology

The disease is caused by *Salmonella enterica* sub sp *enterica* serovar *pullorum* of the family Enterobacteriaceae

It is a highly host specific organism

It is a gram negative, facultative anaerobe, non motile and rod shaped organism

Epidemiology

Source of infection

Infected adult will become a carrier

The organism persists in the ovaries and excreted in the ova

Fluff from chicks, heavy, dry and the bacteria rapidly spread in the incubator

Transmission

Vertical transmission/Transovarian transmission from parents to chicks. The egg transmission may result from contamination of ovum following ovulation, but localization of *S.pullorum* in the ovules before ovulation is likely and probably constitute chief mode of transmission

Horizontal transmission through direct contact with infected birds, cannibalism, through wounds on the skin

Mechanical transmission occurs through, **rodents**, wild birds, mammals and flies

Rapid lateral spread from chick to hatcheries vice versa

Host affected

All avian species are susceptible but predominantly affects chickens, turkeys, and pheasants

All chickens less than 3 weeks are mostly susceptible

The heavy breeds (brown egg layers) Rhode Island Red (RIR) are highly susceptible than light breed leg horn birds

Pathogenesis

It is a systemic infection in both domestic and wild species are affected

S.pullorum has tropism for reproductive tract and developing eggs following infection with particularly high numbers in the oviduct at the point of lay

Morbidity is 10-80%; mortality is increased in stressed or immuno compromised flocks and may be up to 100%

Clinical Signs

Chicks and Poults

Initially there is an excessive number of dead in shell or dead chicks in the incubator, shortly after hatching has been noticed

Somnolence, weakness, depression, loss of appetite, tendency to huddle near the heat source and adherence of white chalky material on the vent and mortality of 100% in severe condition

In some cases clinical signs are not observed until 5-10 days after hatching

Laboured breathing or gasping may be observed as a result of extensive involvement of lung

Survivors are shown retarded growth and small in size, poorly feathered. These birds may not mature into vigorous or well developed laying or breeding birds and frequently asymptomatic carriers with localized infection of the ovary. Some of the egg laid by such hens hatch and produce infected progeny

Swelling of the hock joint due to synovitis

Growing and mature birds

Infected birds may not exhibit any clinical signs and cannot be detected by physical appearance

In some cases anorexia, depression, diarrhoea, dehydration, drop in egg production, decreased fertility and reduced hatchability may be observed

Turkeys

Thirst, inappetence, listlessness, a tendency to segregate from healthy birds, and **green to greenish yellow diarrhoea**. However death may occur without prior clinical symptoms.

Necropsy finding

Acute form in chicks

Chicks die of acute phase show no specific lesions.

Peritonitis with an inflamed **un-absorbed yolk sac**

Gray nodules in the liver, spleen, lungs, heart, gizzard and intestine

Firm cheesy material in the caeca (**caecal cores**) due to inflammation of caecum (typhilitis)

Sub-acute form in growers and mature birds

Inflammation and enlargement of hock joints due to presence of excess lemon or orange colored gelatinous material around the joints

Nodular pericarditis, fibrinous peritonitis or hemorrhagic, atrophic and regressing ovarian follicles with caseous contents

Diagnosis

Based on clinical signs and necropsy findings

Isolation and identification of organism by culture and staining

Serological test (Rapid whole blood agglutination, rapid serum agglutination (RST), tube agglutination, micro-agglutination test and ELISA) are used

Serological tests are best applied as a flock test as results for individual birds will vary according to the stage of infection. It is therefore necessary to take sufficient individual samples to determine infection in the flock. The number of samples will depend on the expected prevalence and level of confidence desired

The flock was screened by rapid whole blood agglutination test and positive reactions confirmed by the more specific tube agglutination test

If the test is to be used for detecting individual infected birds for culling, it should be repeated at least twice and preferably until the whole flock has given at least two negative test

Ideally serum samples tested by any method should be tested within 72 hours of collection as nonspecific reactions may increase in older samples

Molecular diagnosis by PCR

Identification of strain by restriction fragment length polymorphism (RFLP)

Sample collection

Live birds-cloacal swab, whole blood and serum

Dead birds - organs like ovaries/oviduct are collected at post mortem.

Differential diagnosis

Fowl typhoid

Fowl cholera

New Castle disease

Treatment

Sulfamethazine, sulfamerazine, and sulfaquinoxaline have been used successfully to reduce the death losses in chicks and poults

Furazolidone, a chemical derivative of furfural, chicks losses in acute outbreaks.

It may be incorporated in the feed at a low level for continuous feeding to prevent outbreaks and spread of the infection

Prevention

Establish breeding flocks free of *S. pullorum* by routine serological screening and elimination of carriers.

Serological evidence of infection should be confirmed by bacteriological examination of one or more reactors.

Chicks and poults should be obtained from sources free of pullorum disease and should not be mixed with other poultry or confined birds

Chicks and poults should be placed in clean and sanitized environment

Chicks and poults should receive pelletized, crumbled feed to minimize the introduction of *S.pullorum*

The ingredients used for preparation of feed should be free of salmonellae

Maintenance of high biosecurity measures

Poultry house should be bird proof

Rats, rabbits, dog, cat and pest may be carrier of salmonellae, that should be controlled

Provision of chlorinated drinking water.

Proper disposal of dead birds.

Vaccines are not normally used because of the interference in the serological testing and elimination of carriers.

Control

Sanitation and disinfection are important in hatching and brooding sections.

The organism is easily destroyed by disinfectant, heat, and direct sunlight.

After each hatch, incubator egg trays and the other readily removable parts should be taken out, washed free of dirt and foreign matter, and then scrubbed with a 2-percent lye solution or other acceptable disinfectant, such as a 3-percent solution of cresylic acid. Lye and caustic soda are good disinfectants.

Precautions are necessary while using these disinfectants.

Incubator rooms should be cleaned brushed and washed down and cobwebs from ceilings and walls and from entire farm house.

Formaldehyde gas is a good disinfectant for incubators during incubation or following the removal of the hatch.

Following removal of reactors to the blood test, poultry houses and equipment should be cleaned and disinfected

Fowl typhoid

Fowl typhoid is an acute septicaemic disease of chicken and turkeys. Which is more often observed in the late growing period and in mature stock. The disease may be severe in commercial laying flocks often characterised by rapid spread with high morbidity and mortality.

Etiology

The disease is caused by *Salmonella enterica* sub sp *enterica* serovar *gallinarum* of the family Enterobacteriaceae

It is a gram negative, non-capsulated, non-sporogenic, non-motile, rods shaped and flagellated organism

Epidemiology

Source of the infection

Recovered birds are acts as carriers and excrete the organism in faeces

Transmission

Vertical transmission/Transovarian transmission from parents to chicks. Horizontal transmission through direct or indirect contact with an infected bird, cannibalism, through wounds on the skin and ingestion of contaminated feed or water

Mechanical transmission occurs through, **rodents**, wild birds, mammals, flies and fomites

Red mites may be involved in the transmission of disease and persistence in poultry houses

Host affected

Adult and growing birds are mostly affected with serious problems

It affects other avian species like turkey, ducks, pheasants, guinea fowl, peafowl, grouse, quail, game birds, sparrows, parrots, canaries and bullfinches

Clinical signs

Acute form in young chicks

Initially there is an excessive number of dead in shell or dead chicks in the incubator, shortly after hatching has been noticed

Somnolence, weakness, depression, loss of appetite, tendency to huddle near the heat source and adherence of white chalky material on the vent and mortality of 100% in severe condition

Laboured breathing or gasping may be observed as a result of extensive involvement of lung

Growing and mature birds

Sudden drop in feed consumption, dropsy, ruffled feathers, pale and shrunken comb

Decreased egg production, reduced fertility and hatchability

Death may occur within 4 days of exposure

Necropsy finding

Acute

Liver is usually enlarged, dark and friable with a distinctive **coppery bronze sheen that may only develop after exposure to air**

The bone marrow is also often dark brown

Chronic

Pericarditis, accumulation of turbid yellow fluid in the pericardial sac, fibrin attachment at the surface of the heart is characteristic for chronic fowl typhoid. Young birds have discrete, necrotic foci in the lung and gizzard and in the testicles of cockrels noticed

Turkeys

Liver exhibits **bronze to mahogany color streak**, necrosis of heart, greyish lungs are pathognomic lesions of this disease

Crop usually contains food which suggests paralysis of digestive tract

Mucous membrane of proventriculus sloughs off readily. Hemorrhagic enteritis (duodenum) intestinal ulceration is remarkable

Diagnosis

Based on clinical signs and necropsy findings

Isolation and identification of organism by culture and staining

Serological test (Rapid whole blood agglutination, rapid serum agglutination (RST), tube agglutination, micro-agglutination test and ELISA)

Serological tests are best applied as a flock test as results for individual birds will vary according to the stage of infection. It is therefore necessary to take sufficient individual samples to determine infection in the flock. The number of samples will depend on the expected prevalence and level of confidence desired

The flock was screened by rapid slide agglutination test and positive reactions confirmed by the more specific tube agglutination test

If the test is to be used for detecting individual infected birds for culling, it should be repeated at least twice and preferably until the whole flock has given at least two negative tests

Ideally serum samples tested by any method should be tested within 72 hours of collection as nonspecific reactions may increase in older samples.

Sample collection

Live birds - Blood, serum and fecal sample

Dead birds- liver, gall-bladder, caecal tonsils as well as ovary and oviduct should be collected.

Environmental sample - Eggs, embryos, fecal droppings and hatchery debris, fluff, dust, broken eggshells and chick box linings also be collected

Differential diagnosis

Typhoid

Pasteurellosis

Colibacillosis

Viral diarrhoea

Parasitic infection

Mycoplasmosis

Treatment

Furazolidone in feed for 10 days at the level of 0.04%.

The antibiotics alleviate the symptoms and lesions, but will not eliminate carrier status.

Prevention

Establish breeding flocks free of *S.gallinarum* by routine serological screening and elimination of carriers

Serological evidence of infection should be confirmed by bacteriological examination of one or more reactors

Chicks and poults should be obtained from sources free of fowl typhoid and should not mix with other poultry or confined birds

Chicks and poults should be placed in clean and sanitized environment

Chicks and poults should receive pelletized, crumbled feed to minimize the introduction of *S. Gallinarum*

The ingredients used for preparation of feed should be free of salmonellae

Maintenance of high biosecurity measures

Poultry house should be bird proof

Rats, rabbits, dog, cat and pest may be carrier of salmonellae, that should be controlled

Provision of chlorinated drinking water

Proper disposal of dead birds is essential

Vaccines are not normally used because of the interference in the serological testing and elimination of carriers

Live attenuated rough Houghton strain of 9R of *S.gallinarum* and Freund's complete adjuvanted vaccine can be given subcutaneously between 10-18 weeks of age found to give long lasting solid immunity and protection for up to 32 weeks

Some vaccinated birds can develop hepatic and splenic lesions without mortality. Few birds develop antibody, detectable by whole blood agglutination test

Control

Sanitation and disinfection are important in hatching and brooding sections

Easily destroyed by disinfectant, heat, and direct sunlight

After each hatch, incubator egg trays and the other readily removable parts should be taken out, washed free of dirt and foreign matter, and then scrubbed with a 2-percent lye solution or other acceptable disinfectant, such as a 3-percent solution of cresylic acid. Lye and caustic soda are good disinfectants.

Precautions are necessary while using these disinfectants

Incubator rooms should be cleaned brushed and washed down and cobwebs from ceilings and walls and from entire farm house

Formaldehyde gas is a good disinfectant for incubators during incubation or following the removal of the hatch

Following removal of reactors to the blood test, poultry houses and equipment should be cleaned and disinfected

Fowl cholera

(Syn: Avian pasteurellosis)

It is a peracute and acute highly contagious septicaemic diseases of poultry. Acutely, it causes elevated mortality. Chronically, it causes lameness, swollen wattles (in chickens), pneumonia (in turkeys), and torticollis, but it can also be asymptomatic.

Etiology

It is *Pasteurella multocida*, family *Pasteurellaceae*

It is a gram negative, capsular non motile, rods, exhibit pleomorphism on frequent subculture

It shows non-spore forming, coccobacilli rod shaped organism and bipolar staining in blood smear

P. multocida can be subgrouped by capsule serogroup antigens into five capsular types (A, B, C, D, and F) and into 16 somatic serotypes

P. multocida type A- most frequently isolated serovars in fowl cholera

Pasteurella multocida contains three sub species

- ☒ ***Pasteurella multocida* sub sp *multocida*-most common causes fowl cholera**

- ☒ *Pasteurella multocida* sub sp *septica* and *Pasteurella multocida* sub sp *gallicida* causes fowl cholera-like diseases

Duck-*Rimerella anatipestifer*

highly susceptible to drying and common disinfectants, 3% hydrogen peroxide is the most promising disinfectant.

Epidemiology

Pandemic-World wide distribution

Notifiable disease in many countries

Morbidity and mortality may be up to 100%

Predisposing factor

High density

Concurrent infection with other respiratory disease

Source of infection

Chronically infected and asymptomatic carrier birds are major source of infection

High carrier rate of *P. multocida* sub sp. *multocida* and *septica* exist in apparently healthy poultry flocks (including chickens and ducks) many birds carried *P. multocida* on the cloacal mucosae

Conjunctival, oral and nasal discharge of infected birds, faeces and contaminated soil

Infected carrier birds

Rats, cats, and possibly pigs act as reservoirs

Transmission

Ingestion of contaminated feed and water

By contact with infected materials

Host affected

All avian species are susceptible

Ducks, geese, water fowl and turkeys are more susceptible than chicken

Mostly growers and adult birds are being affected than young ones

Caged chickens are less affected

Pathogenesis

The disease can range from **acute septicaemia** to chronic and localised infections

Following an upper respiratory tract infection *P.multocida* may subsequently spread to the Lungs and multiply before entering the bloodstream. Once in the blood stream, *P.multocida* either multiply rapidly Or localized in the liver and spleen where initial Multiplication occurs before a massive bacteraemia. Death is presumed to be due to the effects of endotoxin as signs of acute fowl cholera

Many of the lesions are related to vascular disturbances

Hyperemia of blood vessels especially evident in the abdominal viscera

Clinical signs

Peracute- no clinical signs and found dead

Acute

The incubation period is 5-8 days.

Affected birds are dull, depressed, and anorectic

Mucous discharge from the mouth

Whitish or greenish foetid diarrhoea

Increased respiratory rate

Cyanosis just prior to the death

Chronic

In chronic fowl cholera, signs and lesions are generally related to localized infections results in swollen wattle, sternal bursae, joints, tendon sheaths, and footpads due to accumulation of fibrinosuppurative exudates,

There may be lameness, as well as exudative conjunctivitis and pharyngitis, tracheal rales and dyspnea

Torticollis may result when the meninges, middle ear, or cranial bones are infected.

Necropsy findings

Acute

Congestion of carcass

Multiple petechiation in all visceral organs

Multiple pin-point necrotic foci in the liver

Free yolk found in the body cavity in layer birds

Increased amounts of peritoneal and pericardial fluids

Liver may be swollen with multiple, small, necrotic foci

Petechial and ecchymotic hemorrhages are common in sub epicardial and serosal surface

Subacute

Turkey

Edematous lung and focal perihepatitis and yolk peritonitis

Consolidation of lung with cooked appearance

Purulent pneumonia

Cellulitis of face and wattles

Purulent arthritis

Chronic

Caseous arthritis of hock and foot joints

Swelling and induration of both wattles

Caseous exudate in the middle ears

Sample collection

Live birds- Swab from nasal cleft, cloaca and pharynx

Dead birds- Smear from heart blood and heart blood swab, Impression smear from liver, heart, lung and spleen

Diagnosis

- Based on clinical signs and necropsy finding
- Identification of bi-polar organism in blood smear by new methylene blue staining, Leishman staining and liver imprints by Wright's or Giemsa stain
- Isolation and identification of the organism by culturing in blood agar, dextrose starch agar, trypticase soy agar, which is improved by 5% heat inactivated serum
- Mice inoculation test- Intra peritoneal inoculation of suspected broth culture 0.2-0.5 ml, if *P. Multocida* is present, the mice will die within 24-48 hrs

- Immunofluorescent and in situ hybridization technique
- Molecular diagnosis by PCR

Differential diagnosis

- Pullorum disease
- Fowl typhoid
- Infectious coryza
- Avian influenza
- New castle disease
- Infectious laryngotracheitis

Treatment

Peracute case- no response to treatment

Administration of **sulphaquinoxaline in food / water for turkeys and tetracycline for all avian species are effective**

If tetracyclines are used, the **addition of citric acid to the drinking water** and reduction of calcium use will increase the efficacy of the antibiotic

Recurrence of infection is common after treatment

Long-term or periodic use of antibiotic is essential to control the recurrence of infection

Sulfamethazine doses are 0.5% to 1% in food or 0.1% in the water for at least 5 days

Turkeys: Penicillin was effective for sulfa resistant infections

In ducks: Combination of streptomycin, dihydro streptomycin can be effective

Use of sodium salicylate in drinking water for 3 days may provide a good recovery from infection

Prevention and control

Good management practices, including a high level of biosecurity, are essential to prevention

Rodents, wild birds, pets, and other animals that may be carriers of *P multocida* must be excluded from poultry houses.

Both attenuated live vaccines and adjuvanted bacterins are available to aid in prevention, and it is sensitive to some antibiotics.

Autogenous bacterins are recommended, when polyvalent bacterins (contain serotype 1, 3 and 4) are found to be ineffective

Attenuated live vaccines are administered in drinking water to turkeys and by wing-web inoculation to chickens. These live vaccines can effectively induce immunity against different serotypes of *P multocida*. They are recommended for use in healthy flocks only.

Avian mycoplasmosis

It is a respiratory disease of chicken characterized by nasal discharge, sneezing, moist rales, open mouth breathing, reduced hatchability, sub optimal egg production and high condemnation in broiler due to poor carcass quality there by impose economic loss.

Etiology

The disease is caused by *Mycoplasma* spp. belongs to family Mycoplasmataceae, order Mycoplasmatale. It is a small bacterium, lack of cell wall, but surrounded by a triple layered plasma membrane

There are 22 species of mycoplasma recovered from avian species, but only four of them are found affecting domestic poultry

Mycoplasma gallisepticum - causes **chronic respiratory disease in chicken** and **infectious sinusitis in turkeys**

Mycoplasma synoviae- causes **infectious synovitis** in chicken and turkeys

Mycoplasma meleagridis - infect only turkeys

Mycoplasma iowae- mostly infect turkeys

Epidemiology

Predisposing factors

Live virus vaccination, cold climate, crowding and stress may initiate the spread of infection.

The disease usually occurs in association with other pathogens such as New castle disease virus, Infectious bronchitis virus, *E.coli* and *Haemophilus paragallinarum*

Source of infection

In intensive method of rearing, the birds establish respiratory disease complex disease in association with other pathogens and predisposing factors. The disease is present in poultry species for many years

Infected birds become carrier

Transmission

Vertical transmissions (in ovo) from an infected breeder flock to their progeny. The highest rate of egg transmission was found during the acute phase of the disease

Horizontal transmission through direct or indirect contact with infected carrier or contaminated debris

Flock to flock transmission occurs readily by direct or indirect contact from the movement of birds, people or fomites from infected to susceptible flocks

Air borne transmission may also occur

Host affected

Chicken and turkey are the natural host for this infection

However, turkey and very young birds are more susceptible to the disease

Intensively reared birds mostly infected

Infection also occur in pigeons, quails, peafowl, partridge, ducks, geese, psittacine birds, and wild house finches

Pathogenesis

The mycoplasma organism have integral membrane surface protein (adhesins) that attached to receptors on host cells, and initial colonization occur on nasal passage, trachea, epithelium of conjunctiva, sinuses and joints in *M. synoviae* infection. However in severe disease, the infection may occur in bronchi, air sacs and occasionally lungs. Once infected, birds may remain carriers for life. There is marked interaction between viruses of new castle disease, infectious bronchitis, *E. coli* and *M. gallisepticum* in the pathogenesis.

Clinical signs

Chickens

Respiratory distress with difficulty breathing, tracheal rales, nasal discharge, sneezing, coughing, reduced feed consumption and loss of body weight. In laying flocks egg production declines, but usually is maintained at a lower level. Male birds may have the most pronounced clinical signs and the disease is often more severe during winter.

In broiler flocks, **most of the outbreaks occur after 4th weeks of age.** The clinical signs are frequently more marked than those of mature flocks.

The severe outbreaks with high morbidity and mortality observed in broilers frequently due to concurrent infection with other pathogens.

Open mouth breathing increase during concurrent infection

Pipped embryo or dead in shells and poor quality chicks

Swelling of hock joint in *M. synoviae* infection in turkey

Turkeys

The clinical signs are **more severe than chickens**

Sinusitis, respiratory distress, depression, decreased feed intake, weight loss and decreased egg production

Nasal discharge and conjunctivitis with frothiness of eye is more severe in turkey than in chickens

In turkeys sinusitis with swelling of one or both of infraorbital sinuses is noticed

Complete closure of eyes

Nasal exudate often join with sinusitis

Soiling of wing feathers with nasal discharge

Swelling of infraorbital sinuses rare in chicks

Reduced feed efficiency and weight gain

Tracheal rales, coughing and laboured breathing may become evident, if tracheitis or air sacculitis is present

Swelling of hock joint in *M. synoviae* infection in turkey.

Necropsy finding

Accumulation of excess mucous or catarrhal exudate in nares, trachea, lungs and edema of the airsac walls

Dilatation of the infraorbital sinuses especially in turkeys due to the presence of excess mucous and later filled with caseous materials

Concurrent infection with *E.coli*, adhesive pericarditis, fibinous perihepatitis, and airsacculitis is observed in 4-10 weeks old chickens

Tenosynovitis, arthritis (*M.synoviae*), salphingitis with caseous exudate occurs in the oviduct

Diagnosis

Based on clinical signs and necropsy findings

Isolation and identification of organism by culture

Rapid serum agglutination test: Serum from infected birds forms a clumps when mixed with antigen given 1-2 min and in turkeys 3 min given

In flocks, in which small numbers of positive reactors to the RSA test persists and from which mycoplasma infection has been confirmed by other serological tests or by cultural examination from live birds

Haemagglutination inhibition test using mycoplasma antigen. The titre of 1:40, 1:80 or above is considered as positive. This is confirmatory test for *Mycoplasma gallisepticum*

Enzyme linked immunosorbent assay: For antibody detection, the test should be repeated in 2-4 weeks time. Frequently they will be found negative

Antigen and antibody detection by immunofluorescence technique using species specific sera

Molecular diagnosis by PCR

Molecular typing method used for differentiation of *M.gallisepticum* strains. **The random amplification of polymorphic DNA (RAPD) technique** is most

commonly used for molecular tracking of spread of infection among flocks and from putative reservoir of infection in commercial poultry

Biological inoculation: Demonstration of the organism or antibody into number of 8-16 weeks old germ free chickens are essential. The inoculated birds are kept in isolation for up to 6 weeks. The serum tested for antibodies and samples tested by culture

Sample collection

Live birds-swab from trachea and choanal cleft

Dead birds- swab from infraorbital sinus, nasal turbinate, choanal cleft, trachea, airsacs and joints All the swab must have been dipped in mycoplasma broth before use

Differential diagnosis

Infectious bronchitis

New Castle disease

Chronic Respiratory disease

Infectious Laryngotracheitis

Treatment

Most strains of *M.gallisepticum* are sensitive to a number of broad spectrum antibiotics.

Tylosin or tetracycline have been commonly used to reduce the egg transmission or as prophylactic treatment to prevent respiratory diseases in broilers and turkeys.

The antibiotics may alleviate the clinical signs and lesions but do not eliminate infection.

Prevention

Natural immunity provides some degree of immunity

Both live and killed bacterin can be used

Live vaccines are relatively low virulence for chickens

The vaccine strains like F, 6/85 and ts-11 strain are commonly used for vaccination.

The live F strain of *M. gallisepticum* is low to moderate virulence and transmissibility and shown to be effective in preventing egg transmission and administered at 14 weeks of age. The F strain is too virulent for use as a vaccine in broiler and turkeys and capable to induce respiratory signs

A temperature sensitive mutant mycoplasma vaccine (ts-11) shown to give good protection against virulent mycoplasma for the respiratory and oviduct for at least 40 weeks after vaccinations

Bacterin for enhancing egg production may be given although the vaccinated birds do not produce more eggs than unaffected birds

Control

Routine serological screening and elimination of infected carriers in breeder flocks and progeny chicks

In international trade, the *M.gallisepticum* infection status of chicks (or embryos) is usually determined by the testing and certification of the breeder flock of origin

The breeder flock should be tested before onset of egg laying and every 60-90 days thereafter by slide plate agglutination test to detect IgM antibodies

The progeny chicks are tested by slide plate agglutination test to detect IgG antibodies. This passes from the yolk to embryo during the last 5-6 days of embryo development and to the chicks for approximately 2 days post hatching, at which times serum level of IgG will be peak

Immersion of egg in antimicrobial solution (egg dipping) at 37°C for 15 -20 min.

Maintaining high level of bio security measures in breeder flocks

Rearing of birds in all in all out system

Mycotoxicosis

Mycotoxins are the poisonous condition resulting from consumption of these toxic chemicals released by fungus that grows in food grains (cereals), hay, straw, pastures, or any other fodder prior to harvest, during storage or after inclusion in finished feed stuff

Mycotoxins can have additive or synergistic interactions with other natural toxins, infectious agents, and nutritional deficiencies

Mycotoxins produced by fungus are stable in nature and their toxic effects are not eliminated following milling or storage

The toxin producing capacity and the amount produced may vary depends on the factors such as strain of fungus, temperature of the atmosphere, where in the fungus growing, moisture, types of fungus, degree of stress on the host plant etc

S. No	Mycotoxin	Fungus	Disease	Pathogenesis	Clinical syndromes
1.	Aflatoxin	<i>Aspergillus flavus</i>	Aflatoxicosis	Hepatic necrosis	Blindness, circling, convulsions, teeth

					grinding, knuckling off etlocks
2.	Citrinin	Penicillium citrinum	Citrinin poisoning	Nephrotoxic	Uremia, depression, recumbency
3.	Ochratoxin	<i>A. Ochraceus</i>	Ochratoxicosis	Nephrotoxic	Diarrhoea, polydipsia, polyurea
4.	Ergot alkaloids	Claviceps purpurea	Ergotism	Arteriolar spasm	Lameness, Gangrene of lower limbs, tails, ear tips, loss of tail switch
5.	Ergot alkaloids	Neotyphodium coenophilum	Fescue summer toxicosis	Depression of prolactin	Hyperthermia, hypersalivation
			Fescue foot	Local vasoconstriction of blood vessels impairs blood supply	loss of tail switch, lower limbs, tails, ear tips
6.	Fumonisin	Fusarium moniliforme	Equine encephalomalacia	Necrosis and haemorrhage in cerebral white matter	Tremor, staggers, unable to swallow, recumbency
7.	Phomopsis	Phomopsis rossiana	Lupinosis	Hepatitis, skeletal myopathy	Jaundice, photosensitive dermatitis
8.	Sporidesmin	Pithomyces chartarum	Facial eczema	Obliterating cholangitis	Photosensitive dermatitis, conjunctivitis, stomatitis and jaundice
9.	Zearalenone	Fusarium roseum, F. culmorum	Vulvovaginitis	Vulvovaginitis	Infertility, anestrus, swelling of vulva, vagina and rectal prolapse
10.	Macrocyclic trichothecenes (Patratoxin, Verrucarins, Roridin)	Stachybotrys atra	Stachybotrytoxicosis	Agranulocytosis, bone marrow suppression, Immunosuppression	Fever, diarrhoea, dysentery, necrotic ulcers and haemorrhages on oral and nasal mucosa, epistaxis, periorbital dermatitis
11.	Non-macrocyclic trichothecenes (T2 toxin, Deoxynivalenol, Diacetoxyscirpenol)	Fusarium sporotrichoides, Fusarium roseum, F. culmorum	Fusariotoxicosis	Immunosuppression and haemorrhagic diseases	Vomiting, feed refusal, diarrhoea, mucosal ulcers and haemorrhage

Aflatoxicosis, ochratoxicosis and trichothecene mycotoxicosis are the most common mycotoxicoses affects commercial poultry

Aflatoxicosis

Aflatoxicosis is caused by aflatoxins, produced by ***Aspergillus flavus***, ***Aspergillus parasiticus*** and *Aspergillus nominus*

Aspergillus sp are ubiquitous in nature and produce aflatoxin in warm (30-35°C) and high humidity (90-99%)

Aflatoxins are commonly found in maize, peanuts, cottonseed, sorghum, miliet and other food grains while grow in them

There are five important aflatoxins, called aflatoxin B₁, B₂, G₁, G₂, and M₁. Aflatoxin M₁ is a metabolite of aflatoxinB₁ found in milk and urine. It is not found in feed

Highest concentration of Aflatoxin B₁ is most frequently found in naturally contaminated feed

Epidemiology

Outbreaks are usually seasonal and sporadic occurrence

Occurrence of vaccination failure

Prevalence of infection

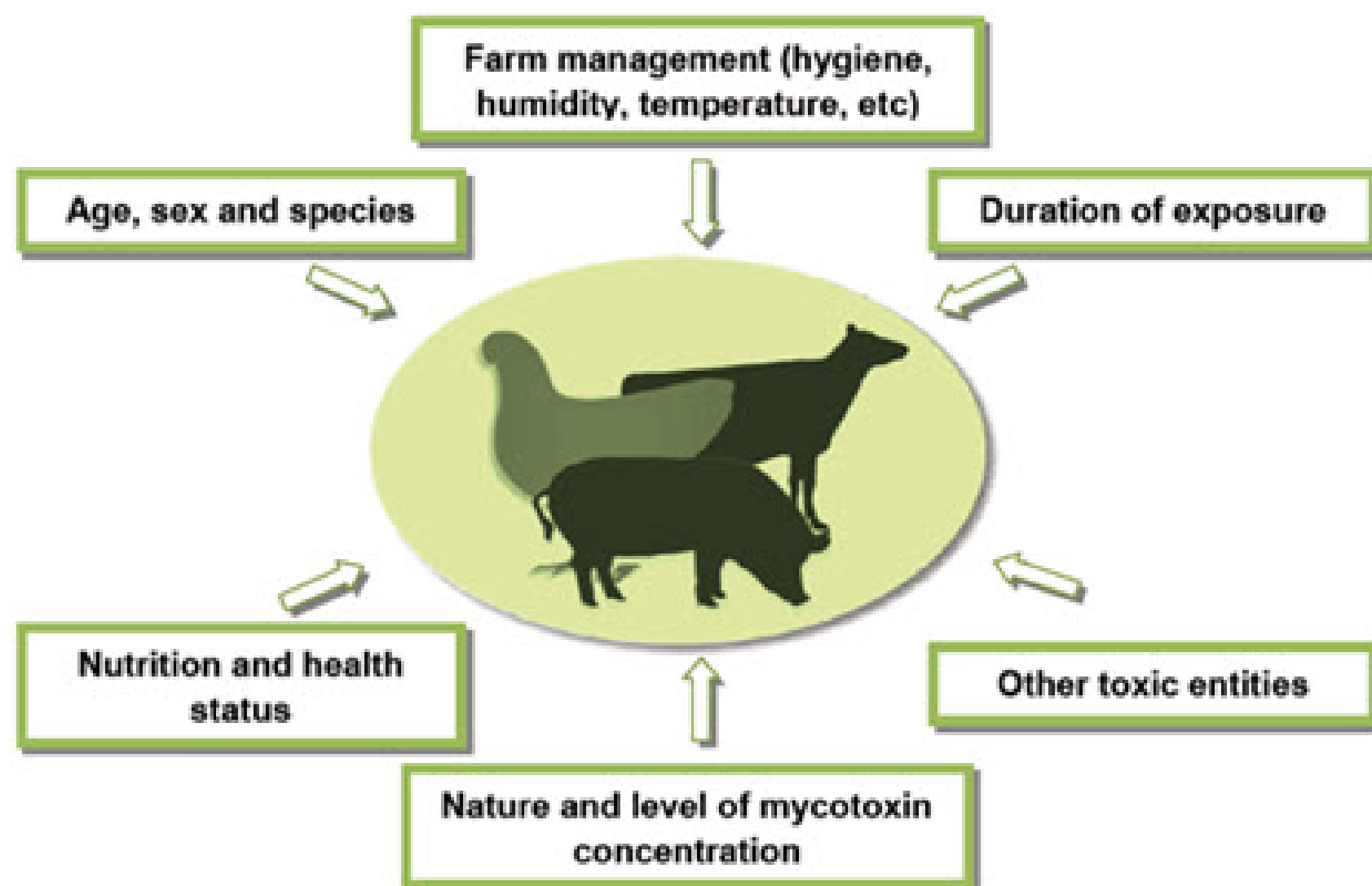
Problems occur worldwide, but especially **climates with high temperature and humidity** and where **grain is harvested with high water content**

Predisposing factors

High grain humidity

Grain damage due to insects

Poor storage conditions



Epidemiology - Predisposing factors

The following factors increases the release of Aflatoxin in grain

- o Grain damage by insects
- o Poor nutrition
- o Delayed harvesting of grain
- o Storage of grains or feeds in moist area
- o B₁, B₂, G₁, G₂ but B₁ are the toxins naturally produced Aflatoxins.
- o B₁ toxin present in highest concentration and is the most toxic not degraded by food processing and storage
- o ***Young birds are more susceptible to Aflatoxin than adults***
- o **Ducks are 10 times more susceptible than chickens.**
- o Turkeys are moderately affected than chicken and ducks

Transmission

Ingestion of feed or grain contaminated with toxins

Host affected

Duckling, turkey poult, calves, pigs and dogs are sensitive to toxic effect of aflatoxin

Sheep and adult cattle are more resistant

Aflatoxicosis is uncommon in horse and goat

Pathogenesis

Mycotoxins are ingested and then absorbed through gastrointestinal tract and metabolized in liver

The Metabolites of aflatoxin bind to DNA and RNA genetic materials thereby reduce the synthesis of protein, induce carcinogenesis, teratogenesis and

depression of immune response (both cell mediated and humoral immunity) is due to aplasia of thymic cortex

Clinical findings

Acute aflatoxicosis-Birds

It is associated with ingestion of high concentration of toxin in diet

Ataxia, opisthotonos and sudden death in ducklings

Hepatopathy

Subcutaneous haemorrhage

Haemorrhagic diathesis is due to hepatopathy is a characteristic of acute toxicity in chicken and turkeys

Widespread haemorrhage may have been due to combined effect of aflatoxin and cyclopiazonic acid, which are simultaneously produced by *A.Flavus*

Subacute aflatoxicosis

It is associated with prolonged exposure to low concentration of toxin

Slow developing ill thrift, reduced growth rate, steatorrhoea (fat in feces), bruise and immunosuppression, with increasing prevalence of endemic infections and vaccination failure

Impaired production efficiency can be a clue to a mycotoxin problem

Oral ulcers and crusts occurring on the palate or tip of the tongue occur with exposure to mycotoxins, including aflatoxin and mycotoxins produced by *Fusarium*

Calves

Blindness, circling, diarrhea and convulsion

Dogs

Depression, anorexia, vomiting, diarrhoea, jaundice and epistaxis

Subacute aflatoxicosis

It is associated with prolonged exposure to low concentration of toxin

Slow developing ill thrift, reduced growth rate, steatorrhoea (fat in feces), bruise and immunosuppression, with increasing prevalence of endemic infections and vaccination failure

Swine

Anorexia, unthriftiness, slow growth, icterus, mild anemia, ascites, increased susceptibility to infection

Cattle

Slow rumen motility for 24 to 48 hours,

Necropsy finding

Acute exposure: Hemorrhages and ascites

Chronic exposure: Pale, soft, clay-colored liver, mild anemia, icterus and ascites

Enlargement of liver, kidney and spleen

Enlarged liver are friable and yellow colour

Atrophy of bursa, thymus and testes

Histopathology: Hepatocyte degeneration and necrosis; centrilobular hemorrhagic hepatic necrosis

Diagnosis

Based on clinical signs and blood biochemical parameters such as **anemia, elevated liver enzymes, serum bile acids, and albumin: globulin ratio; prothrombin** activity

Tissue or fluid analysis: Aflatoxin M1 present in milk or urine- TLC, HPLC, UPLC and ELISA

Parent compound may be present in kidney or liver

Grain or feed analysis: Aflatoxins are present in corn, peanuts or cotton seed.

Sometimes more than one mycotoxin may be present in feedstuffs

Decreased performance may occur at concentrations as low as 200 ppb in young, sensitive species

At 200 ppb immunity may be impaired

At 200 to 400 ppb hepatic lesions become evident. At 400 ppb clinical illness become obvious

Violative residues in milk can occur at concentrations at or near **50 ppb aflatoxin M1**

Biological assay in duckling

Sometimes more than one mycotoxin may be present in feedstuffs,

Differential diagnosis

Other mycotoxicoses.

Other fungal infections.

Treatment

- Remove toxic feed and replace with unadulterated feed
- Concurrent diseases should be treated to alleviate disease interactions and correct substandard management practices
- Provide supportive treatment with vitamins, trace minerals (selenium), and increased dietary protein
- inactivation of the preformed toxin in grain or feed, and adsorption or inactivation of the toxin in the GI tract with activated charcoal

Prevention and Control

Mycotoxicooses may be prevented by careful choice of raw materials, **reduction in moisture content of the raw materials and hygienic storage**

Addition of **acid additives (propionic acid) in food grains @ 500– 1,500 ppm (0.5– 1.5 g/kg)** during storage to **control mold growth**, but do not destroy preformed toxin

Periodical testing of aflatoxin in feed

Feeding of aflatoxin level in feed should be <20 ppb

Monitoring of batch of feed for aflatoxin contamination

Decontamination of feed with ammonia gas at high temperature and pressure

Antimycotic feed additives may also be used but may not deal with toxins already formed

Feeds with high levels of fish meals are particularly susceptible and **should not be stored for more than 3 weeks.**

Pelletising feed may reduce fungal counts but does not affect toxins.

Certain minerals additives have been shown to bind mycotoxins and reduce their effects.

Addition of **hydrated sodium calcium aluminosilicate (toxin binder)** in feed effectively adsorb the aflatoxin, but it is little or not effective for Trichothecene mycotoxins, including deoxynivalenol.

Zeolite, Sodium bentonite is an effective adsorbent for aflatoxins in cattle and poultry but appears ineffective for trichothecenes and zearalenone

Glucomannan containing polymers extracted from yeast wall **high adsorptive capacity for binding of mycotoxin** including **aflatoxin B1, ochratoxin, T-2 toxin, and zearalenone at low concentration in the feed**

Cholestyramine has been an effective binder of fumonisins and zearalenone in vitro

and for fumonisins in animal experiments

Good stock control, management of feeders and bins, and avoidance of feed spillage are all important

Cleaning of equipments and sanitize with hypochlorite to reduce fungal contamination

Chemical detoxification of feed with acid and alkali

Alkali treatment of feed with calcium hydroxide, sodium hydroxide and ammonia acidifying agents like benzoic acid and copper sulphate @ 0.04-0.05%

Ochratoxicosis

Ochratoxicosis occurs seldom in poultry

More lethal in acute toxicity

Interferes with functions of kidney, proventriculus and gizzard

Ochratoxin is carcinogenic to humans and type A is weakly mutagenic by induction of oxidative DNA damage

It causes **immunosuppression** and immunotoxicity in animals

Etiology

Ochratoxin is produced by the fungus ***Aspergillus ochraceous*** (commonly called as **storage fungi**)

This toxin is also produced by *Penicillium verrucosum* *P. Viridicatum*

Toxin is derivatives of dihydroisocoumarin

There are four ochratoxins, A, B, C and D

Ochratoxin A is most common and most toxic

Commonly found in cereal grains such as Barley, Oats, Rye, Wheat and Maize, animal products, grape juice and wine

Epidemiology

Problems occur worldwide, but especially climates with high temperature, high humidity and where grain is harvested with high moisture content and storage of feed under moist condition

Found in raw and processed food commodities all over the world

Predisposing factor

Ochratoxin is produced at favourable temperature (13°C to 37°C) and high grain humidity

Grain damage is due to insects

Poor storage conditions

Host affected

Pigs, poultry and dogs are most often affected

Chickens, turkeys, ducklings are susceptible

Young birds are mostly susceptible to Ochratoxin

Ducks are seven times more sensitive to the acute effects of the toxin than chickens

Quails and turkeys are more sensitive than chickens

Ruminants are high tolerance to Ochratoxin A

Pathogenesis

OTA is a **potent nephrotoxin** and causes both acute and chronic effects in the kidneys of all mammalian species

A level of 200 µg/kg in feed over three months is sufficient to cause acute damage to the kidneys of pigs

Ochratoxin inhibits protein synthesis, produce acute proximal tubular epithelial necrosis in kidneys

- It causes renal carcinoma, teratogenesis and immunosuppression

Clinical signs

Pigs

Inappetence, depression, weight loss, **polydipsia and polyuria**

Poultry

Renal disease, hepatic damage and immunosuppression

Decreased egg production, poor quality egg and reduced growth rate ,
Diarrhoea and Paralysis or incoordination

Decreased resistance to infections such as Salmonellosis, Coccidiosis,
Infectious bursal disease and candidiasis

Poor feed conversion (>1.0 ppm).

Increased condemnation during processing >0.5 ppm)

Loss of pigmentation & increased bruising (>0.5 ppm)

Necropsy finding

Affected kidneys are white to tan, **swollen, hard** and may have **pin- point urate crystal** on the surface

White pasty urate crystals on pericardial, perihepatic, peritoneal and articular surfaces

In surviving birds after renal failure, kidneys is enlarged, fibrotic and pale

Liver is enlarged and yellowish in color

Bursa and thymus reduced in size

Diagnosis

Based on clinical signs

Blood biochemical parameters such as anemia, elevated liver enzymes, serum bile acids, alteration of albumin:globulin ratio, prothrombin activity

Differential diagnosis

Aspergillosis

Other mycotoxicosis

Treatment

The most effective treatment is removal of the source of toxins

Addition of antifungal feed preservatives is also helpful

Increasing protein level in the feed

Administration of fat soluble vitamins and selenium (0.2 ppm) along with finely divided copper sulphate in the feed 1kg/ton for 7 days

Prevention and control

OTA is destroyed by acid and alkaline hydrolysis

Careful selection of raw materials (avoid use high moisture content of raw materials)

Hygienic storage

Antimycotic feed additives may also be used but may not deal with toxins already formed

Feeds with high levels of fish meals are particularly susceptible and should not be stored for more than 3 weeks

Pelletizing of feed may reduce fungal counts but does not affect toxins

Addition of toxin binders in feed

Tricothecene Mycotoxicosis

The tricothecene mycotoxins constitute a large and diverse group of sesquiterpenoid compounds produced by *Fusarium sp*, *Myrothecium sp*, *Trichoderma sp*, *Cephalosporium sp*, *Verticimonosporium sp*, and *Stachybotrys sp* and *Trichothecium sp*

On the basis of molecular structure, the tricothecenes are grouped as non-macrocylic and macrocylic compounds like

- Deoxynivalenol - Non macrocylic
- T-2 toxin - Non macrocylic
- Diacetoxyscirpenol– Non macrocylic
- Stachbotryotoxicosis- Macrocylic
- Myrotheciotoxicosis- Macrocylic

Tricothecene are **powerful inhibitors of protein synthesis** in host

Tricothecenes are different from most other potential toxins because **they can act through the skin**

Compared with some of the other mycotoxins such as aflatoxin, the tricothecenes **do not appear to require metabolic activation to exert their biological activity**

Predisposing factor

Wet season and warm weather

Delayed harvesting

Harvesting of grain in winter months

Storage of grain in cold conditions

Deoxynivalenol toxicosis

Deoxynivalenol toxin (Vomitoxin) is produced by *Fusarium graminearum* and *F. culmorum*

Commonly found in cereals

Host affected

Pig, dog and cat-susceptible

Ruminant are resistant

Transmission

Ingestion of toxin contaminated feed

Clinical signs

Pig

The toxin have potent emetic activity causes vomiting, , feed refusal, digestive disturbance and weight loss

Poultry

Decreased feed efficiency

T-2 Toxicosis

T-2 toxin is produced by *Fusarium sporotrichoides* and *F. poae*

Toxicosis is common in pig, poultry and cat

Toxicosis is associated with **ingestion of mouldy wheat** contaminated with *Fusarium* sp

Clinical signs

Pig

Mucosal lesion, toxic effect on bone marrow, thrombocytopenia and haemorrhage

Immunosuppression

Poultry

Beak lesions, abnormal feather formation, drop in egg production, tissue haemorrhage and increased susceptibility to infectious disease

Cat

Vomiting, dysentery, ataxia and dehydration

Diacetoxyscirpenol toxicosis

Diacetoxyscirpenol toxin produced by *Fusarium tricinctum* and other fusarium sp on cereals

Common in cattle, pig and poultry

It causes necrotic lesions, mucosal haemorrhage, salivation and depression, vomition and necrotic lesions in the alimentary tract and haemorrhage in the skin, oral cavity, intestine and major organs

Stachbotryotoxicosis

It is caused by Satratoxin and Roridins produced by *Stachbotrys chartarum*

Commonly found in stored cereals, straw and hay

Toxicity occur in horse, cattle, sheep and pigs

Cytotoxicity, coagulopathy and immunosuppression

Stomatitis, necrotic lesions in alimentary tract and haemorrhages and agranulocytosis

Myrotheciotoxicosis

Roridin produced by *Myrothecium verrucaria* and *M.roridium* on ryegrass, rye stubble and straw

It affect sheep, cattle and horse

Inflammation of many tissues, pulmonary congestion and sudden death

Prolonged exposure to sub lethal dose may causes weight loss and unthriftiness

Diagnosis, treatment and control measures similar to aflatoxicosis

Aspergillosis

(**Syn:** Brooder Pneumonia, Mycotic pneumonia and pneumomycosis)

Aspergillosis is primarily respiratory disease of chickens, turkeys, and less frequently ducklings, pigeons, canaries, geese, and many other wild and pet birds. In chickens and turkeys, the disease may be endemic on some farms; in wild birds, it appears to be sporadic, frequently affecting only an individual bird. Severe outbreaks usually occur in birds 7– 40 days old. . **Newly hatched chicks poults and duckling** including other avian species are highly susceptible.

Etiology

☒ ***A.fumigatus***, *A. flavus*, *A. parasiticus*, *A. niger* , *A.terreus*’

☒ ***A.fumigatus* is common cause of disease in chicks**

☒ Aspergillus species are common inhabitant in soil

Fungi grow rapidly in Saboraud’ s agar at 27°C and sporulated to produce fuzzy white colonies and golden brown to greenish colonies

Fungi have got septate mycelium, the hyphae are 4-6 mm in diameter

The colonial heads are columnar and compact with parallel chains of conidia

Their growth is inhibited in media containing cyclohexamide

Epidemiology

Prevalence of infection

Avian aspergillosis distributed through out the world

High mortality rates are seen in chicks and poults that inhale large numbers of spores during hatching in contaminated incubators or when placed on mold-bearing litter.

Acute respiratory disease causes 5-50% mortality in first 1-3 wks of age of birds

Chronic pulmonary insufficiency, ascites, blindness, neurological fungal metastasis causes up to 5% mortality

Predisposing factors

Cold, stress, high ammonia content of the farm

Dusty environment increase incidence and severity

Adult birds are resistant to new infection

Adults receives infections only when individual birds exposed to high level mycotoxins or under immunosuppression or nutritional insufficiency

Source of infection

Moist grains favour the growth of *Aspergillus* sp

Spores liberated in large numbers from infected eggs opened at incubation or hatching

Wet litters favour the growth of spores and present in aerosol when litters are dried

Contaminated feed or poultry house litter

Transmission

Inhalation of spore from aerosol

Host affected

All birds are susceptible for infection

Chicken, turkeys, ducks, quails are mainly affected

Newly hatched chicks, poults and duckling

Pathogenesis

Acute form

It is common in brooding chicks

In birds, acute Aspergillosis is known as “ Brooder pneumonia” and causes heavy mortality in them

The Conidia present in aerosol or environment spread through conjunctival, nasal, tracheal, para bronchial and air sac epithelium and form discrete, **granulomas** of 1-15 mm diameter, containing **white plaques or nodules at these sites.**

Fungi tend to proliferate within the granuloma and rarely invade adjacent tissues in immuno competent birds

Chronic

In turkeys often pulmonary blood flow impeded by enlarging pulmonary granulomas and this cause **right ventricular dilatation or ascites**

In chronically infected birds **accumulation of exudate in the trachea** or syrinx producing acute respiratory embarrassment

After inhalation, spores are disseminated rapidly through **haematogenous route** to other tissues and cause lesions in **brain, pericardium, bone marrow and kidneys** etc.

Lesions in meninges, leading to large superficial **white plaques over the caudal cerebral cortex** and in pecten and iris leading to fungal ophthalmitis and iridocyclitis

Clinical signs

Brooder pneumonia

At hatchery

Dyspnoea, hyperpnoea, open mouth breathing/gasping due to progressive air way obstruction and increased thirst was observed in first 3-5 days old chicks

Adult birds

Birds remain sub clinically infected for sometimes

Slow progressive respiratory embarrassment

Emaciation

The encephalitic form is most common in turkeys, torticollis

An ocular form is seen in chickens and turkeys as mycotic keratitis, in which large plaques may be expressed from the medial canthus.

Pet birds

Tracheitis in immunocompromised birds.

Aspergillus granulomas often form in the syrinx of psittacine birds and raptors

Changes in vocalization may occur before dyspnea is observed, and often these birds will stretch out their necks while breathing.

airsacculitis

Necropsy finding

Birds

Pulmonary lesions are characterized by white to yellow plaques and nodules a few mm to several cm in diameter; occasionally, mycelial masses may be seen within the air passages

The plaques and nodules also may be found in the syrinx, air sacs, liver, intestines, and occasionally the brain.

In acute mycotic pneumonia, serosanguineous fluid in pleural cavity and fibrinous pleuritis was noticed

Pet birds

Granulomas of the air sacs, usually in the caudal thoracic or abdominal air sacs

Sample collection

Granulomatous lesions in lung,

Diagnosis

Based on clinical signs, and necropsy findings

Radiography of air sacs

Identification of organism: Examination of branched septate *Aspergillus* hyphae. Add one or two drops of 10% KOH in smears taken from lesions and heat it gently and examine under microscope.

Tissue sections are stained by methenamine silver or Periodic acid Schiff (PAS) staining method may reveal hyphal invasion

Routine observation of hyphae stained with H and E

Isolation of organism in sabouraud dextrose agar containing antibiotics

Serological tests are based on growth phase or hyphal specific antigen of *Aspergillus fumigatus*

Differential diagnosis

Infectious bronchitis

New castle disease

Infectious laryngio-tracheitis

Mycobacteriosis

Colibacillosis

Nutritional encephalomalacia

Treatment

Treatment of affected birds is generally ineffective.

Spontaneous recovery from pulmonary aspergillosis can occur if reexposure to the mold is prevented.

Pet birds

Treatment varies depending on the site of infection.

An acute tracheal obstruction by aspergilloma is an emergency that may require placing an air sac tube. After tube placement, syringeal or tracheal plaques are removed with a rigid endoscope.

Amphotericin B can then be instilled intratracheally at 1 mg/kg through the glottis, and treatment with systemic antifungals and nebulization should be started. Amphotericin B is the only fungicidal agent available and can be used in nebulization @ 0.25– 1 mg/mL of sterile water, as a nasal flush, intratracheally,

Nasal and sinus flushes are generally more dilute at 0.05 mg/mL of sterile water.

Control

It depends on reducing exposure to fungus

Avoid use of contaminated or cracked eggs for hatching

Contaminated hatcheries should be thoroughly cleaned and fumigated with enilconazole or formaldehyde

Avoid use of moldy feed and litters

Aspergillosis in livestock

(Syn: Mycotic abortion)

It is primarily a respiratory infection that may become generalized; however, tissue predilection varies among species. The most common forms are pulmonary infections in poultry and other birds; mycotic abortion in cattle; guttural pouch mycosis in horses; infections of the nasal and paranasal tissues, intervertebral sites, and kidneys of dogs; and sinonasal, sino-orbital, and pulmonary infection in domestic cats.

Etiology

Among 300 species of *Aspergillus* identified only a few have been considered as potential pathogens for animals and humans

Common species of *Aspergillus* affecting human and animals are as follows:

- ☒ ***A.fumigatus***, *A. flavus*, *A. parasiticus*, *A. niger* , *A.terreus*
- ☒ *A.ruber*, *A. wentil*, *A. glaucus*
- ☒ *ostianus*, *A. ocharaceus*, *A. terrus*

☒ *A. nidulans* *A. tamari*, *A. clavatus*

Epidemiology

Predisposing factors

Prolonged use of antibiotics and corticosteroids,
exposure to radiation, neoplastic diseases, malnutrition
tuberculosis makes the animals more prone for infection

Source of infection

Aspergillus has been isolated from placenta of aborted cows, uterine discharge
of repeat breeder and semen of bull

Moist grains favour the growth of *Aspergillus* sp

Transmission

inhalation of spore from aerosol and ingestion of *Aspergillus* sp contaminated
fodder due to delayed harvesting

Infection may also spread through abraded skin, mucous membrane and setup
lesions in the embedded organs and systems

Clinical manifestation

Ruminants

Aspergillosis may be asymptomatic

The bronchopulmonary form causes mycotic pneumonia, which may be
rapidly fatal. Signs include pyrexia; rapid, shallow, stertorous respiration, nasal
discharge, and a moist cough.

In the absence of pneumonia, infected cows generally have no signs except
for abortion occur at 6-9 month of gestation, retained placenta

**Aborted fetal skin may have soft elevated red to gray discrete foci that
resembles ring worm lesion**

Bloody diarrhoea

Mastitis

Horse

Guttural pouch mycosis

Epistaxis, dysphagia, laryngeal hemiplegia, periauricular swelling, unilateral
nasal discharge may follow accumulation of inflammatory exudates in the
pouch

Allergy is due to inhalation of spores of *A. fumigatus* causes high temperature,
asthma, malaise and prostration

Locomotor and visual disturbances, including blindness, may occur when the
infection spreads to the brain and optic nerve

Dogs

Nasal aspergillosis

It is common in dolicocephalic breeds like Dobermans, German shepherds and Labrador

Clinical signs are often unilateral

Nasal pain, ulceration of the nares, sneezing, unilateral or bilateral sanguinopurulent nasal discharge, frontal sinus osteomyelitis and epistaxis

The mucosa of the nasal and paranasal sinuses may be covered by a layer of necrotic material and white to gray-white fungal growth

Otorrhoea is due to growth of *Aspergillus* sp in cerumen

Disseminated disease

It is usually seen in middle-age, female German Shepherds infected with *A terreus*, *A deflexus*, and *A niger*

Clinical signs include lethargy, lameness, anorexia, weight loss, muscle wasting, pyrexia, hematuria, urinary incontinence, generalized lymphadenopathy and neurologic deficits

Disc spondylitis is common

Cats

Sinonasal and sino-orbital disease is most often seen

Sinonasal disease can present similarly to the disease in dogs,

The sino-orbital disease can be very aggressive, often causing severe facial swelling. In some cases, a mass in the pterygopalatine fossa or ulceration of the hard palate may be seen. The organism may invade CNS and causes neurologic signs

Necropsy findings

Pulmonary form

The lungs are firm, heavy, and mottled

In subacute to chronic mycotic pneumonia, the lungs contain multiple discrete granulomas, and the disease grossly resembles tuberculosis

Horses

Nodule formation in lungs

Interstitial haemorrhage with caseous nodules in the kidneys

Sample collection

Granulomatous lesions in lung, aborted foetus and placenta

Diagnosis

Based on clinical signs, and necropsy findings

Radiography of lung and nasal cavity

Endoscopic examination of guttural pouch to detect lesions

Rhinoscopy and CT scan

Identification of organism: Examination of branched septate *Aspergillus* hyphae. Add one or two drops of 10% KOH in smears taken from lesions and heat it gently and examine under microscope.

Tissue sections are stained by methenamine silver or Periodic acid Schiff (PAS) staining method may reveal hyphal invasion

Routine observation of hyphae stained with H and E

A blood and urine *Aspergillus* galactomannan antigen (GMA) assay for diagnosis of systemic aspergillosis in dogs

Isolation of organism in sabouraud dextrose agar containing antibiotics

Serological tests are based on growth phase or hyphal specific antigen of *Aspergillus fumigatus*

ELISA is commonly used in dogs to detect antibodies to *A.fumigatus*

Differential diagnosis

Brucellosis

Tuberculosis

Sporotrichosis

Coccidioidomycosis

Histoplasmosis

Treatment

Dogs

The topical treatment is considered the treatment of choice for nasal and paranasal aspergillosis

Clotrimazole formulated in a polyethylene glycol base is generally considered the first-line treatment.

It can be administered through indwelling tubes trephined into the frontal sinuses or via the nares as a single infusion

Enilconazole, 10 mg/kg, instilled bid for 7– 14 days, via tubes implanted surgically into the frontal sinuses

Systemic antifungal agents

Fluconazole @ 2.5– 10 mg/kg, divided bid for 6-8 wks

Itraconazole @ 5– 10 mg/kg/day

Ketoconazole @ 5– 10 mg/kg, bid for 6– 8 wks

Voriconazole (3– 6 mg/kg/day)

Horse

Surgical curettage has been used to treat gutturomycosis

Topical natamycin and oral potassium iodide has been reported effective

Itraconazole @ 3 mg/kg, bid for 84– 120 days has been reported effective in *Aspergillus* rhinitis in horses

Control

Control depends on reducing exposure to fungus

Feed, water lines, soil should be disinfected

All attempts should be made to prevent proliferation of aspergilli in animals' sheds on surroundings

Candidiasis

(**Syn:** Moniliasis, Thrush, Crop mycosis, Oidiomycosis)

It is a fungal disease caused by yeast, *Candida albicans*, and has been manifested by the development of mucocutaneous mycoses in skin, hair and nail. Oesophageal or Crop Candidiasis occurs very frequently but clinical signs are rarely exhibited by the affected birds. *Candida* organisms mainly affect **birds below 3 weeks of age** but cause low to nil mortality.

Etiology

Candida albicans is the most pathogenic species followed by *C.tropicalis*, *C.labrata*, *C.krusei*, *C.parapsilosis*, *C.dubliniensis* in human

Candida is a small, oval, thin walled dimorphic, budding type yeast like cells (chlamydospore

Candida species are ubiquitous in the environment

Epidemiology

Predisposing factors

Continued use of antibiotics

Prolonged use of corticosteroids

Vitamin A and B deficiency and high level of glucose in diet

External use of detergents or douches or internal disturbances by hormone can disturb the normal vaginal flora such as lactobacilli

Transmission

Infected faeces loaded with organisms spread the disease

Man acquires the infection by eating contaminated meat and food

Human acquires the infection through in-utero and venereal contact

Host affected

Chickens, turkeys, cattle, buffaloes, horses, goats, sheep, pigs, dogs, cats, including human are susceptible to infection

Pathogenesis

Animals

The candida species produces the enzyme known as leucine amino peptidase which causes damage to the tissues and thereby induce the development of disease

The presence of high levels of antibiotics, maltose and dextrose cause cutaneous and systemic mycoses in pigs and candida endometritis in mares, panleukopenia in cats due to repeated use of antibiotics

The organism also invades the vital organs and set up **diarrhoea, pneumonia, nephritis, endocarditis, mastitis, repeat breeding and abortion**

The organism can be isolated from semen, aborted materials, uterine mucus of repeat breeders and mastitis milk and from the environment

Poultry

The organisms lives in mouth, oesophagus and crop as a normal resident and also acquired by ingestion

Inhibition of normal microflora and immunosuppression allow the yeast to invade and proliferate on the superficial epithelial layers which results in development of **epithelial hyperplasia, pseudo or diphtheritic membrane formation**

Crop appears to have sticky and adhesive multifocal, confluent mats of **white cheesy materials, which may be found less in oesophagus and pharynx.**

Severely affected birds may have ulcers otherwise very mild inflammatory response will be seen in mucosal candidiasis

Inflammatory response to mucosal candidiasis is mild unless ulceration is prolonged

Clinical signs

Mucocutaneous form

The tongue, gums, buccal mucosa of the infected animals are filled up with **whitish creamy pseudomembranous exudate** which contains yeast cells are known as **thrush**

Cutaneous form

Thrush in rumen, sub-clinical and acute mastitis, anorexia, lameness, and decrease milk yield may be developed in adult cattle

In calves, **gastroenteritis, profuse watery diarrhoea, anorexia**, dehydration and death

Mycotic mastitis and abortion in cows at 6-8 months of pregnancy

In pigs, lesions from mouth, esophagus, skin and stomach may spread from vagina of pregnant and non pregnant sows

Systemic form

Cattle and calves suffer from inflammation of the stomach and intestine and leads to profuse watery diarrhoea

Mycotic pneumonia is uncommon in farm animals

Poultry

Candidiasis is the most common infection, nevertheless, clinical signs seen in severely infected birds

Birds having infection in **superficial region of oral cavity, oesophageal and crop become dehydrated and fail to gain body weight**

Candida spp. invades into nervous system, renal, intestinal area but rarely shows symptoms

Necropsy finding

Pseudomembranes and diphtheritic membranes in the crop, oesophagus and mouth are highly suggestive of candidiasis

Visualization of hyphae and pseudohyphae microscopically in hair and/or scrapings.

Diagnosis

Based on clinical signs and necropsy findings

Animal Inoculation: In mice and rabbit One ml antigen in 15ml saline solution is suspended and injected intravenously or intra peritoneally causes death in a rabbit within 4-5 days of injection, because of the abscess formation in kidney.

The kidney smear show budding cells

Antigen and antibody detection by immuno-diffusion, counter immuno-electrophoresis, tube and latex agglutination test can be done to detect candidia antibody

Speciation requires culture in Sabouraudes dextrose agar but many normal birds may be positive

Sample collection

Hair, pus discharges, swab from tissue lesions, blood and serum

Differential diagnosis

Histoplasmosis

Other mycotic infection

Treatment

Most candidial infections respond to the treatment with antibiotics

The **commonly used antifungal drugs are topical Clotrimazole**, Nystatin, **Fluconazole** and Ketoconazole

Diflucan (Fluconazole): A single dose of 150mg tablet/oral give 90% control in a vaginal yeast infection. (It is also essential to take care of persons allergic forazole group of medicines). It is only useful for vaginal yeast infection.

Prevention

Crop mycoses can be best prevented by controlling excess usage of antibiotic and immunosuppressive diseases.

Control

Segregation, clean water and good food provision, without any mould, or fungi

Prolonged use of antibiotics or steroids should be avoided

Maintenance of adequate hygiene is essential

Encouraging cleaning and sanitation

Temporarily controlled with Gentian Violet (8mg/kg) or Nystatin (142mg/kg)

Sporotrichosis

It is a contagious disease of horse, cats, cattle and humans characterized by development of **cutaneous nodules and ulcers on limbs** that may be accompanied by lymphangitis.

Etiology

It is caused by *Sporothrix schenckii*

The organism is ubiquitous and found in soil, vegetation and timber, worldwide in distribution. In USA, the disease is mostly found in coastal region and river valleys

Sporotrichosis has been reported in **dogs, cats, horses**, cows, camels, goats, mules, birds, pigs and people

Cats are most important in zoonotic transmission of sporotrichosis to human through shedding of organism in wound and **faeces**

Transmission

Skin wound infection via contact with plants or soil or penetrating foreign bodies

Inhalation of spore is rare

Pathogenesis

Infection develops via hematogenous or tissue spread from the initial site of inoculation to the bone, lungs, liver, spleen, testes, GI tract, or CNS

Clinical signs

It is divided into three groups namely

Lymphocutaneous form – most common

Cutaneous form remain localized to the site of inoculation

Disseminated form – rare

Lymphocutaneous form Small, firm **dermal to subcutaneous nodules**, 1– 3 cm in diameter, develop at the site of inoculation. As infection ascends along the lymphatic vessels, cording and new nodules develop. Lesions ulcerate and discharge a **serohemorrhagic exudate**. Chronic illness may result in fever, listlessness, depression and respiratory signs may be apparent

In cat, the lesion is often found in head especially on the nasal bridge and pinnae of ear

Diagnosis

Microscopic examination of the exudates or biopsy specimen (samples obtained from unopened lesions). In tissues and exudate, the organism is presented as few to numerous, **cigar-shaped, single cells within macrophages**.

The fungal cells are pleomorphic and small ($2-10 \times 1-3 \mu\text{m}$); buds may be present and give the appearance of a **ping-pong paddle**

Isolation of organism in fungal culture

Differential diagnosis

Epizootic Lymphangitis

Glander

Ulcerative Lymphangitis

Histoplasmosis

Treatment

Itraconazole (10 mg/kg/day) is considered the treatment of choice for sporotrichosis. Treatment should be continued 3– 4 wk beyond apparent clinical cure.

Supersaturated solution of **potassium iodide**, administered PO has been used with some success; therapy is continued **30 days** beyond apparent clinical cure.

Prevention

Disinfection of infected articles, equipments and shed with antifungal agents will prevent the spread of infection in enzootic areas.

Control

Segregation of infected animals

Cleaning of cut wounds, abrasions with tincture iodine

Washing of hands and arms with povidone iodine or chlorhexidine recommended for human handling infected animals or plant materials are effective

Wearing gloves and long sleeves

Histoplasmosis

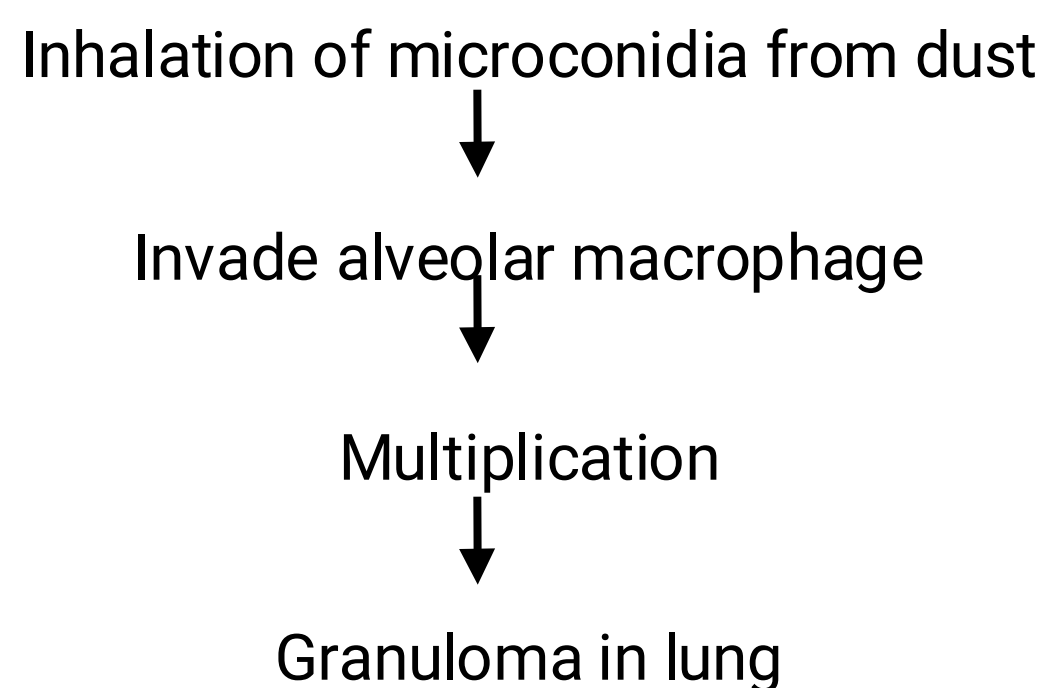
It is a **rare systemic mycosis** in farm animals caused by *Histoplasma capsulatum*

The organism is usually found in soil, bird or bat faeces

The fungus is able to survive for longer period more than 4 months in soil and water

Infection occurs by **inhalation of contaminated dust**, primary invasion is usually takes place in lung

Pathogenesis



Clinical signs

Dog and cat

Anorexia, chronic cough and persistent diarrhoea, ulcerative lesions in the intestine. Ulcerative skin nodules, lameness and emaciation

In Cat- intestinal ulceration is not occur

Horse

Pneumonia, dyspnoea, nasal discharge, hepatic insufficiency with jaundice, anasarca, placentitis with abortion

Necropsy finding

Liver is enlarged with necrotic foci

Pulmonary consolidation

Granulomatous pneumonia

Enlargement of splanchnic lymph node

Diagnosis

Aggregation of fungal bodies in lymphoid tissue

Histopathological examination of affected tissue shows pyogranulomatous foci with yeast like forms in macrophage

Culture of organism in sabourad dextrose agar

Agar immunodiffusion test

Histoplasmin skin test

Treatment

Itraconazole – Drug of choice

Coccidioidomycosis

(Syn: Valley fever)

Coccidioidomycosis is a dustborne, noncontagious infection caused by the dimorphic fungus *Coccidioides immitis*.

Epidemiology

Infection occurs in arid and semiarid regions of the southwestern USA, Mexico, Central and South America.

Host affected

The mostly affected animal is **dog and cat** other animals like **horse**, ruminants and pigs also affected

Transmission

Inhalation of fungal spores from dust particles

Placental transmission occur in horse

Clinical signs

The disease varies from inapparent (cattle, sheep, pigs, dogs, cats) to progressive, disseminated, and fatal form

Coccidioidomycosis is **primarily a respiratory disease** that ranges from self-limiting to chronic

Dogs with disseminated disease may have fever, chronic cough, anorexia, cachexia, lameness, enlarged joints and intermittent diarrhoea

Cats often present with dermatologicals signs like **subcutaneous granulomas**, abscesses, fever, inappetance and weigh loss

The respiratory signs are less common in cat

Horse – severe emaciation, fluctuating temperature, persistent cough, muscle pain superficial abscess in pectoral area, colic is due to internal abscess and osteomyelitis

Ruminants- sub clinical infection with respiratory symptoms only

Sheep – fever and abscess in peripheral lymph node

Lesions

The discrete, variable-sized firm **gray- white nodules** (like tuberculosis) are limited to lung, thoracic lymph node and mediastinum

The nodules are pyogranulomas composed of epithelioid and giant cells, and the centre of some foci may contain purulent exudate and fungal organisms

Diagnosis

Thoracic radiography to identify pulmonary nodules and enlarged lymph node

Demonstration of the spherules in tissue

Isolation of organism in fungal culture

Agar gel immune diffusion test used to detect the IgG and IgM antibodies

Human: Coccidioidin skin test, CFT and immunodiffusion test are used for diagnosis

Treatment

Disease is often self-limiting

For disseminated or chronic respiratory infections treatment should be continued upto 6- 12 months with the following antifungal drugs

Fluconazole (2.5– 10 mg/kg/day)

Ketoconazole (10– 30 mg/kg/day)

Itraconazole (10 mg/kg/day)

Amphotericin B may be the most effective antifungal drug, but it is highly nephrotoxic

Prevention and control

Dust control is a major factor in prevention of human coccidioidomycosis

No vaccine is available

Eradication of spores from soil is not feasible

Halogens (such as iodine, and **chlorine in the form of hypochlorite** [bleach]), phenolics and quaternary ammonium compounds are proven effective against *Coccidioides* spp

Rhinosporidiosis

Rhinosporidiosis is a chronic disease of the nasal mucosa in cattle and nasal mucosa, pharynx, and larynx in horses that causes formation of large polyps or granulomatous lesions and occasionally infect skin of horse, dogs cats and aquatic birds..

Etiology

Rhinosporidium seeberi, is an aquatic protist that typically causes disease in amphibians.

Epidemiology

The disease is most often seen in India, Africa, South America and uncommon in North America

Predisposing factors

Trauma may predispose to infection

Pathogenesis and Clinical manifestation

The disease is evident as single or multiple, pedunculated or sessile, pink to red masses in the mucous membranes of the nose and nasopharynx and large enough to occlude the nasal passages

Severe dyspnea with loud stertor and a mucopurulent or blood-stained nasal discharge

The lesions can bleed and become evident as epistaxis

Other diseases with similar clinical profiles include nasal obstruction associated with the blood fluke *Schistosoma nasale* and chronic allergic rhinitis

Sample collection

Biopsy material from nasal polyps

Diagnosis

Based on clinical signs

Microscopic demonstration of spherules (Sporangia) of *R. seeberi* by staining with Periodic Acid Schiff (PAS) stain

Spherules have thick walls may be numerous, vary in size (upto 300µm)

Histopathology: Multifocal hyperplasia and ulceration of the mucosa, mild to moderate multifocal, lymphoplasmacellular inflammatory infiltrate and multiple spherical to polygonal organisms of variable size, consistent with *R. seeberi* in the lamina propria and mucosa

Histologically, there is a marked eosinophilic reaction, and yeast-like bodies are present in cells or free in the tissue spaces

PCR analysis of affected tissue reveals presence of *R. seeberi* DNA

Differential diagnosis

Nasal schistosomiasis

Nasal granuloma

Treatment

Surgical excision of lesions are standard, treatment, but recurrence is common

Excision of a single mass in the rostral nares of a mule was curative

Amphotericin B and Itraconazole have been used for the treatment, but are generally ineffective

Avian Chlamydiosis

It is also known as **Psittacosis or Parrot fever in psittacine birds**

The disease affecting **other than psittacines birds** is called as **Ornithosis**

Avian chlamydiosis can be an inapparent subclinical or acute, subacute, or chronic disease of wild and domestic birds characterized by respiratory, digestive, or systemic illness.

History

1893- Paris- flu like disease is transmitted from parrot to man (Causative agent: *salmonella psittacosis*)

1929-1930- first pandemic psittacosis was reported in human- USA and Europe

1932- first case of zoonotic psittacosis transmitted from chicken

1942- psittacosis was recorded in pigeon

1966- Chlamydia was identified as causative agent

1999-Chlamydia was renamed as Chlamydophilla

2015 Chlamydophilla was again reclassified into chlamydia

Etiology

It is a gram negative, coccoid obligate intracellular bacteria Chlamydia, family chlamydiaceae, order chlamydiales

Chlamydia psittaci – infect poultry, human and mammals

C. avium-pigeon and Psittacine birds

C. gallinacea-chicken, guinea fowl, turkeys and duck

Chlamydia psittaci contains 8 serotype (A-F) and 9 genotype

Serotype A & D-highly virulent for turkeys

Serotype B & E frequently causes disease in wild birds

Chlamydia depend on the **eukaryotic ATP for their replication** produce RB and EB

The organism is susceptible to quaternary ammonium Compounds, **formaldehyde, glutaraldehyde, 70% isopropyl alcohol**, sodium hydroxide and 1:100 household bleach solution are effective

Epidemiology

The disease is worldwide in distribution and endemic in tropical and sub tropical countries

Mortality-5-40%

Host affected

- C. psittaci has been reported in more than 450 species of birds belonging to at least 30 orders.
- **Pet birds-Psittacine birds and their relatives**
- **Domesticated birds: turkeys, ducks, pigeons, chickens**
- Other birds: geese, game birds, quail, partridges, ratites, ostriches,
- canaries and finches and peacocks.

Young birds are more susceptible than older birds

Predisposing factors

Stress arising from captivity

Transportation

Overcrowding

Changes in diet / environment

Concurrent infection with other organisms such as Salmonella or *Pasteurella multocida*

In Turkey-*Ornithobacterium rhinotracheale* often occur as concurrent infection

Source of infection

Persistently infected birds act as carrier of infection

Excretion of organism in faeces and respiratory secretions

The infectious elementary body is highly resistant outside the host and so able to survive in dried excreta and in manure for longer periods

Transmission

Inhalation of organism from dried faecal dropping

Ingestion of contaminated feed with respiratory secretion and faeces of infected birds

Pathogenesis

After inhalation or ingestion, elementary bodies attach to mucosal epithelial cells and are internalized by endocytosis. Elementary bodies within endosomes in the cell cytoplasm inhibit phagolysosome formation and differentiate into metabolically active, noninfectious reticulate bodies that divide and multiply by binary fission, eventually forming numerous infectious, metabolically inactive elementary bodies. Newly formed elementary bodies are released from the host cell by lysis. via the haematogenous route spread into pericardium, liver, spleen and kidney where they undergo further multiplication and produce reticulate body and elementary bodies.

Clinical signs

- Incubation period: 3-10 days, may longer in older birds

Psittacine birds

- Fever, anorexia, oculo-nasal discharge, conjunctivitis, sinusitis, weight loss, ruffled feathers and polyuria,
- Green to yellow green droppings
- Drop in egg production

Domesticated birds

- Respiratory signs predominate in turkeys and chickens
- Watery diarrhea is often present in ducks, trembling, unsteady gait, anorexia, serous or purulent discharge from eyes or nares noticed in ducks. Pigeon affected by temporary ataxia

Zoonotic risk

Human disease most often results from exposure to pet psittacines

Infection in people varies from asymptomatic to flu-like symptoms and respiratory disease (eg, pneumonia). Rarely, endocarditis, myocarditis, hepatitis, and encephalitis occur.

Immunocompromised people are at increased risk of developing clinical diseases

Gross lesion

serofibrinous polyserositis (airsacculitis, pericarditis, perihepatitis, peritonitis)

bronchopneumonia

hepatic necrosis

hepatomegaly

splenomegaly with grey white necrotic foci

Thickened pericardium with fibrinopurulent exudate

Sample collection

Live birds: conjunctival, choanal, and cloacal swabs, paired sera sample
(Multiple samples for 3-5 days)

Dead birds: trachea and conjunctiva, liver, spleen and lung, impression smear
from liver, spleen

Diagnosis

- Based on clinical signs & PM lesions

Clinical pathology

- **Hematological changes-anemia** and leukocytosis with heterophilia and monocytoysis
- **Biochemical changes-** increased Plasma **bile acids, AST, LDH, and uric acid**
- **Radiograph or a laparoscopy** may reveal an **enlarged liver and spleen and thickened air sacs**
- Isolation and identification of *C psittaci* can be performed in **chick embryo or cell cultures** (BGM, L929, Vero)
- Stains such as Giemsa, Machiavello, Modified Ziehl Neelsen (MZN), are used to clearly visualize the elementary bodies (inclusion bodies) in impression smear
- **Chlamydiae stain purple with Giemsa and red with Macchiavello and Gimenez stains.**
- immunohistochemical staining of impression smears have been used

Serologic test

- Direct and modified direct complement fixation test
- Antibody ELISA
- Indirect immunofluorescence test
- Elementary body agglutination test detects IgM Ab
- The **complement fixation methods are more sensitive** than agglutination methods
- Molecular diagnosis-PCR

Differential diagnosis

Borreliosis

Pasteurellosis

Salmonellosis

Colibacillosis

Treatment

Tetracyclines (chlortetracycline, oxytetracycline, doxycycline) are the antibiotics of choice

Tetracyclines are bacteriostatic and effective only against actively multiplying organisms, making extended treatment times (from 2– 8 weeks, during which minimum-inhibitory concentrations in blood must be consistently maintained) necessary. When tetracyclines are administered orally, additional sources of dietary calcium (eg, mineral block, supplement, cuttle bone) should be reduced to minimize interference with drug absorption. Outbreaks of clinical disease in poultry flocks are not common. Treating infected flocks with chlortetracycline at 400– 750 g/ton of feed for a minimum of 2 weeks has effectively decreased potential risk of infection for plant employees

Chlortetracycline in turkeys- 200-800g / ton of feed or 0.2-0.4gm in one litre of drinking water

Doxycycline by oral route for complete elimination of infection

Medicated corn, mash diets with 1% chlortetracycline (CTC) and 0.7% calcium

White millet seed with 0.5 mg CTC/g of seed, for budgerigars, parakeets and finches only for 30 days

A special diet is necessary for Lories and lorikeets, which feed on nectar and fruit in the wild birds

Doxycycline added to feed or water can also result in adequate blood levels and has less effect on normal intestinal flora than does chlortetracycline.

Doxycycline 40-50mg/kg body weight by oral once a day for cockatiels, cockatoos and parrots

Doxycycline 75-100mg /kg body weight IM for every 5-7 days for the first 4 weeks

Prevention-

Vaccine is not available currently

Prohibit sale of infected birds

Control

quarantine and examination of all new birds

prevention of exposure to wild birds

traffic control to minimize cross-contamination

isolation and treatment of affected and contact birds

thorough cleaning and disinfection of premises and equipment (preferably with small units managed on an all-in/all-out basis)

provision of uncontaminated feed

maintenance of records on all bird movements
continual monitoring for presence of chlamydial infection
Disinfection of premises with, 1:1,000 quaternary ammonium chloride, 1:100 bleach solution, 70% alcohol, etc) but is resistant to acid and alkali

Chlamydiosis in livestock

Etiology

***Chlamydia abortus* - Small ruminants (sheep)- Enzootic abortion**

***Chlamydia pecorum*-Cattle- abortion**

***Chlamydia suis*- Swine-Abortion**

***Chlamydia felis*- Cat- Conjunctivitis**

Chlamydia pneumonia- Human

Chlamydia trachomatis-Human

Chlamydia muridarum-Mouse

Chlamydia caviae- Guinea fowl

Source of infection

- Ewes seldom abort more than once, but they remain persistently infected and shed *C abortus* from their reproductive tract for 2– 3 days before and after ovulation
- Infected ram
- Placenta, vaginal discharge, semen, ocular and nasal discharge
- Infected animal act as carrier for 3 yrs

Transmission

- Inhalation, ingestion, direct inoculation into the eye
- Veneral transmission

Clinical manifestations

Chlamydia abortus

- **Abortion in sheep and goats in late term pregnancy**, 2-3 weeks before expected date of parturition, stillbirths, and the birth of weak or premature offspring
- **Vaginal bleeding is sometimes observed during the 2 weeks before an abortion** particularly in goats

- **Stillborn lambs look relatively normal**, although the abdomen is sometimes distended by fluid.
- **Lambs** delivered by ewes with *C. abortus* infection may develop **acute chlamydial pneumonia**
- **reddish-brown vaginal discharge may be seen for several days after abortion or parturition**
- Occasionally, **polyarthritis, keratoconjunctivitis or respiratory disease**
- Subsequent pregnancy carried to term, there is no abortion

Chlamydia pecorum

- Pneumonia, dry hacking cough, dyspnoea
- Abortion in late pregnancy
- **keratoconjunctivitis,**
- **Encephalomyelitis**, usually occur in calves <6 months
- Stiffness and knuckling at the fetlocks tend to be seen initially, but progress to incoordination, staggering, circling, falling, recumbency.
- **polyarthritis-serositis**
- **pericarditis, hepatitis**
- **Enteritis -mucoid, watery or bloody diarrhea in < 10 days old calves**

Chlamydia suis

- Sows: abortions, stillbirths, early embryonic death and increased neonatal mortality
- **Piglets** : pericarditis, polyarthritis and polyserositis
- Boars: orchitis, epididymitis and urethritis
- conjunctivitis, respiratory disease-rhinitis, cough, dyspnea, pneumonia

Chlamydia felis

- Rhinitis, conjunctivitis and bronchopneumonia

***Chlamydia psittaci* in dog**

- Fever, bronchopneumonia, dry cough, keratoconjunctivitis, vomiting, diarrhea and neurologic signs sometimes

Necropsy findings

- Interstitial bronchopneumonia
- Necrotic damage in the placenta
- **thickened dark reddish-brown cotyledons**
- blood-stained fluid in the abdominal and pleural cavities
- **congestion or pinpoint white foci of necrosis in the liver**

Sample collection

- Placenta

- Nasal, ocular and vaginal swab
- Broncho-alveolar lavage
- Aborted fetus foetal abomasal content, liver and lung
- Impression smear from eye, conjunctiva

Diagnosis

- Chlamydial inclusion bodies may be detected in affected tissues
- Isolation of *chlamydia* sp in cell culture
- Immunofluorescence technique
- Antigen capture ELISA
- PCR

Treatment

- Tetracyclines or fluoroquinolones (eg, enrofloxacin)-drug of choice
- Treatment must start as early as possible and continue for at least 7 days

Prevention

- *C.abortus* live vaccines are available
- Vaccines against *C. felis* also available for pet cats

Control

- Isolating all affected ewes and lambs and treating in-contact ewes with long-acting oxytetracycline or oral tetracycline
- The replacement stock should be purchased from *Chlamydiosis* free source
- Cleaning and disinfection of premises with 1: 1000 quaternary ammonium chloride solution, 70% ethanol, formaldehyde, glutaraldehyde, sodium hypochlorite,
- Maintenance of personal hygiene in farm workers to prevent zoonotic transmission of disease

Zoonotic risk

C abortus is zoonotic, but human cases are rare.

All have involved pregnant women, who developed life-threatening illness.

Pregnant women should not work with pregnant sheep, especially if abortions are occurring.

Q Fever

(**Syn:** Coxiellosis)

Q fever was first described in 1935 in Queensland, Australia, during an outbreak of a febrile illness of unknown origin (Query fever) among abattoir workers. It was

subsequently classified as a “ Category “ B” critical biological agent” by the Centre for Diseases Control and Prevention and is considered a potential weapon for bioterrorism. Q fever is a public health concern throughout the world.

It is an occupational disease of animal handlers (veterinarian, abattoir workers and farm workers). Subclinical disease is more common than clinical infection, and endocarditis is the principal form of chronic disease in human. The infection occurs in any mammals, birds, reptiles and arthropods. It causes a mild disease in ruminants, but can cause abortions and still births in cattle, sheep and goats

Etiology

It is caused by *Coxiella burnetii*

It is an obligate, intracellular gram negative bacteria, non-motile pleomorphic - ovoid to rod shaped morphology

Two different antigenic forms of *Coxiella burnetii* are recognized based on surface variation of lipopolysaccharide (LPS)

Phase I- have a complete LPS on their surface and are virulent bacteria

Phase II antigen bacteria have an incomplete LPS due to a spontaneous genetic deletion of 25,992 bp and are nonvirulent.

The organism is resistant to external environmental influences and **viable in environment for 150 days**

The organism is susceptible to **sodium hypochlorite and formaldehyde fumigation**

Epidemiology

Prevalence of infection

The disease is worldwide in distribution except Newzealand

The rates of infection varies between farm animals, locations, regions and countries

The **prevalence is significantly higher in cattle as compared to sheep and goats.**

The organism prevails in **a wide range of wildlife species and their ecto-parasites (Ticks**

In cattle, the prevalence of 6-82% and 23-96% of herds has been reported

Predisposing factors

Those who with pre existing valvular heart disease and vascular graft

Transplant recipients

Patients with cancer

Those with chronic kidney disease are at risk of developing chronic Q.fever

Source of infection

C. burnetii is shed in milk, urine and feces.

Latent infection in udder and uterus become activated and excreted at parturition

High concentration of organism is present in **placenta, fetal fluids (Amniotic fluids)** and also excreted in urine and milk

The organism excreted through the **faeces of sheep from 11-18 days of post-partum**

Goats excrete the organism in milk and feces for upto 2 weeks to 52 days following kidding

In cattle, the organism are excreted through milk upto 2 yrs

Host affected

Sheep are at greater risk for infection

Cattle, sheep and goats are main reservoir of infection for humans

Non-pregnant animals maintained the silent infection followed by latency until the animals get pregnant

Transmission

Animal

Inhalation of dust contaminated with infective materials and direct contact with an infected animals

Parturient period may be the major period for transmission of infection

Outbreaks typically occur following a birth or abortion where the environment becomes contaminated with birthing fluids

Sexual and vertical transmission of infection is possible

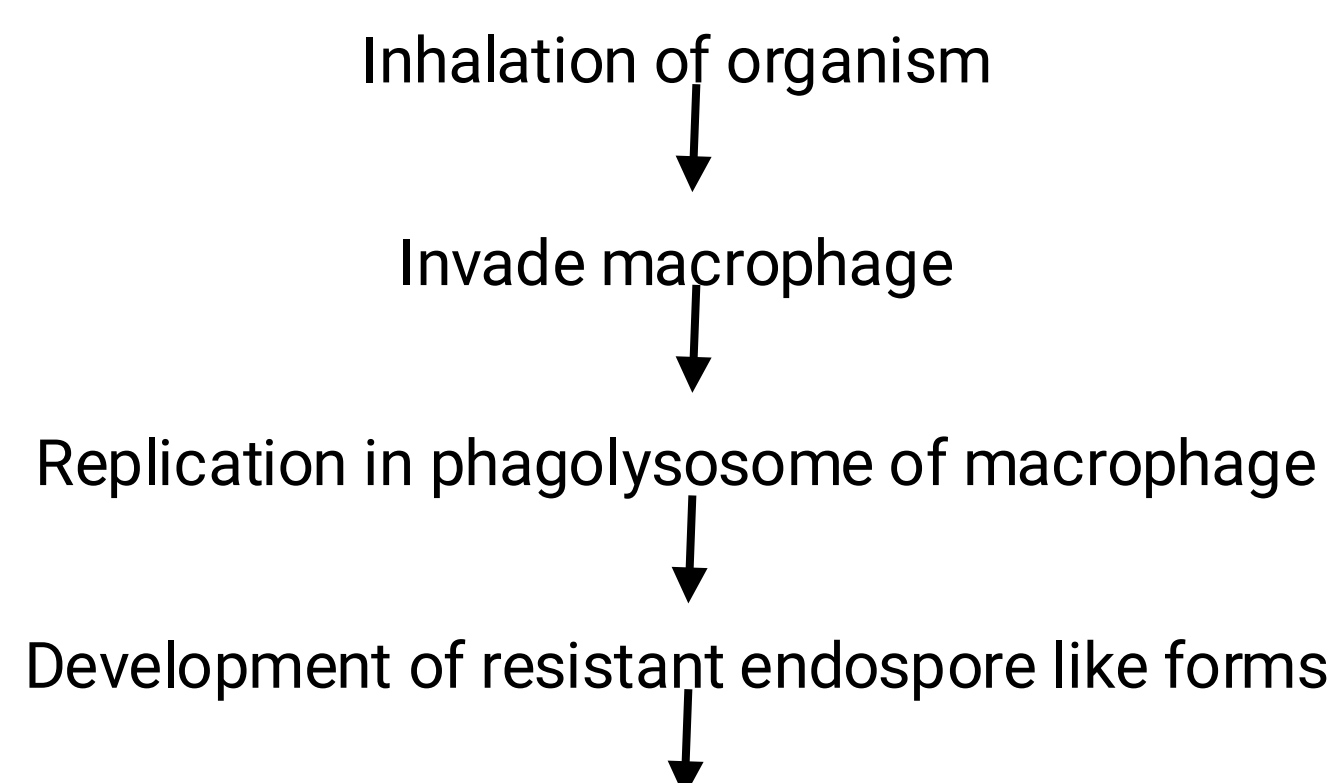
Tick also transmit infection

Human

Inhalation is the main mode of transmission

Ingestion of milk and milk products

Pathogenesis



Localize and replication in the female genital tract and mammary gland



Placentitis/Endometritis



Abortion in sheep and goat

Clinical signs

Ruminants

Infection occur at any age, usually clinically inapparent

Sheep and goat-**Abortion occur during late pregnancy**

Infection ensues abortion, stillbirth, and poorly viable lambs

Lambs born to ewes excreting the organism are found to be clinically normal

Though there is no abortion occurs in successive pregnancy in sheep but resurgence of infection occurs with excretion of organism immediately after the parturition

Cattle- **rarely causes abortion**

Human

Infection is commonly asymptomatic or mild in nature

Fever, general malaise, headache, muscle pain, sore throat, nausea, vomiting, chest and stomach pains and meningioencephalitis

The fever can last for one or two weeks, lead to pneumonia or affect the liver

Endocarditis and hepatitis are manifestation of chronic disease

Immunocompromised persons are at greater risk

Necropsy findings

Placenta from aborting animal is usually thickened

Purulent exudates in intercotyledonary area

Sample collection

Placenta, Aborted foetus, uterine discharge, faeces and milk

Diagnosis

Based on clinical signs

Smear prepared from placental issue and uterine discharge and stained with modified zhel nelson stain (MZN) reveal small clumps of red coccobacillary bodies

Immunoflourescence can be used to demonstrate the organisms in placental tissue

PCR are used to detect small number of organism in aborted material, milk and faeces

Culture of organism by yolk sac route of inoculation

Serological test- CFT, ELISA, indirect immunofluorescence and Competitive immunoassay. The anti-phase II antibodies (IgG and IgM) are found at high levels in acute Q fever, whereas anti-phase I antibodies (IgG and IgA) are found at high levels only during chronic infection.

Differential diagnosis

Chlamydiosis

Leptospirosis

Treatment

Doxycycline within the first 3 days of illness 100mg /oral twice daily for 15-21 days.

Hydroxyquinolone 200mg thrice a day for 3-6months.

Tetracycline 500 mg 4 times a day i/v or oral 2-3weeks.

Combination of Pefloxacin 800mg/day and Rifampicin 1,200 mg /day.

Chloramphenicol 500mg 4 times a day.

Prevention

Human-Vaccine: Australia-Q.Vax: whole cell formalin inactivated vaccine, 0.5 ml S/C upper arm. Immunity lasting for 5 years

In Australia, it is widely used to prevent occupational acquisition of Q fever and is the mainstay for outbreak control.

Embryonated inactivated hens eggs vaccine has been used in US

Vaccine trial with killed vaccine, show good persistent antibody response and suggest that vaccination can limit the excretion of the organism.

Control

Segregation of parturient ruminants

Careful disposal of placentas and aborted fetuses

Cleaning and disinfection of areas where animal gave birth to prevent spread of disease to animals and human

Inactivated egg-yolk vaccines are available for annual vaccination of non-pregnant ruminants

Anaplasmosis

(Syn: Gall sickness)

Anaplasmosis is an arthropod borne rickettsial disease of cattle and other ruminants caused by genus *Anaplasma* and clinically characterized by anaemia, jaundice and death.

Etiology

Anaplasma marginale – Cattle, Sheep and goat, buffalo and wild ruminants

Anaplasma centrale -, Sheep and goat, buffalo and wild ruminants

Anaplasma ovis- Sheep, Deer and Goat causes mild to severe disease same as *A. Marginale*

Anaplasma bovis- mainly infect Cattle and causes monocytic anaplasmosis

Anaplasma phagocytophilum- rarely causes clinical disease in cattle

Anaplasma phagocytophilum formerly known as *E. equi* and the agent of human granulocytic ehrlichiosis causes illness in dogs and horses

Anaplasma platys- mainly infect dogs and causes infectious cyclical thrombocytopenia

Epidemiology

The disease is endemic in tropical and subtropical region of the countries

Anaplasmosis is acutely disease of cattle, severe clinical infections do not occur in animal about 18 months of age

Calves are much more resistant than adult animals

Inapparent infection common in sheep and goat

The animal recovered from acute infection may remain chronically infected carrier and immune to further infection

Relapse may occur in immunosuppressed animal (eg. Corticosteroid therapy)

Transmission

Transmitted by ixodid ticks-*Boophilus* sp, *Rhipicephalus* sp, *Hyalomma* sp and *Dermacentor* sp and Mosquitoes

Transmission of infection in ticks are mainly transtadial and transovarian also possible in boophilus ticks

Transplacental (intra uterine) transmission also possible in animals

Iatrogenic transmission occurs through hypodermic needles, vaccination, castration, spaying, dehorning instruments, blood transfusions, and embryo transplants

Pathogenesis

Anaplasma are obligate intra-erythrocytic bacteria. They infect mature RBC by endocytic process and reproduction occur by binary fission to produce (2-8) initial infective bodies (2-8 cells) which leave the erythrocyte by exocytosis to infect the

other RBC. The number of infected erythrocytes doubles every 24-48 hrs and the infection become patent after 2-6 weeks after infection. The parasitized erythrocytes are removed by phagocytosis in the reticuloendothelial system with release of acute inflammatory reactants and the consequent development of fever

Clinical findings

Incubation period: 3-4 weeks with tick borne infection

Peracute form

It occur when more than 50% of RBC are parasitized by anaplasma sp, occur after 3years of age. The affected animal shows fever, anemia, icterus, severe dyspnoea and death within 24 hrs. The affected animals are hyperexcitable and tend to attack the attendant before death.

Acute form

It occurs when 10-90% of RBC is parasitized by anaplasma sp. The 15% of RBCs are parasitized before exhibiting clinical signs

Fever (fluctuating) temp. 40.5 °C

Anorexia is seldom complete, progressive anemia and icterus are noticed at the time of fever or afterwards

Laboured breathing, constipation, cessation of rumination, decreased milk production and frequent urination and death, **surviving animal shows emaciation**

Blood is thin and watery in nature

Urine may be brown colour in late stages

Abortion is a common feature in advanced case of pregnancy (3rd trimester)

Infected bulls during acute infection show lack of libido with abnormal sperm and transient testicular degeneration

Subacute form

It occurs when 10% of RBC is parasitized by organism, most common in young animals.

Chronic case

Severe anemia and recovery is slow and susceptible to numerous viral, protozoan infections.

Sheep and goat

Infection is usually subclinical, but in some cases particularly in goats a severe anemia may occur. The severe reaction in goat occur when animal suffering from concurrent disease. Goats may show hyperexcitability and may bite inanimate objects.

Post mortem finding

Splenomegaly, hepatomegaly and liver suffused with bile (yellow orange in colour)

Gall bladder enlarged with dark thick bile

Enlargement of lymph node

Pulmonary edema

Petechial haemorrhage in epicardium of heart

Sample collection

Whole blood, peripheral blood smear and serum

Diagnosis

Microscopic examination of giemsa stained blood smear reveal *Anaplasma* sp at the periphery of RBC

Hematological changes- RBC count reduced to 1.5million/ul, Hb-3g/dl and leukocytosis

Serological test like Complement fixation test, Capillary tube agglutination test Card agglutination test and Dot ELISA and FAT used for detection of antibodies in animals

Complement fixation test is the standard test for the detection of carrier animals

DNA probe- used to detect infection in carrier and tick gut

RNA probe-used to detect carrier status

Polymerase chain reaction (PCR)

Differential diagnosis

Theileriosis

Babesiosis

Trypanosomiosis

Eperythrozoonosis

Treatment

Inj. Oxytetracycline @10 mg/kg b.wt for three days or single injection of long acting tetracycline @20 mg/kg.bwt intramuscularly

Blood transfusion in animals with PCV<15%

Imidocarb @ 3mg/kg b.wt

Prophylactic treatment in rest of the herd

Temporary protection in the face of exposure risk is achieved with a single injection of long acting tetracycline @ 20 mg/kg.bwt intramuscularly

Prolonged protection can be achieved by intramuscular injection of long acting tetracycline @20 mg/kg every 28days.

For elimination of infection- Oxytetracycline @10 -30mg/kg b.wt for 10-16 days
or intravenous injection of Oxytetracycline @ 22mg/kg b.wt for five days
Long acting Oxytetracycline @20 mg/kg I/M for every 7days for 2-4 injections

Control

Arthropod control with acaricide

The weekly spraying or dipping of animals in acaricide solution in an endemic area

Chemotherapy for prevention

Vaccination – both live and inactivated vaccines are available

The vaccination induces protective immunity, but will not prevent infection

Prevent introduction of carrier animals by screening

Avoid iatrogenic transmission through use of disposables and sterilization

Ehrlichiosis

Ehrlichiosis is a tick-borne disease of dogs and causes self-limiting to generalized fatal disease and clinically characterized by fever, generalized lymphadenopathy, splenomegaly and thrombocytopenia

Etiology

It is caused by ***Ehrlichia canis***-most common and causes **canine monocytic ehrlichiosis (CME)**

E.canis- can infect cats and rarely causes clinical diseases

E. ewingii- mainly infect **granulocytes** and causes **canine granulocytic ehrlichiosis**

E. chaffeensis causes human monocytic ehrlichiosis in the USA

E.muris-like agent-rarely reported

It is a gram negative aerobic intracellular rickettsial bacteria reclassified into the genera of *Ehrlichia*, *Anaplasma* and *Neorickettsia*, all of which are in the family Anaplasmataceae

Coinfection of *E. canis* and *A. platys* may occur, because the same tick vector is responsible for transmission of both pathogens

Epidemiology

The disease is worldwide distribution including Asia, Africa, Europe, America and Japan

Australia is free from *E.canis* infection

E.canis is most commonly reported in tropical environment

E. ewingii is found in South-Central/South-Eastern USA

E. chaffeensis – found in Southern USA

E. muris-like agent-Midwest USA

CME is most common during summer months due to abundant tick population

Zoonotic risk

E. chaffeensis, *E. ewingii*, and *A. phagocytophilum* are **considered zoonoses**

Host affected

Dog, cat and human are susceptible

Breed- German shepherd and Doberman pinscher are prone to infection

Transmission

E. canis is transmitted by the brown dog tick, *Rhipicephalus sanguineus*, the ticks become infected with *E. canis* after feeding on infected dogs and ticks transmit infection to other dogs during blood meals taken in successive life stages

Transtadial transmission has been reported in ticks

Blood transfusions may also transmit the pathogen

E. chaffeensis and *E. ewingii* have sylvatic cycles in the environment that involve tick species and wildlife reservoir hosts

Pathogenesis

The organism replicates in circulating monocytes and tissue macrophage in affected tissue and multiplies by binary fission and spread throughout the body. The infected monocytes adhere to vascular endothelium and causes vasculitis, which is primary pathogenic mechanism. The Ehrlichia sp (late stage of morulae) released from infected cells by rupture of host cell membrane

Clinical findings

Incubation period: 8-20 days

Many dogs experience self-limiting infection, but some may develop a secondary quiescent phase of disease followed by a severe and life-threatening chronic stage

The clinical symptoms manifested as acute, subclinical or chronic

Acute ehrlichiosis

Fever, reticuloendothelial hyperplasia, generalized lymphadenopathy, splenomegaly and Anorexia, depression, loss of stamina, **stiffness and reluctance to walk, edema of the limbs or scrotum, and coughing or dyspnea, petechial or ecchymotic haemorrhages on skin of ventral abdomen and frequent epistaxis** due to thrombocytopenia

The infection is complicated by co-infection with other pathogens such as *Babesia canis vogeli* and *Hepatozoan canis*

Neurological symptoms in dogs due to meningitis and meningeal haemorrhage, seizures, stupor, ataxia, twitching and vestibular signs with upper and lower motor neuron dysfunction

E. canis could be risk factor for myocardial injury & increased level of cardiac troponin I was observed

Acute disease may last for 2-4 weeks

Death is rare during this phase; spontaneous recovery may occur the dog may remain asymptomatic

Chronic Ehrlichiosis

In chronic cases, the **bone marrow becomes hypoplastic**, and lymphocytes and plasmacytes infiltrate in various organs

Vasculitis and immune-mediated mechanisms induce a thrombocytopenia and a risk of bleeding

marked splenomegaly, glomerulonephritis, renal failure, interstitial pneumonitis, **anterior uveitis** and meningitis (with associated cerebellar ataxia, depression, paresis, and hyperesthesia)

Emaciation

Clinical manifestation of *E. ewingii*-

fever, lethargy, anorexia, **neutrophilic polyarthritis, vomiting, diarrhea, stiff gait, head tilt, anisocoria and tremors**

Necropsy finding

petechial and ecchymotic haemorrhages on serosal and mucosal surface on most of the organs

Generalized lymphadenomegaly, splenomegaly and hepatomegaly

Histopathology: Perivascular plasma cell infiltrate in numerous organs (Lungs, brain, meninges, kidneys, lymph nodes, bone marrow, spleen and skin
CNS-Multifocal, nonsuppurative meningoencephalitis especially brain stem, midbrain and cerebral cortex

Eye- subretinal haemorrhage and retinal detachment

Clinical pathology

Acute case

Hemogram: mild normocytic, normochromic anemia, leukopenia or mild leukocytosis and Thrombocytopenia

Lymph node aspiration reveals hyperplasia

Chronic cases

Hemogram- nonregenerative anemia, severe pancytopenia (mature leucopenia) and severe thrombocytopenia

Prolonged activated partial thromboplastin time (aPTT)

Hypergammaglobuliemia

Aspiration cytology of lymphnode reveals marked plasmocytosis

Sample collection

Whole blood, serum, bone marrow aspirate and splenic aspirate

Diagnosis

Based on clinical signs

Hematology: Anaemia and thrombocytopenia

Demonstration of intracytoplasmic inclusion bodies (morulae) in monocytes of Giemsa stained blood smear and smear from buffy coat

Serological test-enzyme immunoassays and immunofluorescence antibody test is commonly used for. Indirect fluorescent antibody test (**IFAT**) is **gold standard serological test for diagnosis of Ehrlichiosis**

Testing of paired sera and demonstration of four-fold increased antibody titers is recommended to confirm infection when possible

Serologic cross-reactivity is strong among *E canis*, *E chaffeensis*, and *E ewingii*, and among *A phagocytophilum* and *A platys*

PCR can also be used to detect the effectiveness of treatment in clearing infection

Differential diagnosis

Acute stage

Rocky Mountain spotted fever
brucellosis
blastomycosis
endocarditis
systemic lupus erythematosus
lymphosarcoma

Chronic stage

estrogen toxicity
myelophthisis
immune-mediated pancytopenia
multisystemic diseases associated with specific organ dysfunction (eg, glomerulonephritis)

Treatment

Doxycycline is the treatment of choice for all ages because of superior intracellular penetration and bacteriostatic properties against rickettsiae

Doxycycline: 5– 10 mg/kg/day, PO or IV, for 10– 21 days (drug of choice)

Tetracycline: 22 mg/kg PO, three times a day (≥ 2 weeks, acute cases, 1– 2 months, chronic cases)

Imidocarb dipropionate: 2 doses, 5– 7 mg/kg IM, 2 weeks apart (also effective for some strains of babesiosis)

Chloramphenicol 15– 25 mg/kg PO, SC, or IV TID for 14-21 days has been recommended for puppies younger than 5 months

Blood transfusion in severely anaemic animals

Erythropoietin and granulocyte colony stimulating factor with prednisolone may be value in severe chronic ehrlichiosis

Desmopressin (DDAVP) @1µg/Kg S/C for 3 days may given to stop bleeding within one hour

Supportive therapy- short term (2-7 days) therapy with low dose of prednisolone @1 -2 mg/Kg PO may be effective in severe life threatening conditions

In acute cases receiving appropriate antibiotic therapy, body temperature is expected to return to normal within 24– 48 hours after treatment.

In chronic cases associated with *E. canis* infection, the hematologic abnormalities may persist for 3– 6 months, although clinical response to treatment often occurs much sooner

If a dog is thought to have chronic ehrlichiosis, the *E. canis* antibody titer and whole blood PCR should be repeated after 6 months of illness to confirm successful therapy.

Serum titres that persist at lower but positive levels should be rechecked in another 6 months to ensure they are not increasing.

Animals with anemia or severe thrombocytopenia should be treated with platelet or whole-blood transfusions as appropriate, particularly if hemorrhage is extensive. Concurrent broad-spectrum antibiotics may be needed if a dog has severe leucopenia

Prevention

Control of ticks on dogs

Multiple topical medications are available to prevent tick bites- *R sanguineus* infestations can be problematic in kennels and around homes, and long term tick control is needed for management

use of effective long-acting collars on all susceptible dogs might be considered; collars containing **propoxur**, **amitraz (0.025%)**, or **flumethrin** **have proven activity against *R sanguineus***

Flumethrin collars for cats and pyrethroid or amitraz collars for dogs are used for tick control

Prevention of transfusion-associated transmission by screening of donors

Prophylactic administration of tetracycline at a lower dosage (6.6 mg/kg/day, PO) is effective in preventing *E canis* infection in kennels where disease is endemic

In endemic areas, long-term doxycycline 3 mg/kg PO every 24 hours has been used prophylactically to prevent reinfection

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GLOSSARY

Acid- fastness

It is a physical property of bacterium resist decolouration by acids during staining procedures

Acid-fast organism

Acid-fast organism, when stained with carbal fuchsin, it solublizes the lipid material in the mycobacterial cell wall but by application of heat, carbol fuchscin further penetrate through lipid cell wall and enter the cytoplasm. Then after **bacteria or organism appears red** due to resistant to decolouration and not does not allow the counter stain (Methylene blue) to penetrate the cell wall. Example. *Mycobacterium* sp and *Nocardia* sp

Active immunity

Active immunity is defined as **immunity to a pathogen that occurs following exposure to pathogen**. When the body is exposed to a novel disease agent, B cells, a type of white blood cell, create antibodies that assist in destroying or neutralizing the disease agent.

Adjuvant

Adjuvant is an ingredient used in some vaccine that helps create a stronger immune response. Aluminium salts such as aluminium hydroxide, aluminium phosphate, aluminium potassium sulphate (Alum) and oil are commonly used in vaccine.

Aerosol

A liquid agent or solution dispersed in air in the form of a fine mist

Aflatoxin

Toxins produced by fungi *Aspergillus flavus* and *Aspergillus parasiticus* causes poisoning in animals

Alpha toxin

The *C. perfringens* type A produce α -toxin, which possesses phospholipase C and sphingomyelinase activity and consequently hemolytic action. The presence of β -2-toxin in these strains may also contribute to the pathogenicity

Antigen

An antigen has been defined as any substance which when introduced parenterally into the body stimulates the production of an antibody

Anti-tetanus serum

Anti-tetanus serum or Tetanus Antitoxin made up of antibodies against the tetanus toxin is used for prophylaxis and treatment of Tetanus

Beta toxin

Beta toxin (β -Toxin) is a trypsin-sensitive toxin produced by *C. perfringens* types B and C causes hemorrhagic enteritis and ulceration of the intestinal mucosa resulting in diarrhea and dysentery in young lambs, goats, calves, pigs, and foals. β -Toxin is encoded by the gene CPb. Cpb gene has significant homology with alpha toxin, g toxin and leukocidin of *staphylococcus aureus*.

Capsular antigen (K)

It is an important virulence factor in *Bacillus anthracis*, *Pasteurella* sp and *Klebsiella* sp

Haptens

Haptens are incomplete antigen and incapable to induce antibody formation by themselves

Complete antigen

It able to induce antibody formation by themselves

Cell mediated immunity

It is the arm of immune response that does not involve antibodies but rather incorporates the activation of macrophage and Natural killer cells (NK Cells) enabling them to destroy intracellular pathogens

Cytokines

Cytokines are mediators that function as up and down regulation of immunogenic mechanism

Epsilon toxin

Epsilon toxin is produced by *C. perfringens* type D and most important to causes vascular damage and the damage to the nervous system typical of this disease. Proteolysis of epsilon proto-toxin converts it to epsilon toxin resulting in a > 1000 fold increase in toxicity. Epsilon toxin is necrotizing toxin produced by *Clostridium perfringens* type D

Humoral immunity

Humoral immunity is an aspect of specific immune responses directed at particular antigens. It takes the form of unique antibodies produced by B Lymphocytes that have been specifically selected to neutralize the antigen

Hyperimmune plasma

Hyperimmune plasma is plasma that has been collected from horses vaccinated against a specific pathogen to induce high serum concentrations of antibody against gram-negative LPS especially for treatment of *Rhodococcus equi*, *Escherichia coli*, *Clostridium botulinum*, *Salmonella* spp, West Nile virus, *Streptococcus equi subsp equi*, and rotavirus infections in horse

Idiopathic cause

For any consequences / effects for which the cause is not known. Disease or anything which is caused by undefined sources

Interferon

Interferons are group of signalling proteins released by virus infected cells and inhibit viral replication or induce antiviral state within the host cell. There are 3 types interferon alpha produced in the leukocyte infected with the virus. Interferon beta

produced by fibroblast and interferon gamma (IFN- γ) produced by sensitized lymphocytes with antigen

Mycotoxin

Toxin produced by fungi in food grain and causes toxicosis in animals

Passive immunity

Passive immunity refers to the transfer of antibodies from one individual to another, which is either maternal or artificial, but the immunity is short lived and does not require previous exposure to pathogen or disease agent.

Pathogenicity: The property of causing disease

Risus sardonicus

It is a characteristic, abnormal and progressive spasm of the facial muscles that appears to develop grinning. There is a raising of eyebrows and an open grin noticed. This condition is most often seen in Tetanus, Strychnine poison, and Water dropwort herb problem. The water dropwort (*Oenanthe crocata*) has been pronounced as responsible for sardonic grin and hence the plant was ascribed as sardonic herb as this plant initialized the neurotoxic effect, this had been used as a ritual killing product for elderly people in Roman Sardonia. Hence the name arise Risus sardonicus.

Somatic antigen (O)

It is important virulence factor in *E.coli* and *Salmonella* sp

Tetanospasmin

Tetanospasmin is a potent neurotoxin produced by *Clostridium tetani* and inhibits the release of gamma amino butyric acid (GABA) and causes spastic paralysis

Tetanus immune globulin (TIG)

Anti-tetanus immunoglobulin, also known as tetanus immune globulin (TIG) and tetanus antitoxin is a medication made up of antibodies against the tetanus toxin

Virulence

Virulence is defined by the ability of a microorganism to cause disease in the host